Important Notice

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1 General Information

Hamilton's MICROLAB® STAR Line is the next generation pipetting workstation. This Operator's Manual is designed to help you to get the most out of your MICROLAB® STAR Line.

Hamilton's MICROLAB® STAR and MICROLAB® STAR^{let} are in the following summarized as ML STAR.

You should read through the entire manual before beginning to operate your instrument. This first chapter should be read with particular attention. It contains important information about the use of the ML STAR instrument and this manual.

1.1 About this Manual

This manual is meant to help users to operate the ML STAR Line instruments correctly and safely.

To achieve that goal, the manual will describe the different components of the ML STAR Line and how they work. The manual describes both the hardware and software of the ML STAR Line instruments in a depth enabling the user to operate the instrument.

After introducing you to the various parts of the ML STAR Line, we will show you step by step how to operate the instrument. When you have worked through this manual, you should be capable of operating the ML STAR Line.

Warnings and ***notes*** are included in this manual to emphasize important and critical instructions. They are printed in italics, begin with the word "Attention" accompanied by the symbol, or the word "Note" accompanied by the symbol, as appropriate.

[...]	Push buttons and their corresponding description.
"..."	Description for all kinds of entry fields, control fields, check boxes, lists, etc.
___	References to Manuals, figures, sections, etc.

This manual refers to VENUS two Software for the ML STAR.

1.2 Additional ML STAR Line Manuals

For the programmer of the system, the Programmer's Manual (P/N 624043_02) describes all the features of the VENUS Software. Sample methods for typical applications guide you through the programming. A detailed software reference for the ML STAR Line can be found in the Help menu of the VENUS Software. This Help Menu will answer any question you may have about details of the VENUS Software .

The Total Aspiration and Dispense Monitoring (TADM) is an additional safety tool for the pipetting processes. The TADM features allow optimization of the entire pipetting process by reading out the pressure curves of the pipetting process and comparing it to a tolerance band in real-time. The description of its functions and how to work with TADM is described in the TADM Manual (P/N 624031).

The VENUS Dynamic Scheduler is a software tool for organizing and controlling the workflows of a laboratory equipped with the HAMILTON Instruments and other manufacturers Instruments. The use of this tool is described in the VENUS Dynamic Scheduler User Manual (P/N 624030).

The VENUS Data Base Plus is a software tool that allows using additional functions on the VENUS Database. It allows tracking over multiple runs and the use of an SQL Server via the network. The Use of this tool is described in the Help-Menu of the VENUS Database Plus.

1.3 Intended Use of the ML STAR Line

The ML STAR Line is a robotic pipetting workstation, in other words, a sampler used for pipetting liquid samples in an automated process suitable for medium to high throughput with a high degree of flexibility in pharmaceutical, veterinary and genetics applications.

A user will typically wish to carry out low, medium, or high volume contamination-free pipetting with disposable tips or with steel needles.

At the present time, the ML STAR Line is classified as a general laboratory instrument and is not an *in vitro* diagnostic device.

1.4 Operation

The operator of the ML STAR must have attended an appropriate training course. The procedures contained within this manual have been tested by the manufacturer and are deemed to be fully functional. Any deviation from the procedures given here could lead to erroneous results or malfunction.

Training courses will be held by your HAMILTON representative. Please feel free to contact your local representative to arrange for an operator training.

1.5 Safety Precautions and Hazards

The following section describes the main safety considerations, electrical and biological, in operating this product, and the main hazards involved.

ATTENTION

Read the following safety notices carefully before using the ML STAR instrument.

1.5.1 General Precautions

1.5.1.1 Instrument

The ML STAR Line instruments conforms to European norms as regarding the interference immunity. However, if the ML STAR Line is subject to electromagnetic RF fields, or if static electricity is discharged directly onto the ML STAR instrument, it's Liquid Level Detection ability may be negatively affected. It is therefore recommended that the ML STAR instrument is kept away from other equipment that emits electromagnetic RF fields in the laboratory, and that static electricity is minimized in its immediate environment.

During operation, the ML STAR instrument should be shielded from direct sunlight and intense artificial light.

The instrument should be positioned in the laboratory in a way permitting personnel to access the front and sides of the instrument in order to operate, maintain, open and close the protective covers. Accordingly, to calculate how much room is needed, consider the dimensions of the instrument (see chapter 7 "Technical Specifications") and sufficient room for a person to move and work comfortably.

Never lift a fully installed instrument to transport from one place to another. It must be reinstalled in the new work location by an authorized service technician.

The instrument weighs more than 150 kg. Necessary precautions should be taken when transporting the instrument.

Only certified technicians are authorized to perform mechanical maintenance on the ML STAR instrument.

For repair or shipment, all mechanical parts must be put in their initial positions. An instrument sent away for repair, must also be decontaminated (see chapter 5 "Decontamination") if it was in a laboratory environment with infected or hazardous materials. The ML STAR instrument must be repacked in the original shipping crate only by an authorized service technician (contact your local HAMILTON representative). There must be no containers or tips on the ML STAR instrument during transportation.

The service technician and the laboratory share the responsibility for the installation qualification (IQ) and the operation qualification (OQ), i.e. verification and training. The process qualification (PQ) is the sole responsibility of the laboratory.

Only original HAMILTON STAR Line-specific parts and tools may be used with the ML STAR instrument, e.g. carriers, racks, tips, steel needles, and waste containers. Commercially available liquid containers, such as microtiter plates and tubes, may of course be used.

A breakdown of the power supply during a run, may cause a loss of data. If data loss is unacceptable, use an independent power supply.

1.5.1.2 Operating the Instrument

When using ML STAR instruments, Good Laboratory Practices (GLP) must be observed. Suitable protective clothing, safety glasses and protective gloves must be worn, particularly when dealing with a malfunction of the instrument, where the risk of contamination from spilled liquids exists.

During instrument operation, stand clear of any moving parts or the working deck of the instrument. In general, never lean over the instrument when it is working.

1.5.1.3 Method Programming

Perform test runs first with water and then, with the final liquids, prior to routine use. Test for all the liquid classes you are going to use. A newly programmed test method must first be run on the ML STAR instrument with the final liquids, prior to validation of the method and routine use. The method programmer should supervise this run.

Before using any newly created or modified method for routine test purposes, a comparison study between the method previously used and the new one must be carried out by the laboratory supervisor to ensure that the processing and data evaluation of both methods produce equal results.

When working with samples, which will be used in particularly sensitive tests, take into account the evaporation and condensation that may occur while the method is running.

If sampling aggressive liquids, use filter tips. Also use filter tips for tasks, which are sensitive to cross-contamination (aerosols).

The inner diameter of sample tubes, reagent vessels, etc. must be greater than the channel diameter of 9mm, if working with 1000µl Pipetting Heads. It must be greater than 18mm, if working with the 5ml Pipetting Heads.

Liquid level detection needs to be explicitly tested, when working with foaming liquids. Foam may affect the accuracy of liquid level detection.

Never disable any security measure.

1.5.1.4 Loading

Do not exchange positions of sample and reagent tubes around, or switch microplates after they have been identified by the barcode reader. This could result in incorrect test data or instrument crash.

Microplates must be placed on the carrier such that well A1 is in the position defined in the deck layout of the method.

When pouring liquid into the containers, ensure that there is no foam on the surface of the liquid. Note that foam may cause pipetting problems.

Do not overfill reagent containers, tubes, or other liquid containers.

Do not mix tip size and type (e.g. with or without filter, or different volumes) in the same tip rack.

Do not fill up partially consumed tip racks with tips from other racks. Tip Racks should be loaded into the Tip Rack Carriers as they are provided in the original package. The Tips are individually labeled with a barcode for identification.

1.5.1.5 Work Routine

Do not try to open the front cover of the instrument during a run because the system will abort and this may cause a loss of data.

If the system is paused, do not wait too long before resuming the run. Loss of liquid from a full tip may result in invalid data.

Discard used tips. Do not reuse them.

Do not empty the tip waste during a run.

Do not leave tips on the pipetting channels for a long period of time (for example overnight). This may cause damage to the CO-RE O-rings. A daily maintenance procedure will remove the tips.

1.5.2 Biohazard Precautions

If the ML STAR instrument becomes contaminated with biohazardous or chemical material, it should be cleaned in accordance with the maintenance procedures. The procedures are described in the chapter 4 "Maintenance". Observe and carry out the maintenance procedures given. Failure to do so may impair the reliability and correct functioning of the STAR instrument.

If working with bio-hazardous samples, observe and carry out the maintenance procedures, paying particular attention to cleaning and decontamination. Wear gloves when handling the pipetting arm and channels, the carriers, racks, and containers, and the tips. Avoid touching tips discarded into the laboratory-supplied waste container. Any surfaces, on which liquid is spilled must be decontaminated.

Do not use disinfecting materials, which contain hypochlorite (Javel water, Chlorox) or bleaching fluids.

If working with biohazardous or chemical materials, the user does not need to touch them. The instrument will drop its used tips into a waste container, that should be emptied as soon as it is full.

1.5.3 Computer Precautions

Use the necessary precaution to guard against software viruses. Use only manufacturer's original installation DVD/CD-ROM sets for the operating system, and the original HAMILTON VENUS Software.

Running other software parallel to the VENUS Software may negatively affect the running of ML STAR instrument.

Any manipulation of Venus Software data files or other information determining or affecting ML STAR Line functions can result in erroneous test results or instrument failure.

Only the VENUS Software may be used to control the instrument.

For reasons of data security and integrity, the use of an uninterrupted power supply (UPS) is recommended, since a loss of power may cause data to be lost or corrupted.

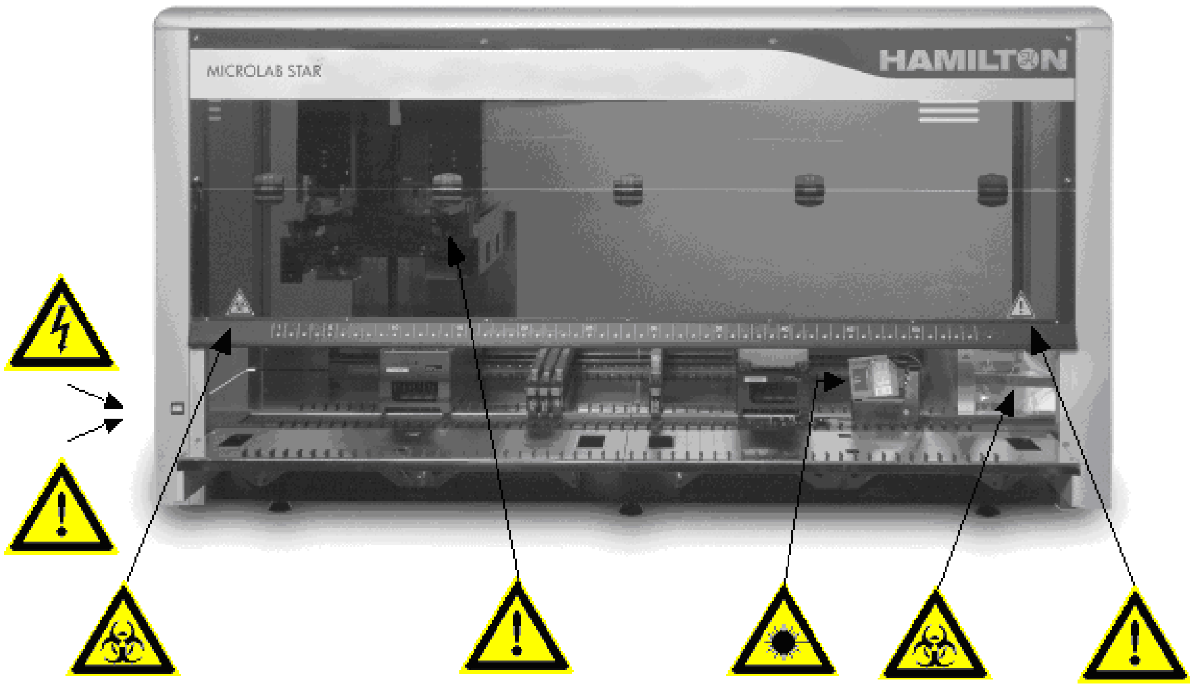
To avoid computer breakdowns, configure a hard disk of sufficient space in the computer. Ensure that there is always enough storage capacity on your hard drive. Delete the log files from time to time. Generated data within the Log files directory, e.g. traces, TADM data and pipetting files, should be backed up on your laboratory's host device and deleted from the control PC's hard disk at weekly intervals.

1.5.4 Electrical Safety Precautions

Before removing a mechanical or electrical component, first the ML STAR instrument must be switched off and disconnected from the main electricity supply and PC.

1.5.5 Hazards

Location and explanation of warning and attention labels:



	Power connection: Connect only to earth-grounded outlet		Barcode reader laser beam: Do not stare into beam of class 2 laser
	Connection to PC: Use only the appropriate shielded cables		Moving parts: Moving arm inside transparent cover. Aborts the run if cover is opened.
	Biohazard warning: Deck may contain biohazardous or chemically contaminated materials		Biohazard warning: Waste may contain biohazardous or chemically contaminated materials
	USB connection Having a total cable distance of more than 5m, signals can be interfered.		

2 Description of the ML STAR Line

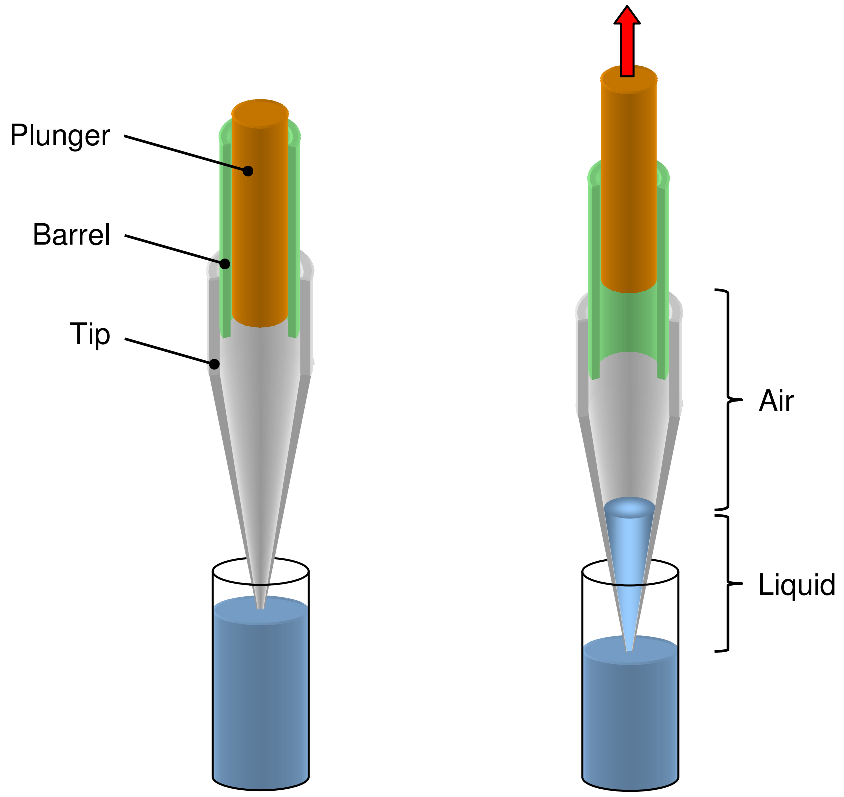
2.1 ML STAR Line

STAR stands for Sequential Transfer and Aliquoting Robot. The ML STAR instruments performs both, pipetting operations on liquids in containers and transferring of microtiter plates placed on its work surface.

Pipetting means transfer of small quantities of liquid from one container to another. A pipetting operation is achieved by aspirating (drawing) liquid from a source container, then transferring and dispensing (dropping) it into a target container.

2.1.1 The Air Displacement Pipetting Principle

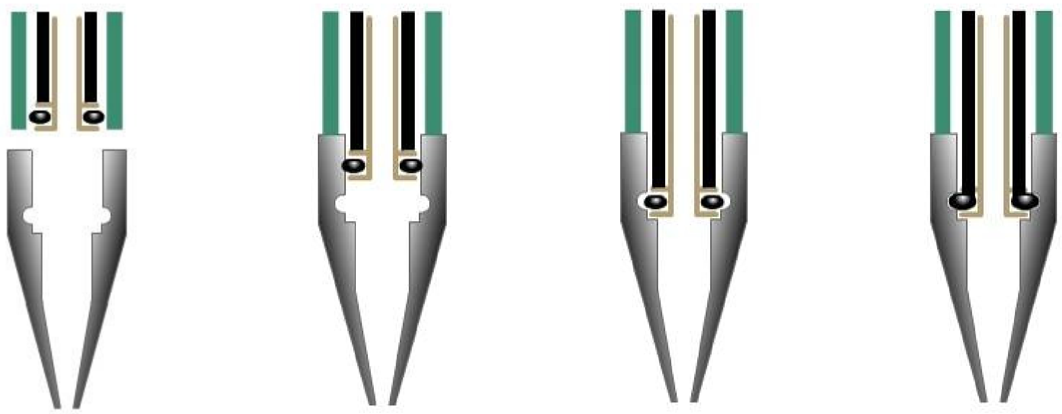
The ML STAR instruments are based on the **air displacement pipetting** principle, comparable to the functioning of hand pipettes. Air displacement means that the liquid is aspirated into and dispensed from a disposable tip or needle by the movement of a plunger. Between the plunger and the liquid surface is air. No system liquid of any kind is involved in the instument.



The Air Displacement Pipetting Principle

2.1.2 Tip Pick-up with the CO-RE Technology

The first task for the ML STAR Pipettor is to pick up a disposable tip or a reusable steel needle. Due to the unique CO-RE (compression-induced O-Ring expansion) technology precise tip attachment and positioning is achieved. The system requires no vertical force for tip attachment or tip ejection, thus eliminating mechanical stress and improving overall system reliability along with pipetting speed and dexterity.



Patent no: *EP 1171240*

US 7033543

The principle of the CO-RE technology has the following advantages:
- Enabling the coupling of disposable tips or washable needles within the same run
- Allowing different sizes of tips on the same pipetting head in the same run.
- Picking up a gripper and other tools.
- Eliminating aerosol production upon tip ejection.

2.1.3 Type of tip recognition

This feature is to increase the instrument's security, especially when more than one tip type is being used, e.g. low- and standard-volume tips. It is available for both disposables and needles and it will be activated during installation by service engineers.

For the recognition of 50µl and 300µl disposable tips a library to determine the tip type is available.

2.1.4 Liquid Level Detection: LLD

The liquid level of the container to be aspirated from can be detected. This can be provided by ML STAR Line's Liquid Level Detection (LLD) feature, based on either capacitive (cLLD) or pressure (pLLD) signal detection. Normally capacitive LLD is used for conductive liquids. The sensitivity of the capacitive LLD that is to be used depends on the vessel size, volume and the conductivity (or polarity) of the liquid that is to be detected.

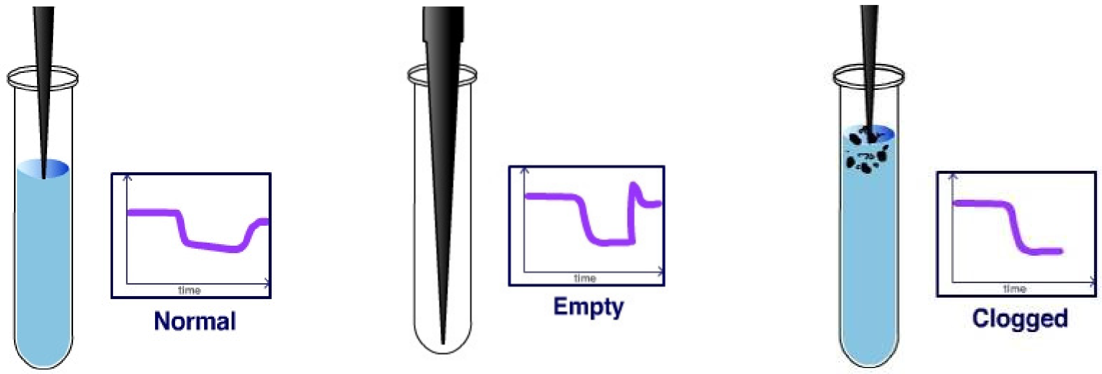
For non-conductive liquids, or in case of an insufficient coupling between container bottom and carrier, pressure LLD is used. Pressure LLD only works with new and empty tips for the aspiration of liquids. The pLLD is available on the individual Pipetting Channels only.

In the case of detecting under demanding circumstances, as e.g. detecting foaming liquids, the capacitive and the pressure liquid level detection can be used at the same time.

2.1.5 Monitored Air Displacement: MAD

The ML Line is equipped with an aspiration monitoring feature. During the aspiration process, the pressure within the pipetting channel is measured in real time. Analyzing the shape of the p(t) curve, the system can distinguish the following situations:

- A correct aspiration takes place.
- Air is aspirated into the tip (because, for example, the container has not been filled properly).
- A clot blocks the tip.



Aspiration monitoring based on pressure.

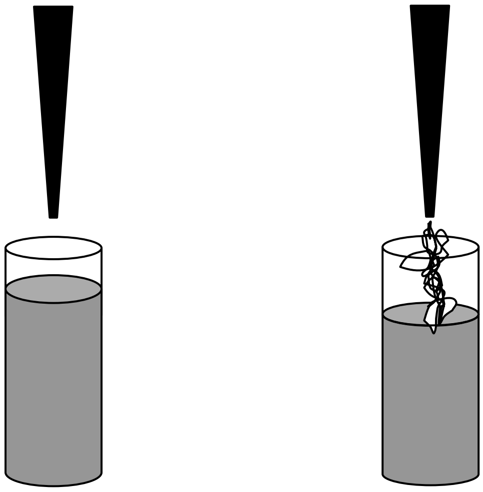
The aspiration monitoring is available on the individual Pipetting Channels only.

The diagram shows the functioning of aspiration monitoring based on pressure.

2.1.6 Capacitance-Based Clot Detection

In addition to pressure-based clot detection, the ML STAR Line is equipped with capacitance-based clot detection. This detection approach works in the case of aspiration with capacitance liquid level detection switched on. The system measures the conductive signal when the tip leaves the liquid after aspiration. Due to the air gap between tip and liquid, the capacitance signal will vanish once a given height is reached. If a clot is present, it bridges the distance and the signal will remain, resulting in an error message. This clot detection is independent of pressure-based monitoring.

The capacitance-based clot detection has to be activated in the Hamilton Service Software.

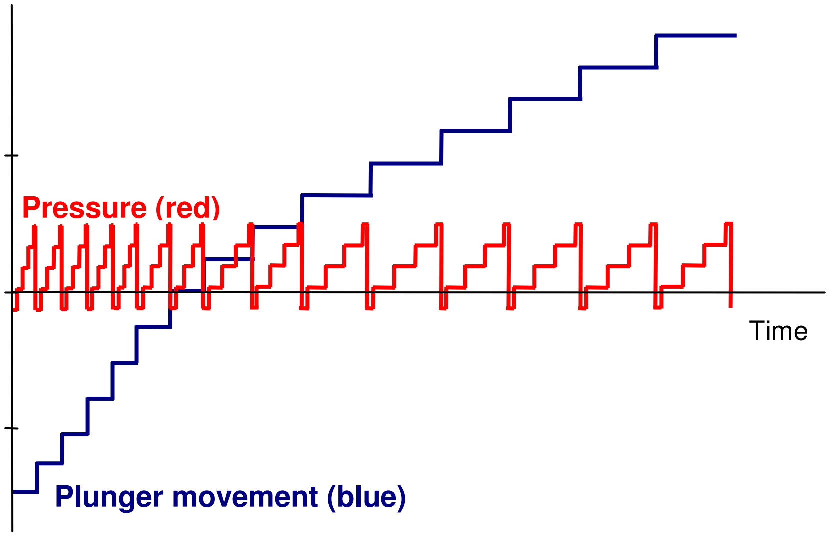


2.1.7 Anti-Droplet Control: ADC

Does your application require pipetting of volatile solvents? The Anti-Droplet Control (ADC) enables you to do this with highest process security. In normal pipettes the high evaporation pressure of volatile solvents causes immediate dripping from the tip. Because the ML STAR Line uses Monitored Air Displacement Technology. It can detect pressure changes following aspiration and compensate for them in real time.

The principle is shown in the illustration bellow: as the evaporation causes a pressure increase (red line) the Pipetting Unit detects the changes and compensates for them with plunger movements (blue line). The liquid remains in the tip.

The Anti-Droplet Control (ADC) has to be activated in the “HSLML_STARLib” Command Library of the VENUS Software.

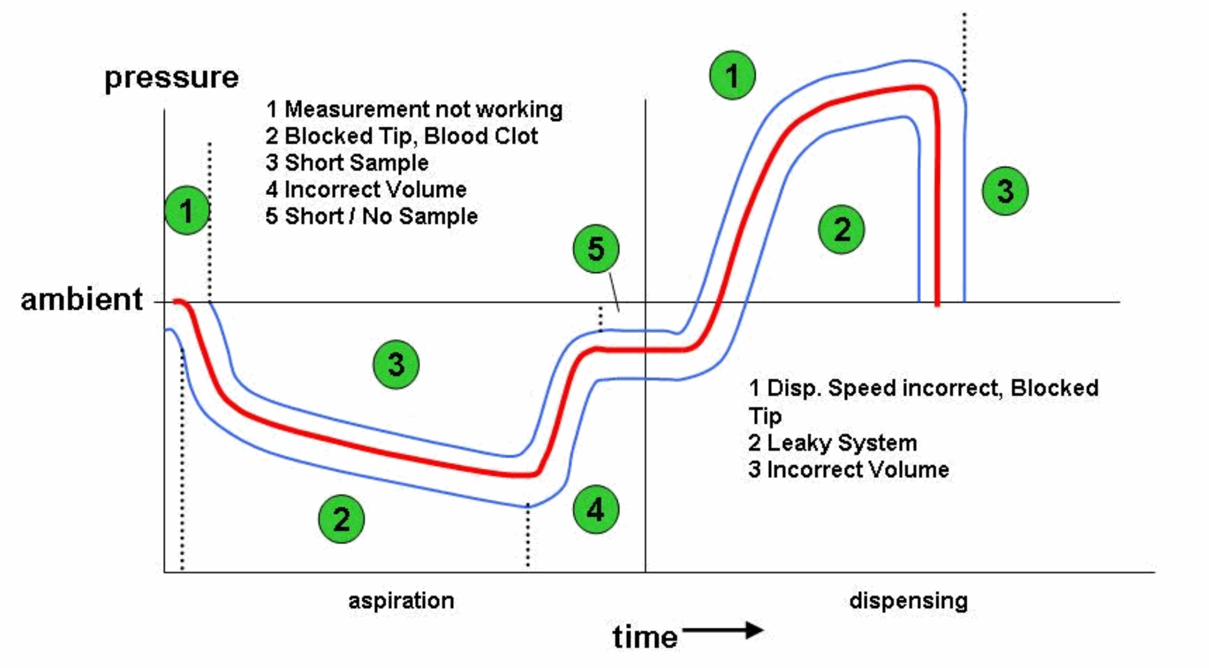


Patent no: *EP 1614468*

US 2007102445

2.1.8 Total Aspirate and Dispense Monitoring: TADM

During crucial sample transfers, the ML STAR Line can monitor aspiration and dispense steps in real time. TADM verifies with a traceable digital audit trail that a sample has been transferred. The functions of TADM is an option and therefore not available by default.

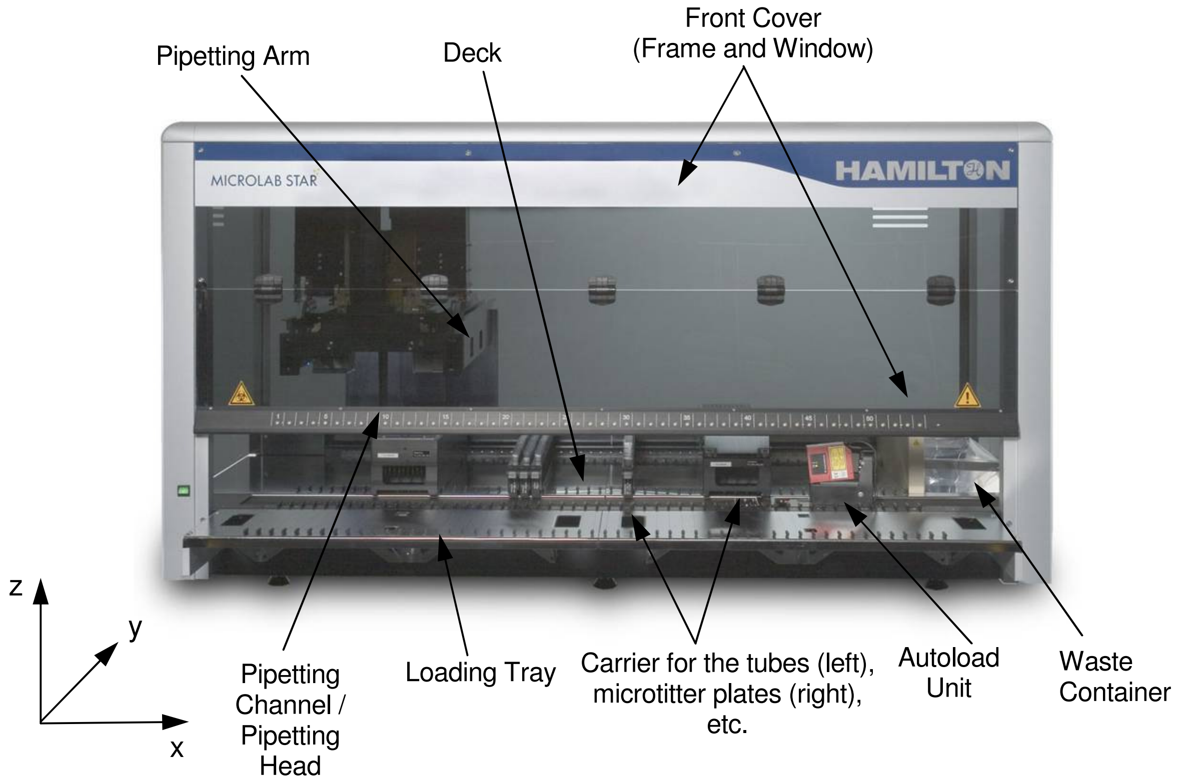


Patent no: *EP 1412759*

US 6938504

2.2 Platforms

The ML STAR instrument work surface, called a "deck", for placing loadable carriers, is available in three different sizes. These carriers hold reagent containers, such as tubes, microtiter plates, or other kinds of labware.



The MICROLAB® STAR

The instrument's deck is divided into equal tracks (T) for loading carriers in predetermined positions. This eliminates the need for precise measurement of positions. The deck has partitions of 22.5mm, which is equivalent to 1-T (track). The labware carriers are adapted to those partitions, e.g. 1-T carriers for sample tubes, or 6-T carriers for microtiter plates or CO-RE tips, etc. An additional partition provides space for the tip waste container.

The instrument's internal coordinate system is shown in the picture above, located at its origin. Please note that the ZERO position is 100mm below the metal deck sheet.

2.2.1 MICROLAB® STAR^{let}

The compact version of the ML STAR Line is called MICROLAB® STAR^{LET}. The MICROLAB® STAR^{LET} is the ML STAR instrument with the smallest deck width for loading carriers. The deck has partitions for a maximum of 30 specialized 1-T carriers for sample tubes, or a maximum of five 6-T carriers for microtiter plates and CO-RE tips. This means that a total of 25 SBS (Standard format of the Society for Biomolecular Screening) positions can be placed onto a STAR^{LET} deck.

MICROLAB® STAR^{LET}

2.2.2 MICROLAB® STAR

The ML STAR's deck is divided into 54 equal tracks (T) for loading carriers in predetermined positions. The deck has partitions for a maximum of 54 specialized 1-T carriers for sample tubes, or a maximum of nine 6-T carriers for microtiter plates and CO-RE tips. This means that a total of 45 SBS (Standard format of the Society for Biomolecular Screening) positions can be placed onto a STAR deck.

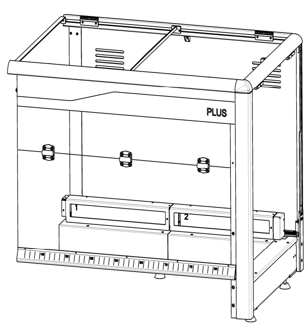
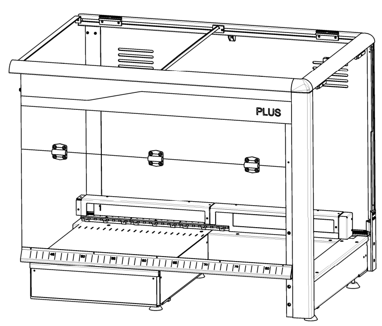
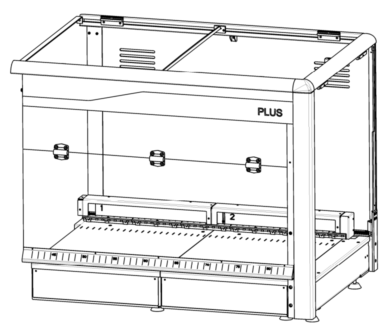
MICROLAB® STAR

2.2.3 MICROLAB® STAR^{PLUS}

The large version of the ML STAR Line is called MICROLAB® STAR^{PLUS}. If equipped with a full deck extension the MICROLAB® STAR^{PLUS} has a deck divided into 71 equal tracks for loading carriers. The deck has partitions for a maximum of 71 specialized 1-T carriers for sample tubes, or a maximum of eleven 6-T carriers for microtiter plates and CO-RE tips. This means that a total of 55 SBS (Standard format of the Society for Biomolecular Screening) positions can be placed onto a STAR^{PLUS} deck.

MICROLAB® STAR^{PLUS}

The MICROLAB® STAR^{PLUS} is the compact MICROLAB® STAR^{LET} extended on site by additional workspace. The workspace can be filled by additional Tracks. The other possibility is to integrate e.g. 3rd party devices, as readers, washers, centrifuges, etc. The extension part of the MICROLAB® STAR^{PLUS} is available in 3 versions:

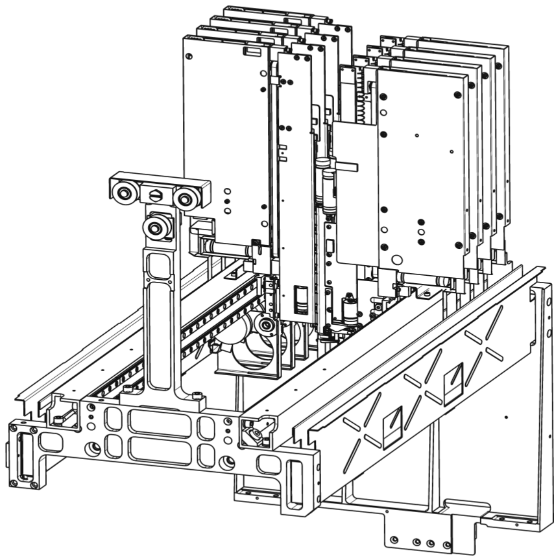
		
Without deck extension: • STAR^{PLUS} has 30 Tracks, • Total of 25 SBS positions	*With left deck extension:* • STAR^{PLUS} has 50 Tracks, • Total of 40 SBS positions	*With complete deck extension:* • STAR^{PLUS} has 71 Tracks, • Total of 55 SBS positions

2.3 Pipetting Arm Configurations

The ML STAR Line offers a selection of a variety of arms, depending on the pipetting units and plate handling modules chosen. The pipetting arm moves in X direction. Whenever higher throughput is required, it is possible to have two arms on the system working in parallel. E.g. while one arm is reserved for the pipetting tasks, the other can transfer plates on the deck or to/from a peripheral device.

2.3.1 Modular Pipetting Arm

The Modular Pipetting Arm typically contains a set of pipetting channels, which work independently. It can be equipped with: up to 16 channels with 1000µl pipetting head, up to 8 channels with 5ml pipetting head, a plate handling tool (iSWAP), a tube handling tool and an imaging channel.



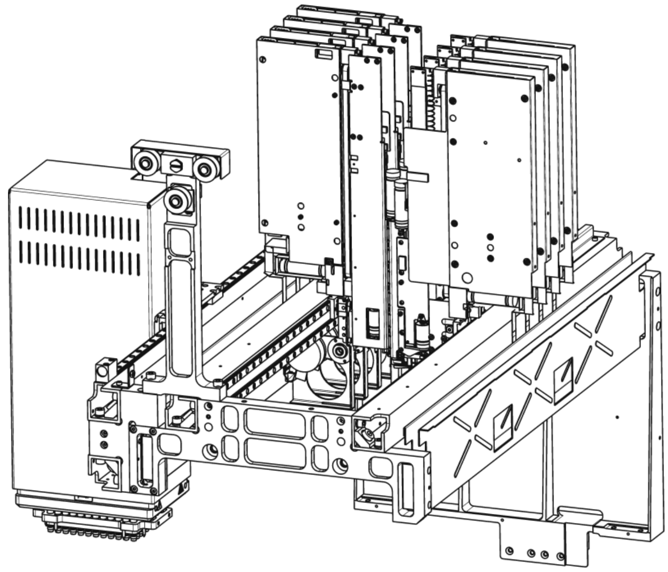
A possible configuration including all the tools is: 6 channels with 1000µl pipetting head, 2 channels with 5ml pipetting head, the tube handler, the imaging channel, with or without the plate handling tool (iSWAP).

The minimum distance between two 1000µl pipetting channels on this arm is 9mm.

The minimum distance between two 5ml pipetting channels on the Modular Pipetting Arm is 18mm.

2.3.2 Modular Pipetting Arm MPH

The Modular Pipetting Arms may also hold the CO-RE 96-Probe Head or the CO-RE 384-Probe Head as an option paired with up to 12 channels with 1000µl pipetting head, up to 8-channels with 5ml pipetting head, a plate handling tool (iSWAP), a tube handling tool and an imaging channel.



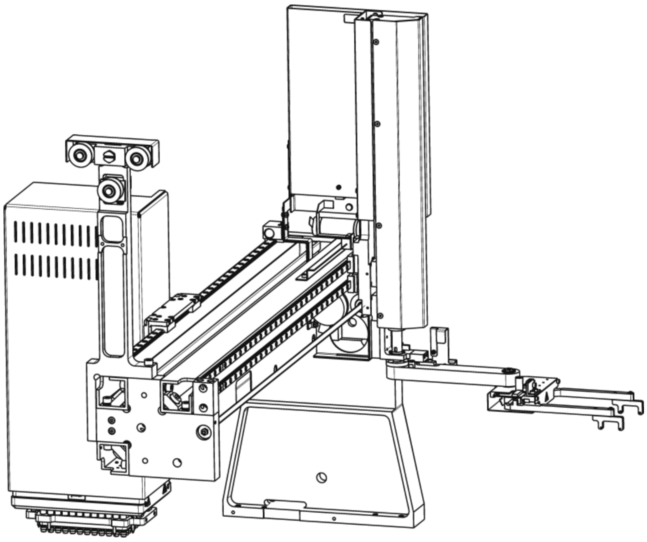
A possible configuration including all the tools is: one Multi Probe Head (CO-RE 96-Probe Head or the CO-RE 384-Probe Head), 6 channels with 1000µl Pipetting Head, 2 channels with 5ml pipetting head, the tube handler, the imaging channel, with or without the plate handling tool (iSWAP).

The minimum distance between two 1000µl pipetting channels on the Modular Pipetting Arm MPH is 9mm.

The minimum distance between two 5ml pipetting channels on the Modular Pipetting Arm MPH is 18mm.

2.3.3 Pipetting Arm MPH/iSWAP

The Pipetting Arm MPH/iSWAP is fitted with the CO-RE 96-Probe Head or the CO-RE 384 - Probe Head on the left. The right side of this arm can be equipped with up to 8 channels with 1000µl pipetting head, up to 4 channels with 5ml pipetting head, a plate handling tool (iSWAP), a tube handling tool and an imaging channel.



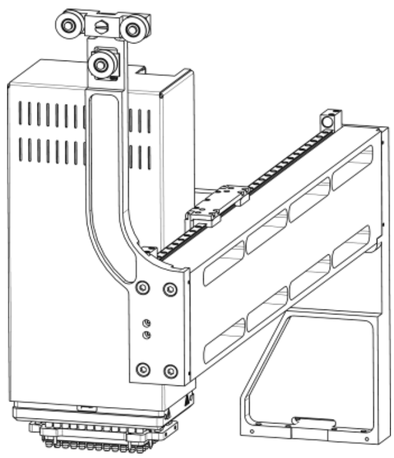
A possible configuration including all the tools is: one Multi Probe Head (CO-RE 96-Probe Head and the CO-RE 384-Probe Head), 2 channels with 1000µl pipetting head, 1 channel with 5ml pipetting head, the tube handler, the imaging channel, with or without the plate handling tool (iSWAP).

The minimum distance between two 1000µl pipetting channels on the Pipetting Arm MPH/iSWAP is 18mm.

The minimum distance between two 5ml pipetting channels on the Pipetting Arm MPH/iSWAP is 36mm.

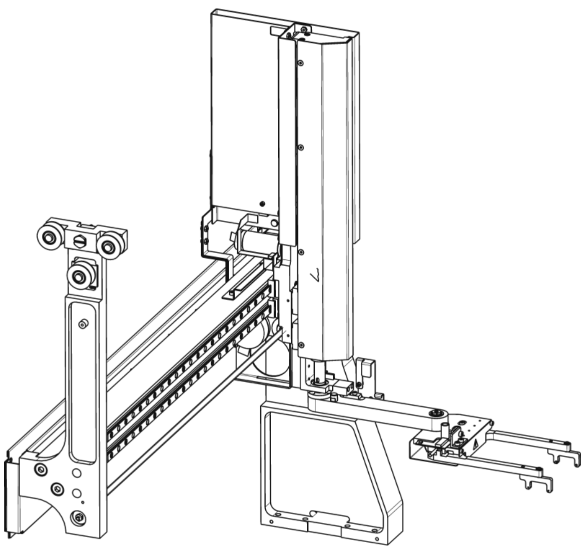
2.3.4 Pipetting Arm MPH

This arm is equipped with the CO-RE 96-Probe Head, the CO-RE 384-Probe Head.



2.3.5 iSWAP Arm

The iSWAP Arm is typically fitted with the iSWAP for dual-arm configurations. This arm can be equipped with up to 8 channels with 1000µl pipetting head, up to 4 channels with 5ml pipetting head, a plate handling tool (iSWAP), a tube handling tool and an imaging channel.



A possible configuration including all the tools is: 2 channels with 1000µl pipetting head, 1 channel with 5ml pipetting head, the tube handler, the imaging channel, with or without the plate handling tool (iSWAP).

The minimum distance between two 1000µl pipetting channels on the iSWAP Arm is 18mm.

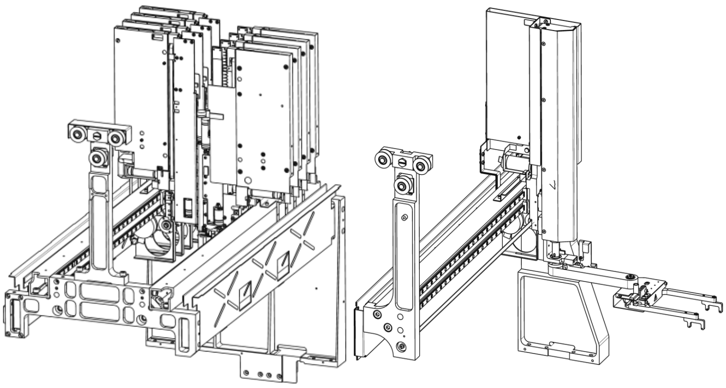
The minimum distance between two 5ml pipetting channels on the iSWAP Arm is 36mm.

2.3.6 Dual-Arm Configurations

When higher throughput is required, it is possible to equip the ML STAR Line with two arms working in parallel. The preferential platform for Dual-Arm Configurations is the MICROLAB® STAR^PLUS. The following examples show some typical dual arm configurations. Several other combinations are possible.

Dual Arm Assay Workstation 8+iSWAP

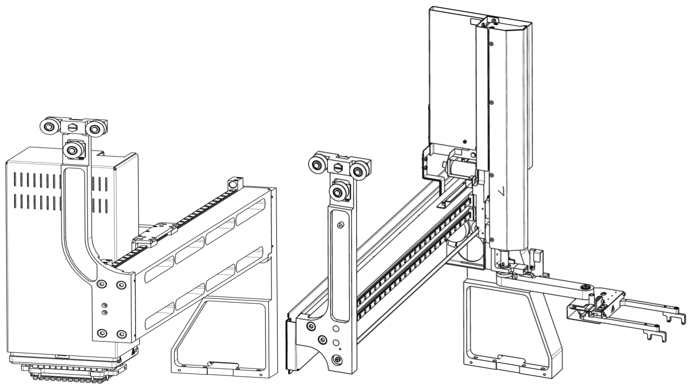
Modular Pipetting Arm with 8 channels combined with the iSWAP Arm.



This workstation for instance makes sense if the sample preparation task (processed by the 8-channels) is isolated from the assay (as incubation tasks, plate washing tasks, analyzing task, etc). The iSWAP is used to transfer the processed microtiter plate to/from the 3rd party equipment as reader, incubator, plate washer, etc.

Dual Arm Assay Workstation 96+iSWAP

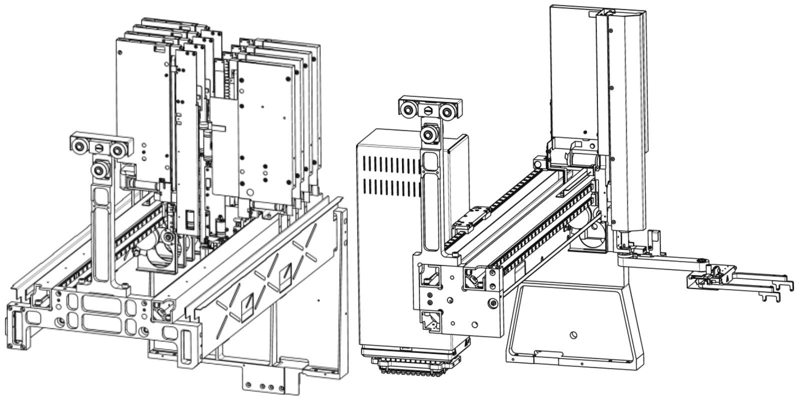
Pipetting Arm 96 with the CO-RE 96-Probe Head combined with the iSWAP Arm.



This workstation for instance makes sense if the pipetting tasks (plate copy, plate reformation, serial dilution) take part in the format of microtiter plates only, and the assay needs no further pipetting steps. In addition, plate transfer from and to other devices is required.

Dual Arm Assay Workstation 8+96/iSWAP

Modular Pipetting Arm with 8 channels combined with a Pipetting Arm 96/iSWAP.



This workstation for instance makes sense if the sample preparation task (processed by the 8-channels) is isolated from the assay part (as incubation tasks, plate washing tasks, analyzing task, etc). The assay part needs pipetting in the pattern of the CO-RE 96-Probe Head (one volume per plate, row or column). The iSWAP is used to transfer the processed microtiter plate to or from 3rd party equipment, such as reader, incubator, plate washer, etc.

2.4 Options

Options are defined as components or configurations that are part of the instrument initially provided to the customer as specified by that customer. Predefined options are as follows: Manual load or Autoload, the type and quantity of Pipetting Arms, the quantity of pipetting channels from 4 to 16, the CO-RE 96-Probe Head, the CO-RE 384-Probe Head, the plate handling tool (iSWAP), the tube handling tool and the camera channel.

Accessories include assemblies such as Wash Stations, Temperature-Controlled Carrier, Basic Vacuum System, Labware Carrier, etc.

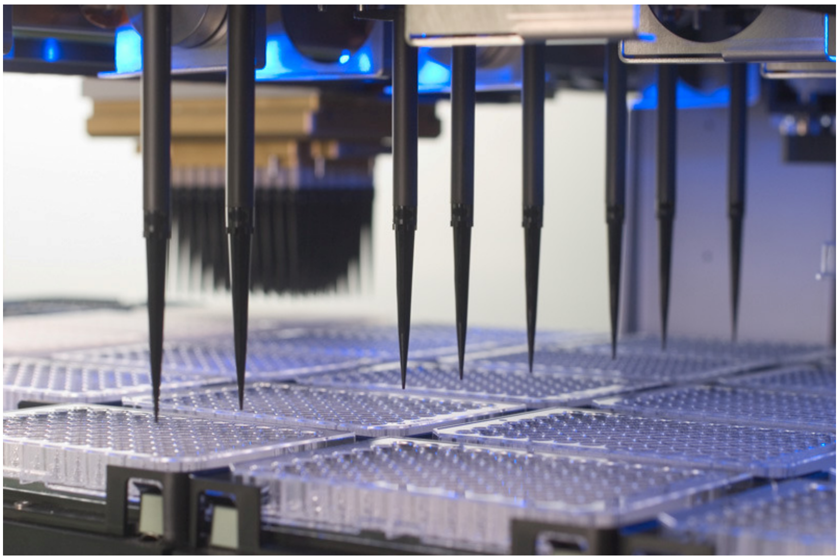
Pipetting Channels, CO-RE 96-Probe Head and Accessories may be ordered as an option for new installation, or added as update kits at the distributor or customer site, when there is a need to upgrade an existing instrument in the field.

The instrument's configuration is set within the Configuration Editor of the VENUS Software.

2.4.1 1000µl Pipetting Channels

The ML STAR instrument comes with 1, 2, 4, 8, 12, or 16 pipetting channels working in parallel for simultaneous transfer of liquids. The Dynamic Positioning System (DPS) of the ML STAR Line moves each pipetting channel independently on the Y-Axis, as well as the Z-Axis. Each channel uses its own high-precision motors and electronics to reach any position on the deck without the need of teaching.

The 1000µl pipetting channels support pipetting with disposable tips or with needles.



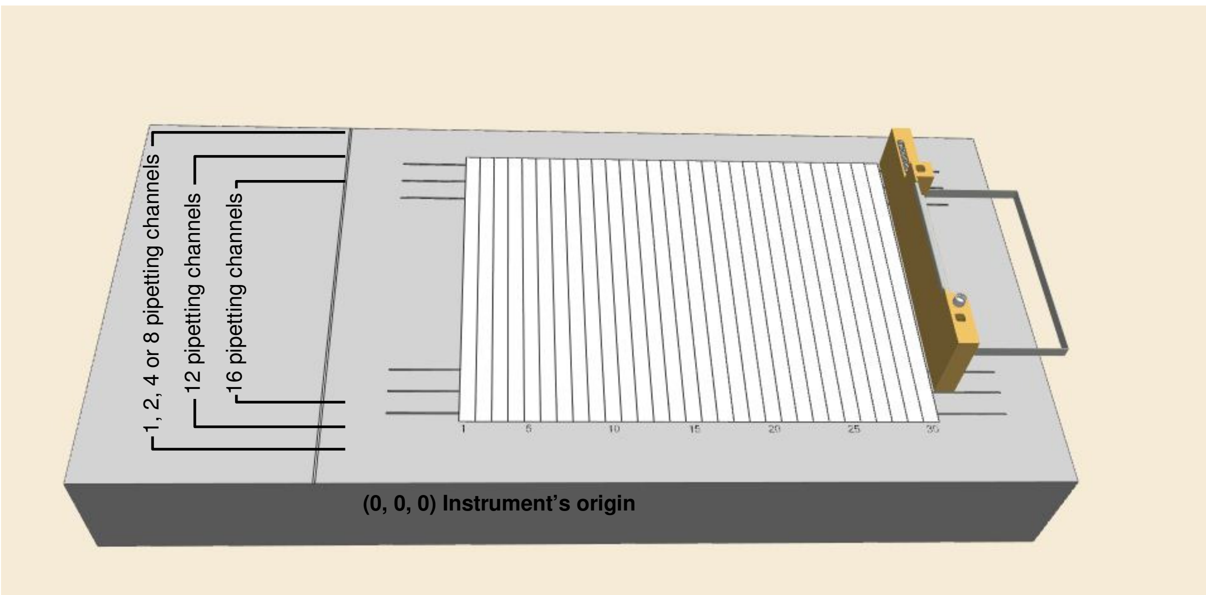
The minimum distance between two 1000µl pipetting channels on the Modular Pipetting Arms is 9mm, on all other arms it is 18mm.

The pipetting channels have a set "traverse height" of 245mm above the origin, or 145mm between the top of the disposable tip and the deck of the instrument. This means that, when a channel is to move from one location on the deck to another, it automatically does so at that particular height. This is a safety precaution, so that channels will not collide with any items that may be on the deck.

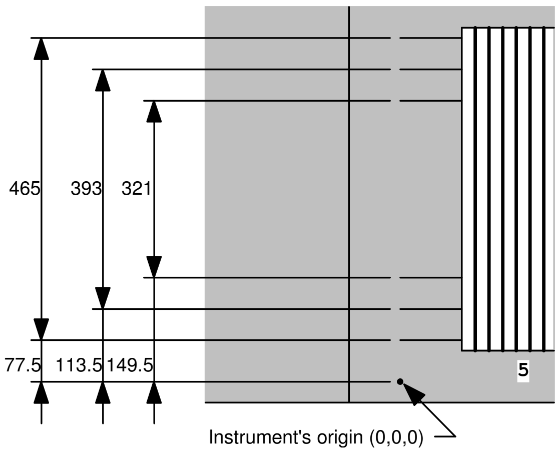
Instruments with 4, 8, or 16 channels are best operated with carriers holding microplates and tip racks in landscape orientation, whereas carriers with portrait orientation for microplates and tips are best suited for the 12-channel ML STAR instrument.

An instrument equipped with 1, 2, 4 or 8 pipetting channels has the largest "random access space" which is the area, where any of the pipetting channels is able to reach. For a 12 or 16 channel equipped instrument, the "random access space" of the instrument is reduced.

The "random access space" of the different numbers of channels is indicated by lines at the left and right side of the deck layout of the VENUS Software, as shown in the screenshot:

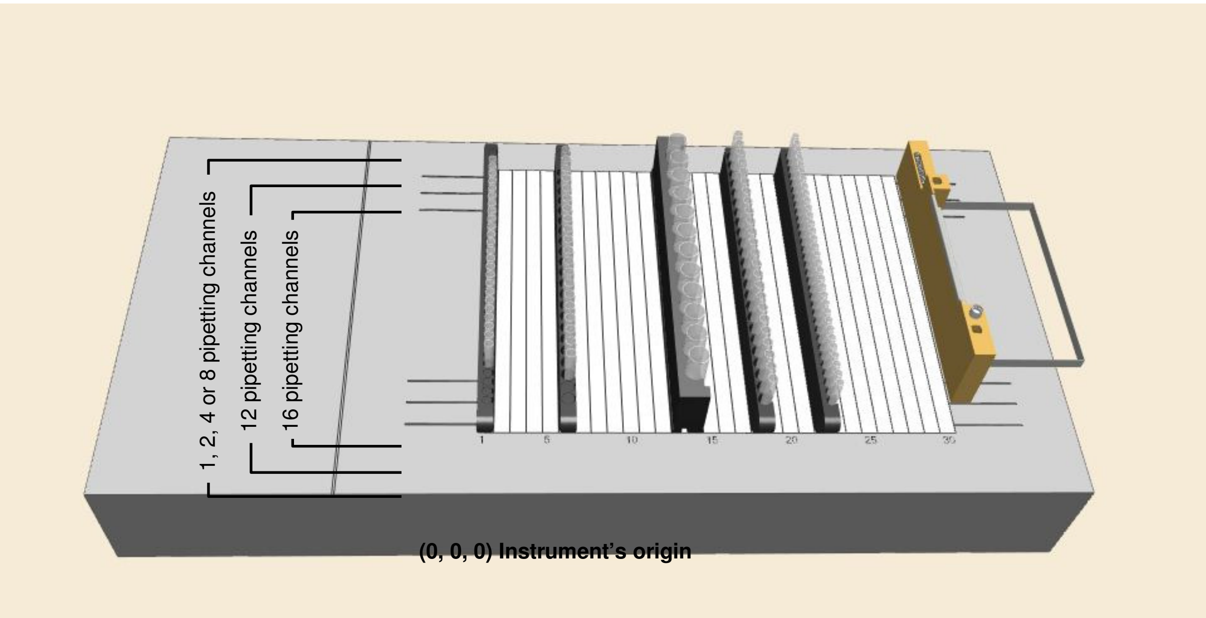


The actual dimensions of random access are shown in the following table:



No. of channels	Y_{min} [mm]	Y_{max} [mm]
	Absolute to Instrument's origin	
1, 2, 4 or 8	77.5	77.5+465
12	113.5	113.5+393
16	149.5	149.5+321

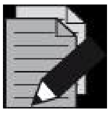
To guarantee random access to sample tube carriers, only the inner tube positions should be used. The number of blank positions in front and rear, to be used for the different number of channels and the different tube carriers, is listed in the following table. The deck layout shows the relevant carriers directly above the columns of the table in each case:



No. of channels	SMP_CAR_12*	SMP_CAR_16*	SMP_CAR_24*	SMP_CAR_32*
1, 2, 4 or 8	0	0	0	0
12	1	2	2	3
16	2	3	4	5

(See chapter 2.5.1 "Carriers" for abbreviations like SMP_CAR_*)

The 16-channel instrument is intended as a batch-type processor, meaning that random access to all positions is neither possible nor required. In contrast to the 8-channel instument, the methods for a 16-channel instrument should be set up in such a way that all positions are processed in batches of at least 8 (or better still, 16, to optimize the pipetting speed) with 8 (or 16) simultaneous aspirations or dispenses at identical X-coordinates.



NOTE

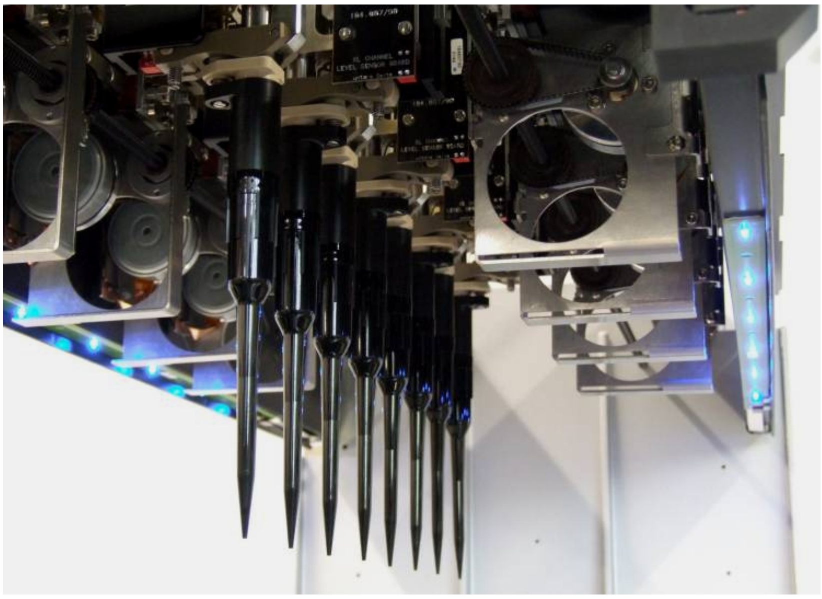
If pipetting positions outside the random access range of the instrument are used, the system reports an error. However, a strictly batch-type process can eliminate these problems.

Low-volume tips do not reach the deck, even at the lowest z-position of the channel.

2.4.2 5ml Pipetting Channels

The ML STAR instruments comes with 1, 2, 4, or 8 pipetting channels working in parallel for simultaneous transfer of liquids. The Dynamic Positioning System (DPS) of the instruments moves each pipetting channel independently on the Y-Axis, as well as the Z-Axis. Each channel uses its own high-precision motors and electronics to reach any position on the deck without the need of teaching.

The 5ml pipetting channels support pipetting with disposable tips.



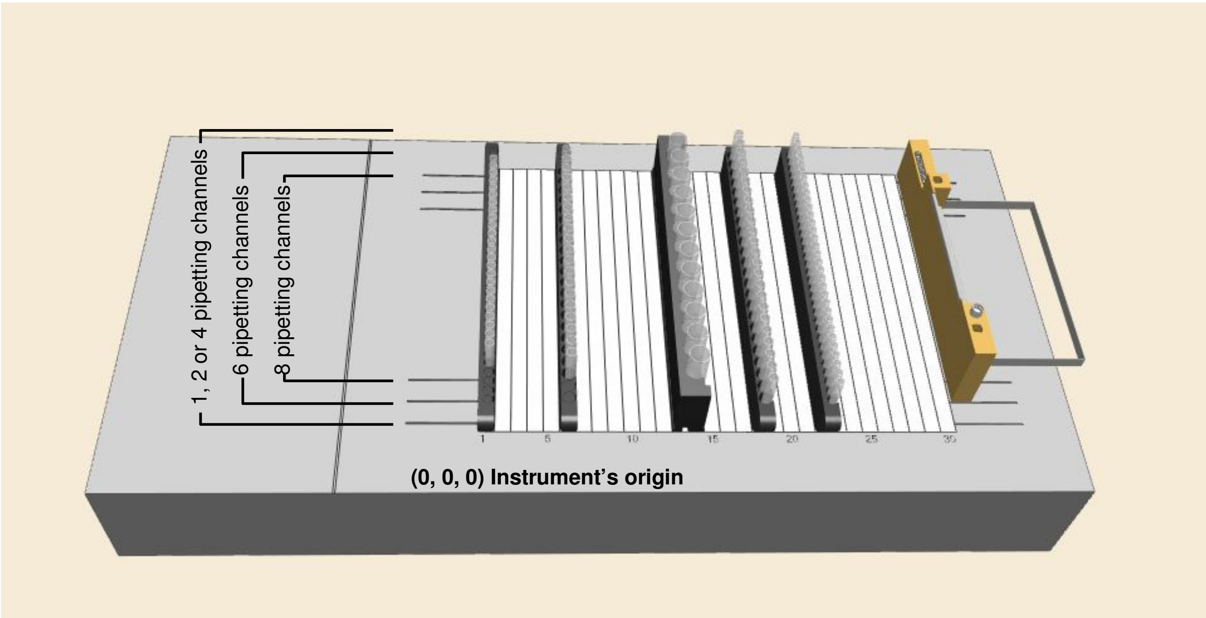
The minimum distance between two 5ml pipetting channels on the Modular Pipetting Arms is 18mm, on all other arms it is 36mm.

The pipetting channels have a set "traverse height" of 245mm above the origin, or 145mm between the top of the disposable tip and the deck of the instrument. This means that, when a channel is to move from one location on the deck to another, it automatically does so at that particular height. This is a safety precaution, so that channels will not collide with any items that may be on the deck.

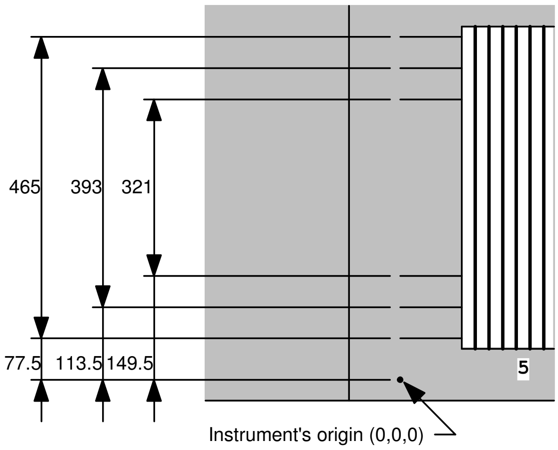
Instruments with 4 or 8 channels are best operated with carriers holding microplates and tip racks in landscape orientation, whereas carriers with portrait orientation for microplates and tips are best suited for the 6-channel instrument.

An instrument equipped with 1, 2, 4 pipetting channels has the largest "random access space" which is the area, where any of the pipetting channels is able to reach. For a 6 or 8 channel equipped instrument, the "random access space" of the ML STAR instrument is reduced.

The "random access space" of the different numbers of channels is indicated by lines at the left and right side of the deck layout of the VENUS Software, as shown in the screenshot:

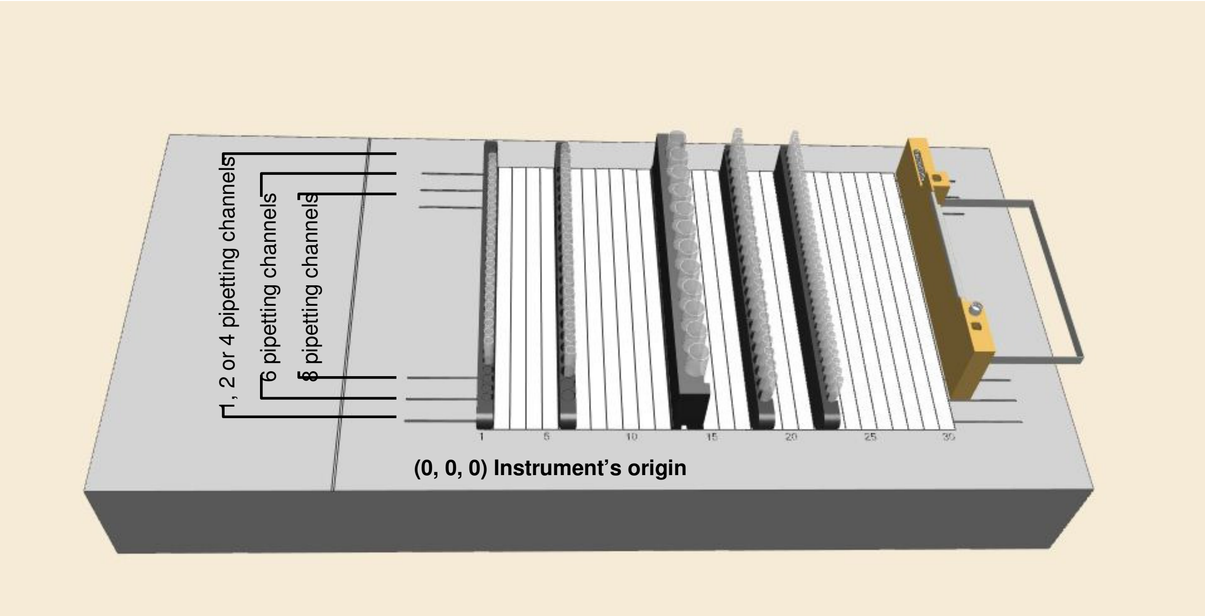


The actual dimensions of random access are shown in the following table:



No. of channels	Y_{min} [mm]	Y_{max} [mm]
	Absolute to Instrument's origin	
1, 2 or 4	77.5	77.5+465
6	113.5	113.5+393
8	149.5	149.5+321

To guarantee random access to sample tube carriers, only the inner tube positions should be used. The number of blank positions in front and rear, to be used for the different number of channels and the different tube carriers, is listed in the following table. The deck layout shows the relevant carriers directly above the columns of the table in each case:



No. of channels	SMP_CAR_12*	SMP_CAR_16*	SMP_CAR_24*	SMP_CAR_32*
1, 2 or 4	0	0	0	0
6	1	2	2	3
8	2	3	4	5

(See chapter 2.5.1 "Carriers" for abbreviations like SMP_CAR_*)

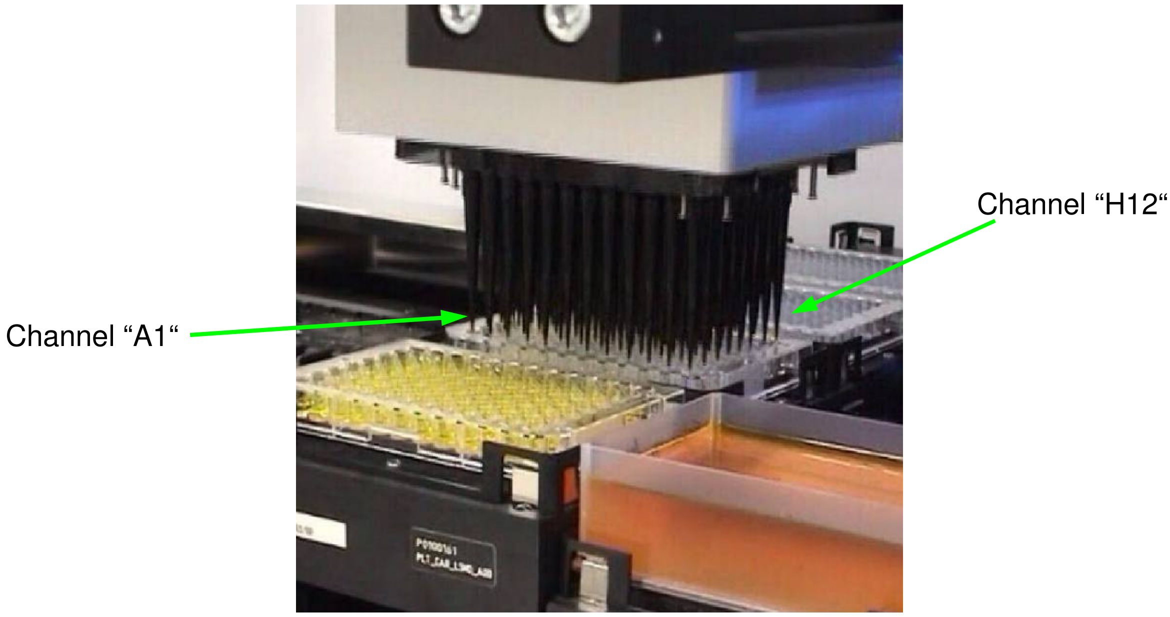
The 8-channel instrument is intended as a batch-type processor, meaning that random access to all positions is neither possible nor required. In contrast to the 4-channel instrument, the methods for a 8-channel instrument should be set up in such a way that all positions are processed in batches of at least 4 (or better still, 8, to optimize the pipetting speed) with 4 (or 8) simultaneous aspirations or dispenses at identical X-coordinates.

NOTE

If pipetting positions outside the random access range of the instrument are used, the system reports an error. However, a strictly batch-type process can eliminate these problems.

2.4.3 CO-RE 96-Probe Head

The CO-RE 96-Probe Head is a high-throughput dispenser built with the CO-RE technology as a ML STAR pipetting channel. The CO-RE technology guarantees fast and accurate pick-up and release of disposable tips. cLLD (capacitive liquid level detection) is available on the two special channels (A1 and H12).



View from the left side of the instrument

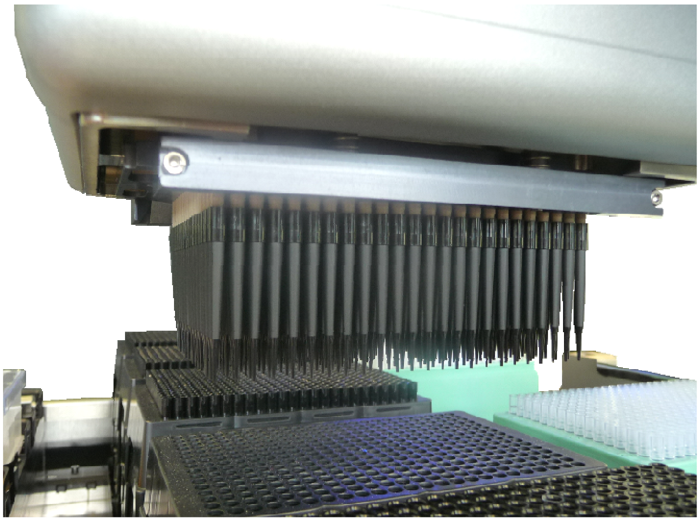
The CO-RE 96-Probe Head is available with 96x 1000µl Pipetting Channels. With both options, a block of up to 96 can be picked up all at once. The 96 channels always work simultaneously with the same volume. In combination with the tip rack adapter, picking up a single tip row/column or a single tip is feasible.

1000µl CO-RE 96-Probe Head

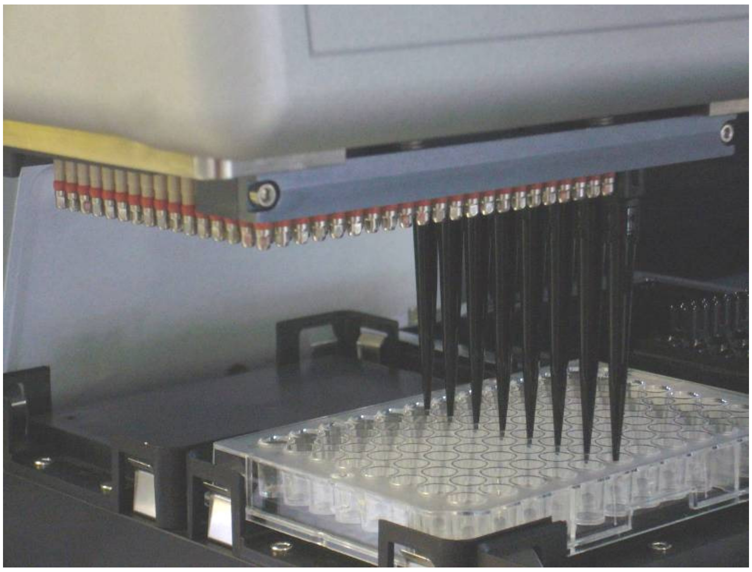
The 1000µl Head supports pipetting with low-volume (10µl), intermediate volume (50µl), standard (300µl) or high volume (1000µl) disposable tips.

2.4.4 CO-RE 384-Probe Head 50µl

The CO-RE 384-Probe Head is a high parallel dispenser. The 384-Probe Head uses the CO-RE technology as the individual channels and the 96-Probe head do, offering accurate and gentle tip pickup. The positioning accuracy allows to pipette into 1536, 384 and 96 well microtiter plates. The 384 channels always work simultaneously with the same volume. The pipetting range is between 0.1µl and 50µl. cLLD (capacitive liquid level detection) is available on two special channels (A5 and P24).



The CO-RE 384-Probe Head is able to pick up one or more columns, rows, a single quadrant or even a single tip together with the Tip Support.



The use of HAMILTON's Rocket-Tips converts the 384 into a 96 channel pipetting head with a volume range of 2µl - 300µl.

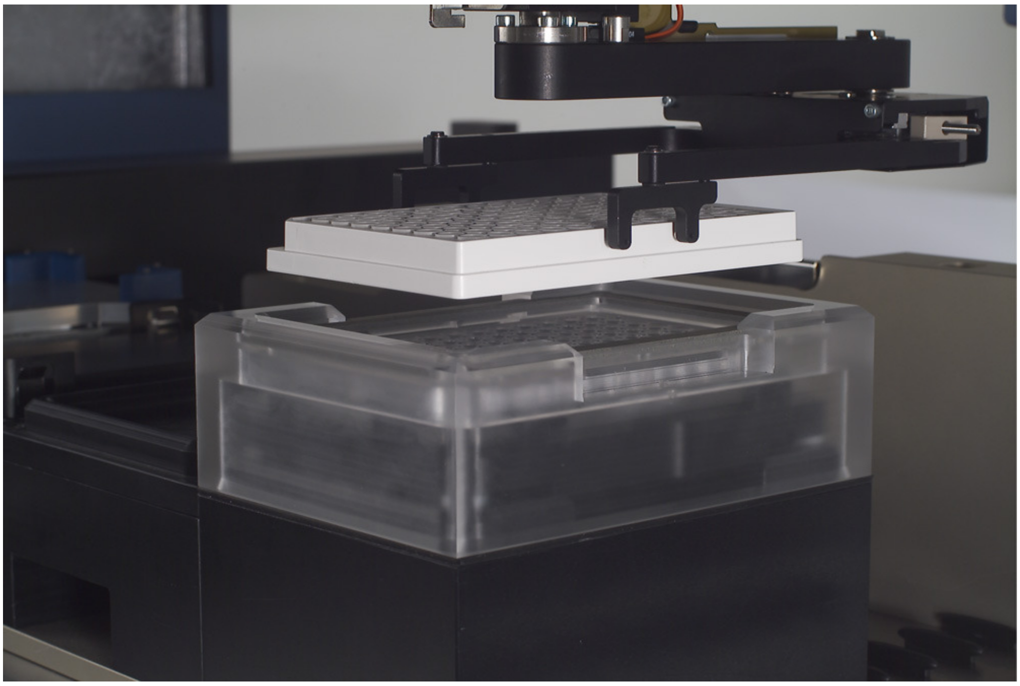
NOTE

Special 50µl tips were designed for pipetting with the CO-RE 384-Probe Head.

2.4.5 iSWAP

iSWAP (internal Swivel Arm Plate Handler) is a robotic arm that transports microplates, lids of microplates, archive plates, filter plates to and from positions on the instrument deck. Plates can be placed in landscape or portrait orientation or rotated by 180 degrees. Typical handling tasks like loading/unloading of plates outside the instrument becomes very simple with the iSWAP. The iSWAP also can stack plates or tips with specially provided carriers.

Among the special features of iSWAP is a sensor which signals to the robot how tightly it is gripping a labware object. The newest generation of iSWAP monitors the presence of an object. Like ML STAR pipetting channels, iSWAP has a "traverse height" of 145mm above the deck (245mm above the origin).



iSWAP robotic hand at work transporting microplates to CVS.

The iSWAP is mounted on the pipetting arm. It does not affect the movement of the pipetting tools.

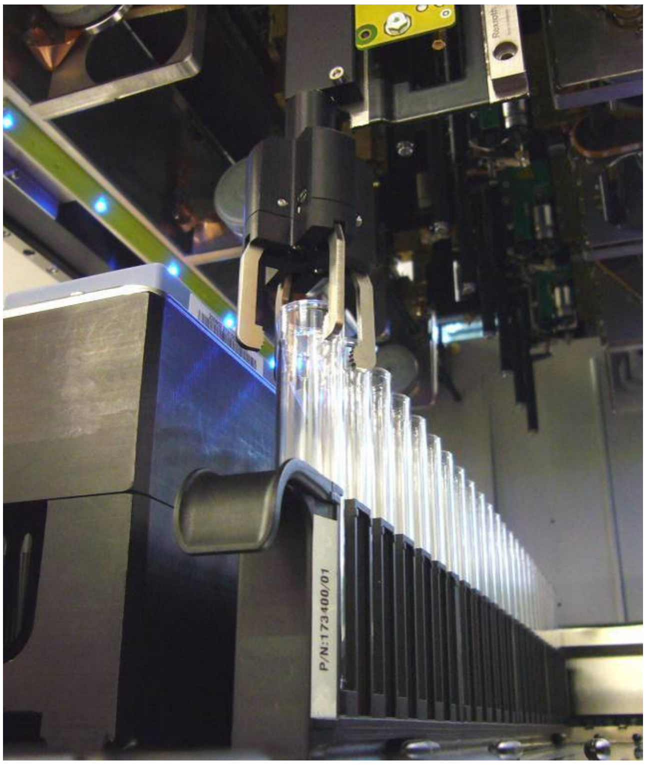
iSWAP can be chosen by the customer as an ordering option, but can also be installed later by a service technician on the customer's request.

2.4.6 Tube Gripper

The Tube Gripper transports tubes to and from positions on the insturments deck. The Gripper Tools is mounted to a separate Channel. It can handle tubes with diameters from 8 up to 20mm and up to a length of 120mm.

The Dynamic Positioning System (DPS) of the ML STAR insturment moves the tube handler independently on the Y-Axis, as well as the Z-Axis. Because the use of high-precision motors and electronics the Tube Gripper reaches any position on the deck without the need of teaching.

Like ML STAR pipetting channels, the Tube Gripper has a "traverse height" of 145mm above the deck (245mm above the origin) between the tube and the deck of the instrument. This means that, when a tube is moved from one location on the deck to another, it automatically does so at that particular height. This is a safety precaution, so that the transported tube will not collide with any items that may be on the deck.



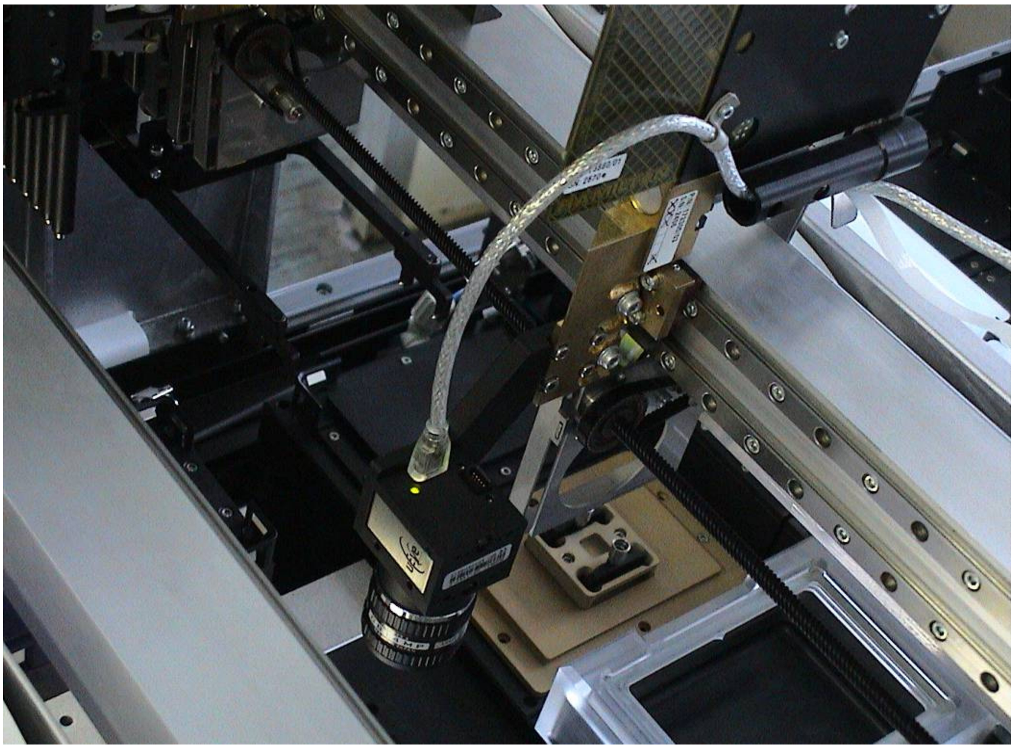
The Tube Gripper can be chosen by the customer as an ordering option, but can also be installed later by a service technician on the customer's request.

2.4.7 Imaging Channel

The purpose of Imaging Channel is to enquire the digital image from any object all over the instruments deck. A high-resolution CCD-Camera is fixed on a separate channel and allows to picture target which is as big as the SBS format. The pictures are sent then to the image analyses software for further investigation.

A typical application is the image analyses of bacteria colonies. In the proprietary EasyPick software, the user defines parameters and determines the weighting for the typical criteria to identify colonies: coordinates, size, shape (circularity), colour, proximity to next colony, distance to the margin, etc.

Reliable automation requires proper quality of the analyzed images. Appropriate lighting of the object of interest is fundamental. In case of image analyses of bacteria colonies a Back-Light Table is used in the system. For other applications different lighting options can be applied.



The Imaging Channel can be chosen by the customer as an ordering option, but can also be installed later as an upgrade option by a service technician on the customer's request.

2.4.8 CO-RE Grip

The CO-RE Grip is the plate handling tool picked up by two pipetting channels during a run. This tool is only available if your system has at least two individual pipetting channels.

The CO-RE Grip transports microplates, covers microplates, archive plates, filter plates, etc. to and from positions on the insturments deck. Plates can be gripped in landscape or portrait format within the working area. Rotation of plates is not an option.

The "traveling height" of the channels with the gripping jaws is the same as with tips: 145mm above the deck. Given that there is no sensor working, you must ensure that CO-RE Grip does not grip plates too tightly, causing them to buckle.

CO-RE Grip for the 1000µl Pipetting Channels (inner positions) and the 5ml Pipetting Channels (outer positions): the "jaws" in their holder

CO-RE Grip for the 1000µl Pipetting Channels

The holder for the two gripping jaws can be mounted on any standard plate carrier and needs one position, which can be freely selected. Another position for the holder of the two gripping jaws is on the waste block. In this case, the CO-RE Grip does not consume one plate position.

The CO-RE Grip is an additional ordering option for the ML STAR instrument, that can also be installed later by a service technician on the customer's request.

2.4.9 Autoload Option

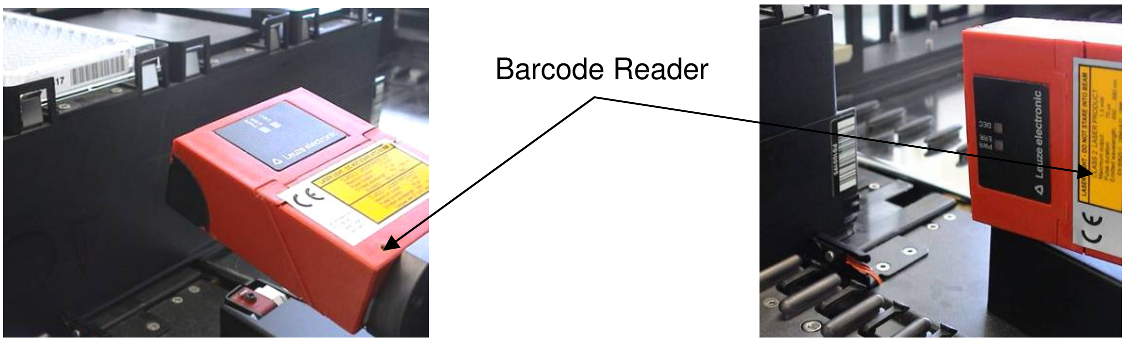
The Autoload option of the ML STAR configuration must be initially specified by the customer and cannot be added later.

The Autoload unit is a device enabling automatic loading of carriers onto the ML STAR instrument deck. It contains a loading head that

- moves in X direction,
- shunts carriers onto and off the deck, and
- reads the barcodes of carriers, tubes, and microtiter plates.

There is a presence sensor that identifies the tubes present on a sample carrier.

Carrier identification by barcodes, and reading of barcodes on plates and tubes, is only possible in conjunction with the Autoload option.



The Autoload option including the barcode reader, which is shown reading horizontally for plates and vertically for carriers and tubes.

Equipped with the Autoload option, the following barcode types can be read:

- ISBT Standard
- Code 128 (Subset B and C)
- Code 39
- Codabar
- Code 2 of 5 interleaved
- UPC A

HAMILTON recommends using the barcode type Code 128 (Subset B and C).

NOTE

In addition, barcodes must have a minimum readability (i.e., good contrast, size, correct orientation and distance between bars) to be fully functional.

Ensure the correct barcode orientation for tubes and plates (see specifications).

For details of barcodes, see the specifications given in chapter 7.1.8 "Autoload Option: Barcodes and Reader Specifications".

2.5 Accessories

Accessories are defined as additional automation components. They provide a high degree of adaptability and permit customizing for multiple applications. These components can be ordered later by the customer and installed by HAMILTON-authorized personnel.

Note that the availability of accessories is subject to change. Please ask your local HAMILTON dealer for an up-to-date list, or consult www.hamiltonrobotics.com.

2.5.1 Carriers

The labware such as plates or tubes is placed on special carriers that are loaded onto the ML STAR insturments deck. HAMILTON provides a wide range of standard carriers for microtiter plates, tubes, tips etc. All standard carriers can be loaded to the deck manually or by the Autoload option.

The naming of carriers follows a systematic nomenclature "**X_CAR_Y_Ann**" where

X represents the type of labware placed on the carriers, e.g., TIP (= tips), PLT (=plates), SMP (=samples), RGT (=reagent)

CAR carrier

Y describes the labware details, e.g.

L: landscape orientation

P: portrait orientation

Number: number of items placed on the carrier (plates or tips)

MD: medium density (96- or 384-well microplates)

HD: high density microplates (1536)

AC: 96-well archive plates

Ann identifies the part number revision (e.g. A00)

A: variant

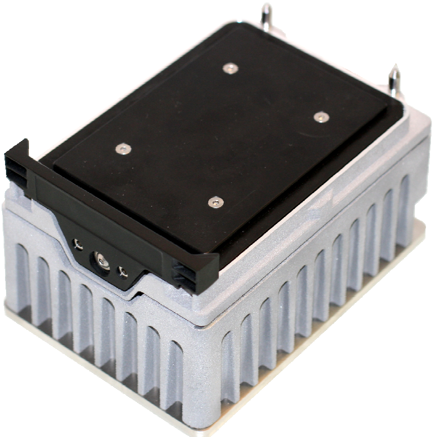
nn: revision

Example: PLT_CAR_L5MD_A00 is a plate carrier for 5 medium density (96- or 384-well) microplates in landscape orientation.

A carrier must always be identified in the VENUS Software (e.g. in deck layouts and methods) by the unique description with which it is tagged.

2.5.2 Microplate Shaker

The microplate shaker carrier base holds up to four HAMILTON Heater Shakers (HHS). The HAMILTON Heater Shaker is designed to shake standard SBS plates in microtiter or deep well format. Especially designed adapters are available to shake plates with different shapes on the HHS.

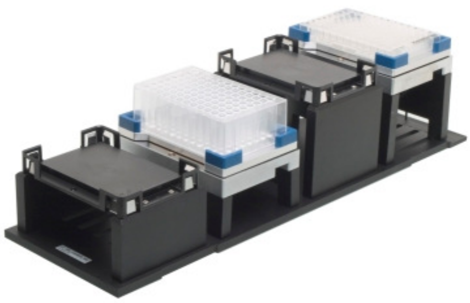


The maximum shaking speed (MTP/DWP: 100 – 2500/2000rpm) depends on the labware used. Shaking can be performed clockwise or counter-clockwise (Available amplitudes: 0.5 / 2.0 / 3.0mm). The HHS can be heated between RT+5°C and 105°C. The temperature is constantly measured by two sensors, which are located in the middle and one corner of the adapter. Before shaking is started, the plates are locked and positioned in the centre of the heater shaker. When shaking has finished the plates are unlocked and can be easily removed from the heater shaker. Each shaker is independently controlled and can be run in parallel with the other functions on the instrument. Up to 8 HAMILTON Heater Shakers can be integrated.

Instead of microplate shakers, however, the shaker carrier base can be equipped with ordinary microplate modules. You can find the list of available modules in Appendix B "Ordering Information".

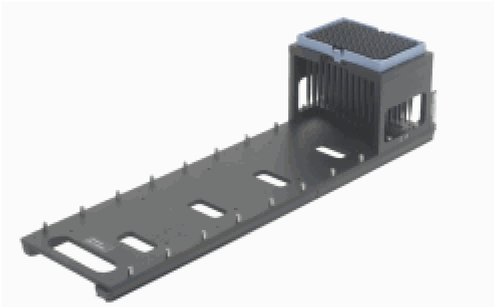
This carrier base will occupy 7 tracks of your ML STAR instruments deck.

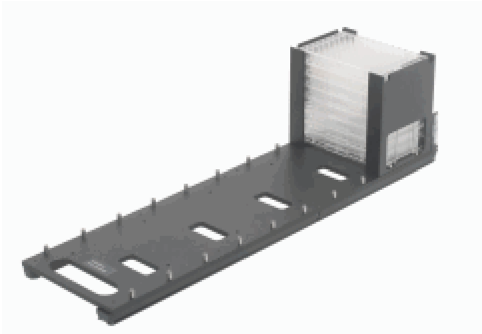
Optional H+P (Shaking: 100 - 2000 1/min, Orbital: 2mm) or Shaker Heater CAT (Shaking: 200 - 1200 1/min, Orbital: 2mm, Temperature: from RT+5°C - 90°C) modules are available.



2.5.3 MultiFlex Carrier

The MultiFlex Carrier consists of a multiple-use carrier base offering space for up to 5 modules. You can choose from several modules like: Tip rack module, microplate module, plate stacker module, module for heating or cooling labware, reagent trough module, tube or cup module, etc. You can find the list of available modules in Appendix B "Ordering Information".





Note that the availability of accessories is subject to change. Please ask your local HAMILTON dealer for an up-to-date list, or consult www.hamiltonrobotics.com.

The modular design of the MultiFlex Carrier allows space optimizing and customizing your instrument. This carrier will occupy 6 tracks of your instruments deck.

2.5.4 MultiFlex Heating and Cooling Module

The MultiFlex Heating and Cooling Module heats or cools plates. The temperature range for the Heating Module is from Ambient +5°C to 65°C. The temperature range for the Cooling Module is from 15°C to 4°C. A temperature selection dial is located on the top of the module for setting the desired temperature. The status LED "**OFF**" indicates the MultiFlex Heating/Cooling Module is coming to temperature. When the LED is "**ON**", the temperature of the Labware adapter is within ±1°C of the desired temperature. It is possible to place up to 4 Multiflexcarrier on one ML STAR instrument.

ATTENTION

Caution – Hot Surface. Avoid contact. Surfaces of the Heating Module are hot and may cause personal injury if touched.

Do not clean the Heating Module when it is still hot! Wait until the Heating Module has cooled down to room temperature before starting the cleaning procedure.

ATTENTION

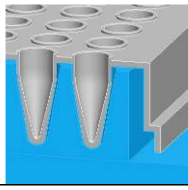
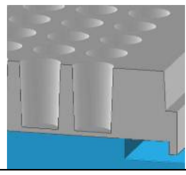
When placing the MultiFlex Heating/Cooling Module on the Deck, one track position must remain empty on each side of the module for air circulation. The modules must not be placed next to each other on the Deck. Extra spacing between the modules is necessary to allow proper ventilation.

NOTE

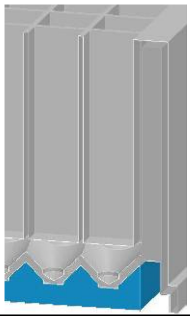
An efficient heating/cooling of the liquid is only achieved if the appropriate adapter for the labware is used. The heating/cooling time and the maximum/minimum temperature, which can be reached for the liquid inside the labware is dependent on the amount of liquid, used type labware, the used adapter and the connection between the labware and the adapter.

The temperature is measured on the labware adapters. Because of the liquid containers and the ambience there is a temperature deviation (Δ T) between the labware adapter and the liquid.

Typical cooling / heating deviation in the liquid (water) compared to the desired temperature:

Labware	**Adapter**	**Δ T cooling** (4°C – 15 °C)	**Δ T heating** (25°C – 60 °C)
PCR plate		≤ 0.5°C	≤ 1.0°C
Eppendorf tubes		≤ 0.5°C	≤ 1.0°C
MTP (flat bottom)		≤ 3.0°C	≤ 5.0°C

Typical time (at 40% rel. humidity, $T_{ambient} \approx$ 22°C, $T_{liquid} \approx$ 22°C) to heat (Ambient +5°C to 65°C) and cool (15°C to 4°C) is between 5 and 20 minutes.

Labware	**Adapter**	**△ T cooling** (4℃ – 15 ℃)	**△ T heating** (25℃ – 60 ℃)
DWP (U-bottom)		≤ 5.0℃	≤ 7.0℃

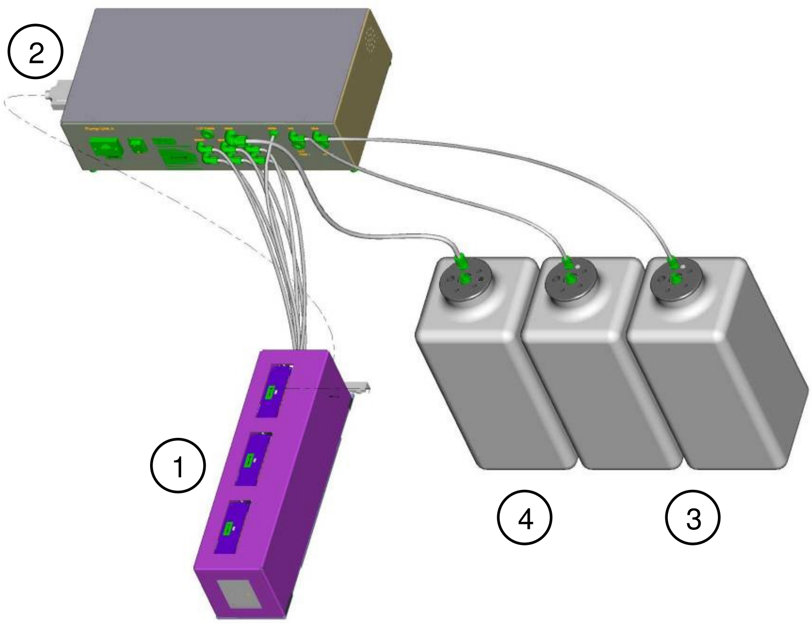
Typical time (at 40% rel. humidity, $T_{ambient} \approx 22$℃, $T_{liquid} \approx 22$℃) to heat (Ambient +5℃ to 65℃) and cool (15℃ to 4℃) is between 5 and 20 minutes.

The MultiFlex Lid Park Module can be used for parking the Heating/Cooling Module lid.

2.5.5 CR Needle Wash Station

The CR needle wash station is a device for the washing of up to 24 steel needles in parallel to the pipetting process. The wash station has the width of a normal microplate carrier (6T), and up to two needle wash stations can be mounted on the instruments deck of the ML STAR. The carry-over of the wash station depends on the wash settings. Typical values are 10^{-5} to 10^{-6}. A set of default parameters is given for the wash process within the relevant dialog boxes of the VENUS Software. Wash parameters can be adapted.

Along with each wash station (item 1) comes a pump station (item 2) with two reservoir containers (item 3) for wash solutions and a waste container (item 4). The wash and the waste containers have a capacity of 12 liters. The pump station is placed below the bench of the instrument.

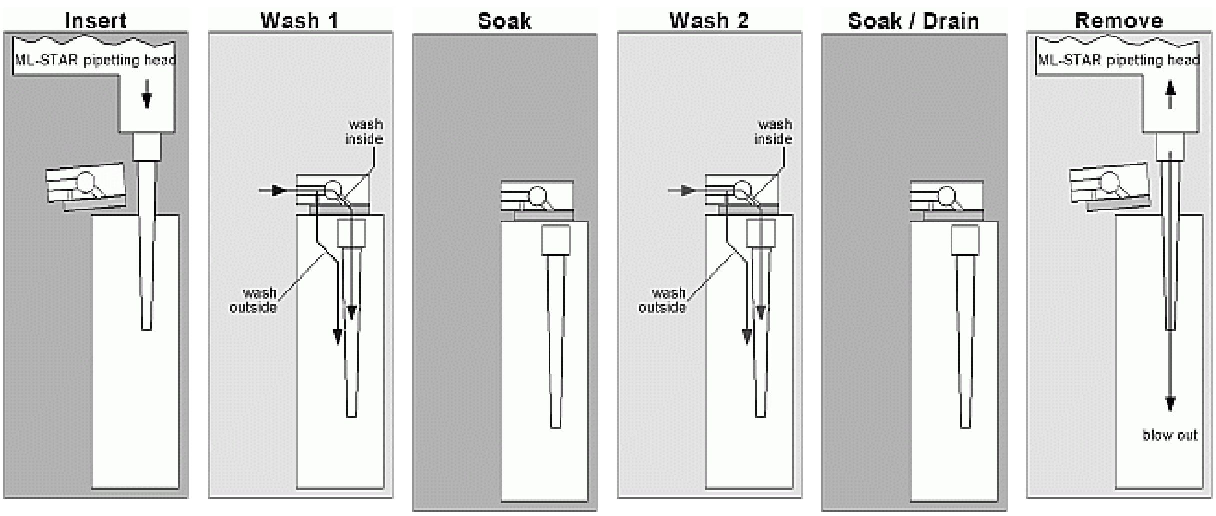


The wash station consists of three individual 8-fold wash modules. The modules are able to wash the 1000µl needles, 300µl needles or the 10µl needles. Within one wash station module all 3 needle types can be washed.

The wash cycle works in parallel to the pipetting steps. In case of an 8-channel instrument, the following steps take place:

1. The needles from the first module are picked up, used for pipetting, and then placed back into the same module. Needle washing starts.
2. While the needles of the first module are washed, the needles from the second module are picked up. After pipetting these needles are placed back into the second module. The wash cycle of this module is started parallel to the next steps.
3. While the instrument starts pipetting with the needles of the third module, the first module has finalized needle washing. After pipetting, the needles are placed back into the third module. Then the wash cycle of this module can start.
4. Clean needles are picked up from the first module again. The process is repeated again.

The principle of the wash station is illustrated in the following figure, where a typical procedure is shown.



Schematic drawing of needle wash process: needles are placed into the wash module, washed from inside and along the outside with 1) wash and 2) rinse solution. The pipetting channels blow air through the needles to expel any residual liquid.

NOTE

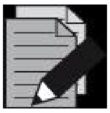
Do not use disposable tips in the Needle Wash Station.

HAMILTON recommends to replace the needles every 6 month.

For a 16-channel instrument, two independent wash stations are necessary to enable high-throughput pipetting with one needle type.

To refill wash solution, remove the lid of the empty container.

To empty the waste container, remove the lid first. Empty the container and put the lid back.



NOTE

A table of chemical compatibilities with the wash station can be found in Appendix A "Chemical compatibility"*. The information listed there is based on laboratory tests with raw materials and should be interpreted as a guideline only.*

Consider local regulations for handling and storage of wash liquids as regards toxicity, contamination, fire protection, etc.

ATTENTION

The needle wash station for ML STAR is not explosion-proof. When working with flammable or explosive fluids or vapors, the necessary precautions must be taken.

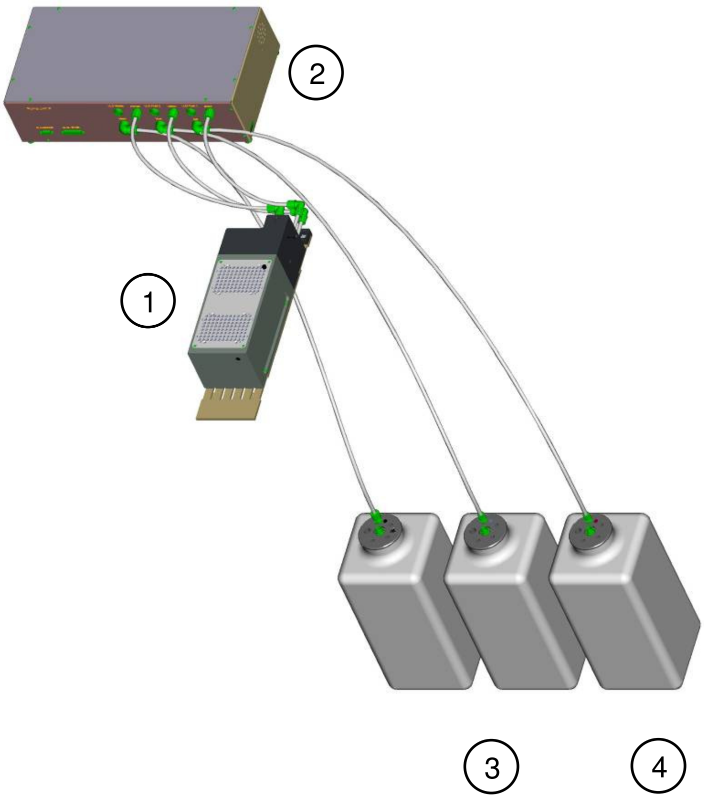
A trained HAMILTON service engineer must perform the installation of the Needle Wash Station.

2.5.6 Wash Station 96 or 384 Dual

The Wash Station 96 or 384 is an optional device for washing 96 or 384 disposable tips in parallel. Washing takes place by aspiration/dispense cycles with the CO-RE 96-Probe Head or the CO-RE 384-Probe Head respectively.

The wash station has the width of 6 Tracks, and is mounted on the deck of the instrument. Two wash chambers make the process of washing tips efficient. Each chamber can be filled individually with wash liquid out of two source containers.

Along with each wash station (item 1) comes a pump station (item 2) with two reservoir containers (item 3) for wash solution and a waste container (item 4). Both the wash and the waste container have a capacity of 12 liters. The pump station is placed below the bench of the instrument.



The pump unit includes two wash pumps and a waste pump. The wash pumps fill the wash chambers in the washer unit. The waste pump empties the wash chamber and/or the overflow chamber.

NOTE

When re-filling the wash liquid container the waste liquid container must be emptied as well.

When re-using washed tips, pipetting precision may decrease by a factor of 3.

ATTENTION

Only use water or DMSO max. 30% as wash liquid for the 96 or 384 Wash Station.

ATTENTION

The 96- or 384- Wash Station for ML STAR instruments is not explosion-proof. When working with flammable or explosive fluids or vapors, the necessary precautions must be taken.

NOTE

If 50µl tips of the 384-Probe Head needs to be washed, the 384 Dual Chamber Washstation must be used.

If Rocket tips need to be washed, the 96 Chamber washstation must be used.

If both tip types are used and need to be washed, both washstations are necessary.

2.5.7 Temperature-Controlled Carrier (TCC)

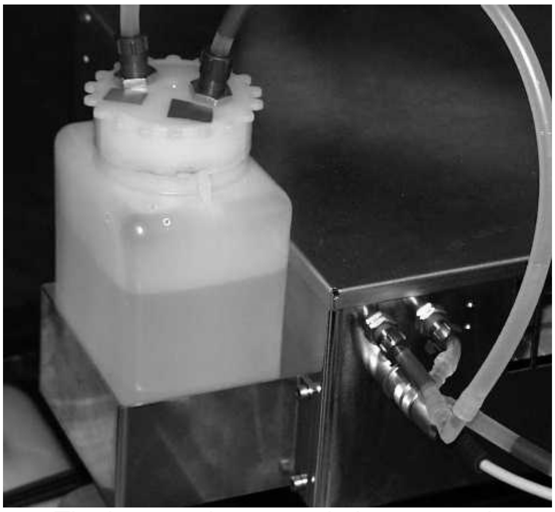
The Temperature-Controlled Carrier (TCC) is a device for heating and cooling microplates. The TCC has a capacity of four positions for microplates. The proceeded temperature is the same on all positions. The heating function of the TCC reaches temperatures of up to 60 C, while cooling microtiterplates down to 22°C below ambient temperature is possible. Up to 2 TCCs can be installed on one instrument.

TCC consists of the heating/cooling plate carrier fix-mounted on the instruments deck. The external heat exchanger device is placed below the bench. The working fluid in the refrigeration cycle is a synthetic liquid.

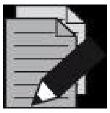
Typical times (at 40% rel. humidity) to heat and cool a microtiter plate are ($T_{ambient} \approx 20°C$):

$T_{ambient}$ to 60°C	20 min
60°C to $T_{ambient}$	20 min
$T_{ambient}$ to 4°C	15 min
4°C to 60°C	25 min

The temperature-controlled carrier (TCC)



The heat exchanger solution reservoir.



NOTE

Ensure there is always enough liquid (1l) within the reservoir.

Allow air exchange between the exchanger and ambient air.

The default position of the TCC is loaded on the instrument deck. Always ensure that the TCC is loaded on the deck at the beginning of the loading process. Never leave the TCC unloaded on the loading tray.

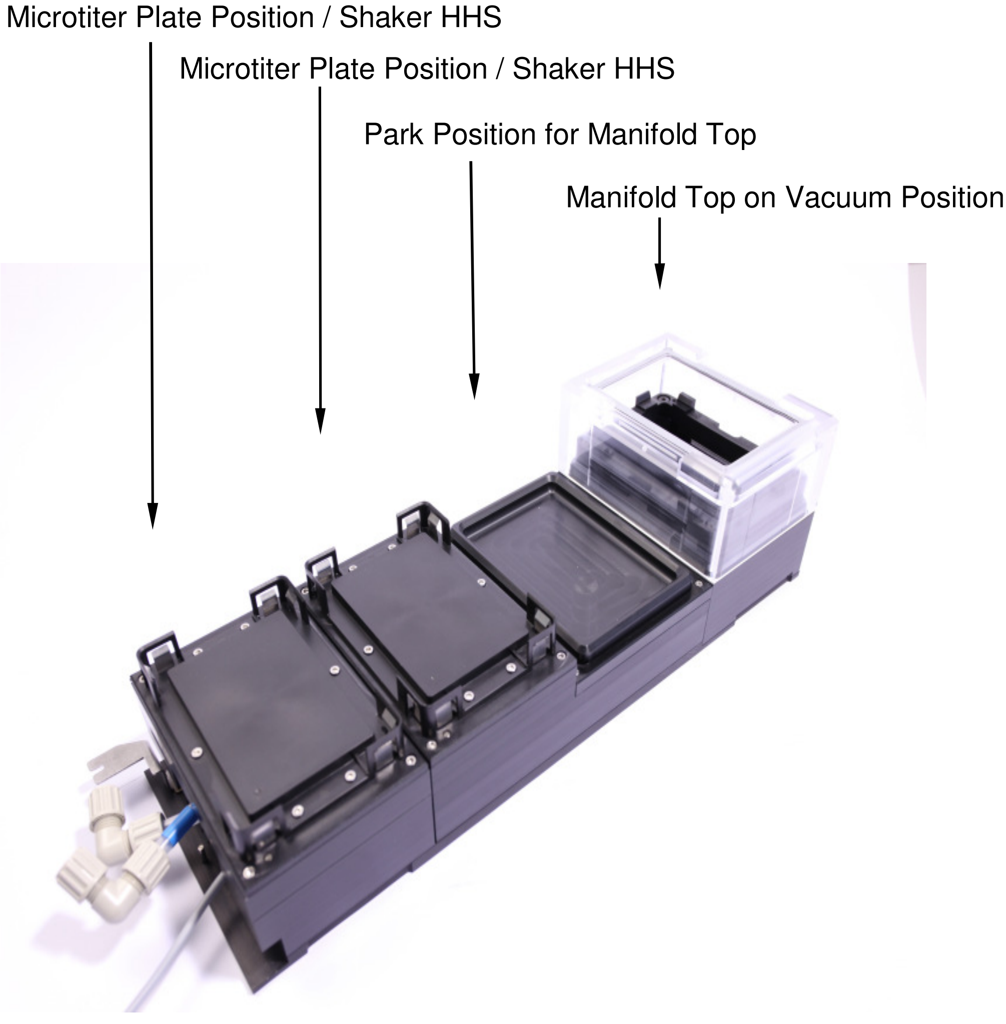
Given the TCC's fix-mounted chassis and movable carrier slide, do not use SMART Steps to load a TCC with the Autoload function. The Autoload slide may collide with the TCC carrier slide.

The installation of the Temperature-Controlled Carrier (TCC) must be performed by a trained HAMILTON service engineer.

2.5.8 CVS Crystal Vacuum System

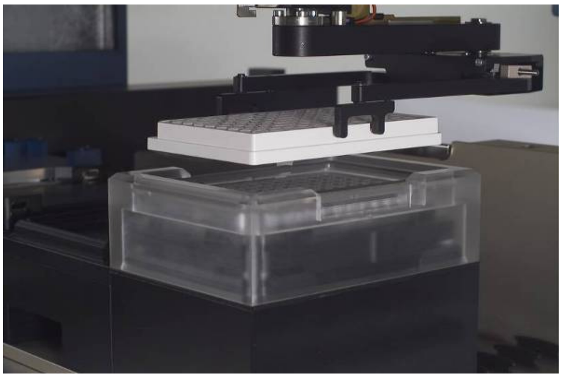
The CVS Vacuum System allows automation of vacuum based kits for SPE, LC-MS, genomics and proteomics. CVS vacuum system consists of a 7-track-wide carrier base equipped with the vacuum manifold, the park position for the manifold top and two microtiter plate positions. By default the manifold top and insert for standard filter plates is included. For processing deep well filter plates the deep well kit is required. You can find the DWP Kit in Appendix B "Ordering Information".

Optionally, the two microtiter plate positions can be replaced by microplate shakers (H+P Variomag). Or the rear microtiter plate position can be replaced by a microplate shaker heater (CAT). You can find the list of available modules in Appendix B "Ordering Information".



The CVS Vacuum System carrier is mounted on the instruments deck.It is possible to place up to 4 CVS carrier on one instrument.

The manifold top can be handled either by the iSWAP (see chapter 2.4.5) or the CO-RE Grip (see chapter 2.4.8). If working with the iSWAP, four tracks next to the CVS carrier have to be empty, usually on the right side.



The iSWAP in action to load a filter plate to the CVS

The CVS Vacuum System accommodates a wide variety of 96-well and 384-well filter plates. With the height adjustable inserts inside the vacuum chamber, almost any kind of collection plates can be used for the elution. Four sets of nuts and bolts are included for adjusting the height as required.

"standard" formats:

deep formats:

NOTE

Because of the hardware adjustment, the labware file in the Software requires an adjustment as well.

A trained HAMILTON service engineer must perform the installation of the CVS Vacuum System.

The vacuum is generated with the Vacuubrand pump (ME4C 230 V). The pump unit is fully software- integrated. For precise control the vacuum inside the chamber of the CVS vacuum system can be monitored.

Vaccubrand Pump

The vacuum inside the chamber of the CVS Vacuum System is generated with the chemical-resistant Vacuubrand Pump. The maximum flow rate of the pump is ~5 m^3/h, and the possible final vacuum ~80 mbar. There is a built-in sound absorber that reduces the noise level of the pump considerably.

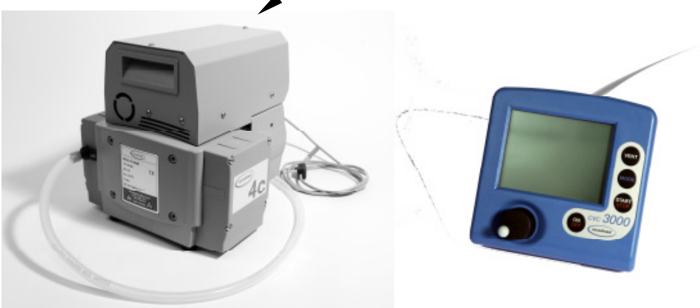
The vacuum-controller CVC 3000 regulates the pump. Communication with the computer is done via RS232 cable, and its software integrates seamlessly with VENUS Software.

NOTE

A trained HAMILTON service engineer must perform the installation of the Vacuubrand Pump.

Vacuubrand Pump

Controller CVC 2000



The air-bleed valve and the pressure sensor are both mounted inside the CVS carrier.

Waste Bottle for BVS / CVS

The liquid that is extracted during the vacuum step is collected in a waste bottle. The waste bottle can be connected to the vacuum chamber and to the pump unit.

BVS/ CVS Waste Bottle 4 liter and 2 liter

Two sizes of waste bottles are available, with volumes of two or four litres. The waste bottle should be placed below the instrument.

2.6 Tips and Needles

2.6.1 Tips and Needles for 1000µl Single channels and CO-RE 96-Probe Head

Disposable CO-RE tips come in 4 sizes with or without filter: low volume (10µl), 50µl, standard-volume (300µl), and high-volume (1000µl) for contamination-free pipetting. CO-RE tips are produced under cleanroom conditions (ISO 14644-1, class 8) and Pyrogen-, RNAse- and DNAse-free.

In combination with a needle wash station, reusable tips (1000µl needles, 300µl needles or the 10µl needles) can be used instead of the disposable tips.

HAMILTON disposable tips and steel needles for the ML STAR Line.

Left to right: low volume (10µl), 50µl, standard-volume (300µl), and high-volume (1000 µl) disposable tips; 1000µl, 300µl, 10µl steel needles.

ATTENTION

Only HAMILTON needles and disposable tips should be used for coupling to the pipetting channels of the ML STAR instrument. Non-HAMILTON tips may cause contaminated or damaged pipetting channels.

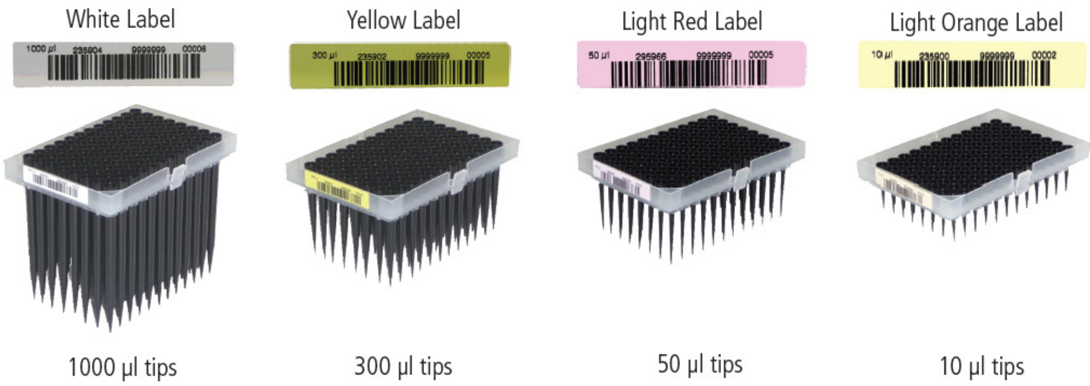
Caution if using the 50µl and 300µl CO-RE tips on the same system. Do not mix up one tip type with the other during the process of loading the deck. The wrong tip is causing either bad pipetting results or damaged pipetting channels.

NOTE

The tip type is printed in plain text on the barcode label of the tip rack and the barcode label is colored for visual identification. E.g. "50ul" for the 50µl CO-RE tips.

HAMILTON recommends to replace steel needles every six month.

All the CO-RE disposable tip types are available in racks of 96 tips each. The tip racks are barcode labeled for automatic identification during the loading process. All instruments equipped with a barcode reader are able to check for the proper tip type loading.



One blister pack contains 5 racks of 96 tips, giving a total of 480 tips. CO-RE tips are available in boxes of 3840 high-volume tips and 5760 standard volume (300µl), 50µl or low-volume tips (10µl).

10µl, 50µl and 300µl CO-RE disposable tips without filter are stackable. These tip types are available in the nestable tip rack (NTR). Because of the higher packing density and availability of tips longer runs are possible without reloading procedures.

One pack tray contains 5 stacks each with 4 NTRs of 96 tips, giving a total of 1920 tips. CO-RE tips in nestable tip racks are available in boxes of 11520 standard volume (300µl), 50µl or low-volume tips (10µl).

2.6.2 Tips for 5ml Pipetting channels

The 5ml CO-RE tip without filter guarantees contamination-free pipetting. CO-RE tips are produced under cleanroom conditions (ISO 14644-1, class 8) and Pyrogen-, RNAse- and DNAse-free.

ATTENTION

Only HAMILTON disposable tips should be used for coupling to the 5ml pipetting channels of the ML STAR instruments. Non-HAMILTON tips may cause contaminated or damaged pipetting channels.

NOTE

The tip type is printed in plain text on the barcode label of the tip rack for visual identification. E.g. “5ml” for the 5ml CO-RE tips.

The 5ml CO-RE disposable tip is available in racks of 24 tips each. The tip racks are barcode labeled for automatic identification during the loading process. All Instruments equipped with a barcode reader are able to check for the proper tip type loading.

One blister pack contains 5 racks of 24 tips, giving a total of 120 tips. 4 ml CO-RE tips with filter and 5ml CO-RE tips are available in boxes of 720 tips.

For ordering information see Appendix B “Ordering Information”, or contact your local HAMILTON dealer.

2.6.3 Tips for CO-RE 384-Probe Head

Special 50µl tips were designed for pipetting with the CO-RE 384-Probe head. These tips are available in two different formats: Either a 384 box containing 384 disposable tips or a 384 box containing 96 disposable tips. With this second option, it is also possible to use the CO-RE 384-Probe head in the 96 well format.

Due to the unique 4to1 tip-adapter the 50µl CO-RE 384-Probe head can be turned into a CO-RE 96-Probe head on the fly. With the 300µl Rocket tips volumes from 2µl to 300µl can be pipetted.

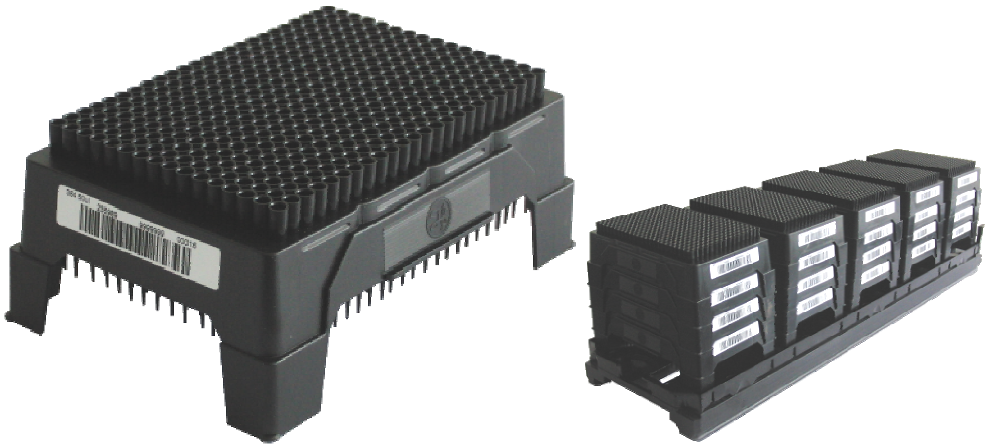
ATTENTION

Only use disposable tips recommended by HAMILTON for coupling to the pipetting heads of the ML STAR instruments. Any other tips may cause contaminated or damaged pipetting channels.

50µl CO-RE tips

384 CO-RE tips are available in the nestable tip rack (NTR). Because of the higher packing density and availability of tips, longer runs are possible without reloading procedures.

One pack tray contains 5 stacks, each with 4 NTRs of 384 (96) tips, giving a total of 7680 (1920) tips. Each lot of 384 CO-RE tips is produced under cleanroom conditions and is certified to be Pyrogen-, RNAse- and DNAse-free.



For ordering information see Appendix B "Ordering Information", or contact your local HAMILTON dealer.

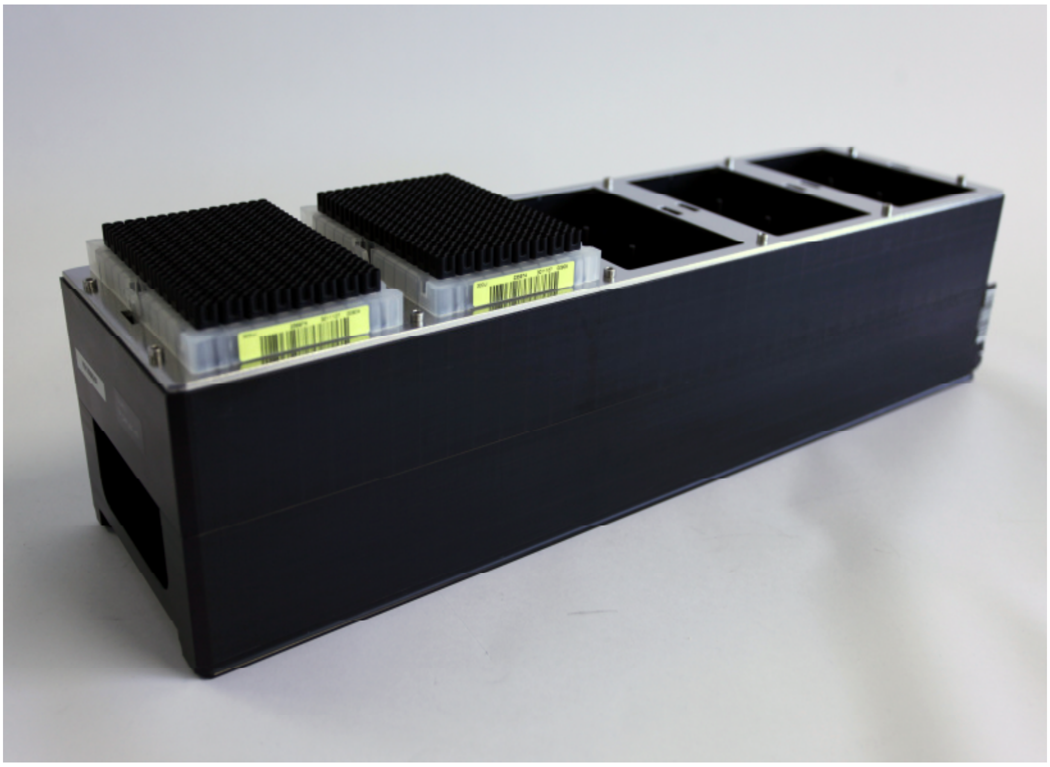
300µl Rocket tips

With the 300µl Rocket tips the 50µl CO-RE 384-Probe head can be turned into a CO-RE 96-Probe head on the fly. Rocket tips can be used with deep well plates and microtiter plates with well diameters ≥ 3.7mm.

300µl Rocket tips are available in racks of 96 tips each. The tip racks are barcode labeled for automatic identification during the loading process. ML STAR instruments equipped with a barcode reader are able to check for the proper tip type loading.

One blister pack contains 5 racks of 96 Rocket tips, giving a total of 480 tips. 300µl Rocket tips are available in boxes of 4800 tips (10 blister, 5 racks per blister, 96 tips per rack).

For ordering information see Appendix B "Ordering Information", or contact your local HAMILTON dealer.



2.7 Computer Requirements

The ML STAR insturments are controlled by a dedicated VENUS Software which controls all functions for daily work routine, method programming, running methods, and other services.

The ML STAR instruments and the computer controlling may be linked in two different ways:

- by a serial interface (RS-232C), or
- by a Unified Serial Bus interface (USB).

The communication interface used on the PC has to be set by the Configuration Editor. For the recommended PC model refer to the chapter 7.1 "Basic ML STAR Line".

To avoid data loss, we recommend a UPS (uninterruptible power supply) for the PC.

2.8 VENUS Software

The VENUS Software provides everything to control the ML STAR instruments.

It is a Windows™-based, menu-driven interface allowing the user to define deck layouts and methods, and then to run the instrument.

The VENUS Software allows programming and running different methods for aspirating and dispensing liquids, also to control accessories such as a wash station, etc.

NOTE

ML STAR Line's functioning has been verified using Windows XP, Windows Vista and Windows 7 exclusively. Running the instruments under any other operating system may lead to severe problems and/or malfunction.

Each programmed method should be validated by the programmer.

For more details refer to the VENUS two Software Programmer's Manual (P/N 624119_02).

The customer needs only to ensure that a suitable control PC is available for installation of the VENUS Software. The ML STAR instrument will be unpacked and installed and the initial set-up will be performed by a trained HAMILTON technician according to IQ form 610911.

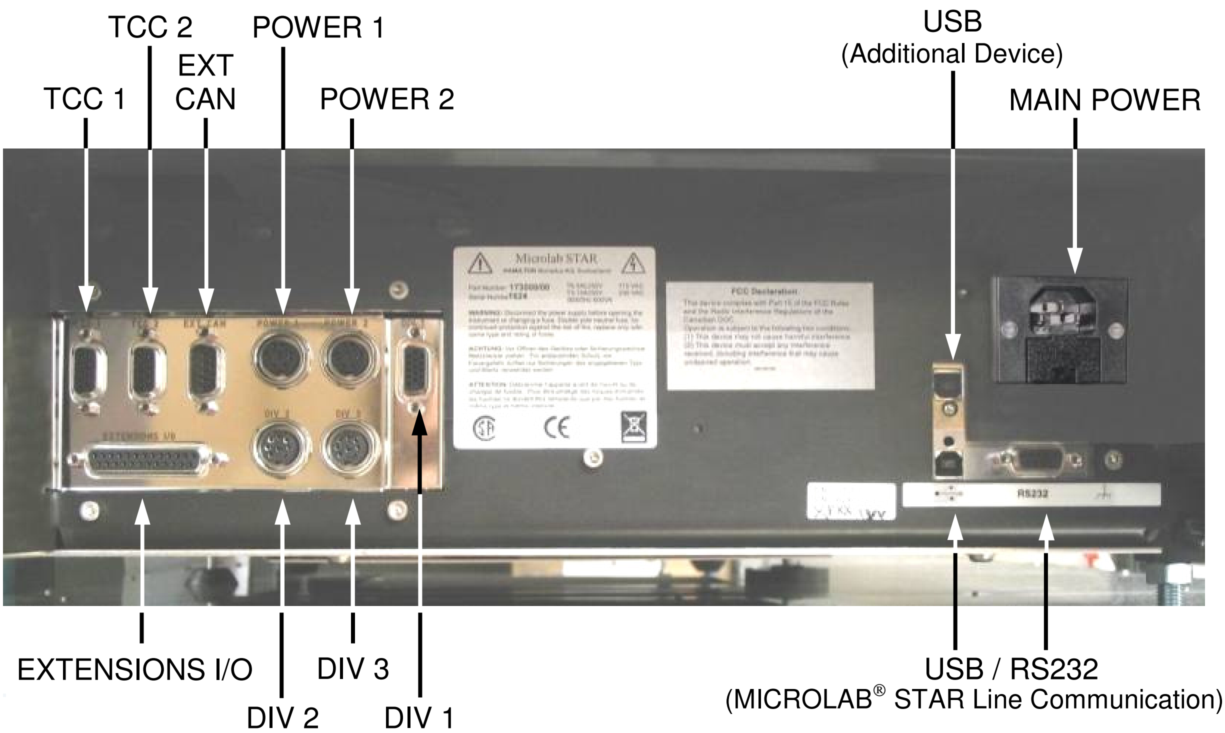
2.9 Electronics

The instrument is fully covered by a Plexiglas hood. The front cover consists of a hinged transparent window made of Plexiglas. This window is equipped with a magnetic switch that is monitored during a run. Opening the cover aborts a run.

ATTENTION

An aborted run (stopped by opening the front cover) cannot be restarted. If you need to open the window during a run, click [Pause] in the run screen, wait until the instrument stops and then open the window.

All the electrical connections are placed on the left side of the instrument, as shown below:



Near the front of the instrument (on the right side of the picture above) you will see the main power connection, and below it the communication connections to the PC: the ML STAR Line can communicate either via USB (the preferred option) or via RS232.

ATTENTION

Never use both connections together!

A second USB connector is reserved for an additional device, e.g. a CCD Camera (optional).

Various connections for external devices are available. Up to two TCCs (temperature-controlled carriers) can communicate and be powered via the connectors labeled "TCC 1", "TCC 2" and also HAMILTON Heatershaker.

The CR Needle wash station is being connected with the connector "EXT CAN".

The connectors labeled "POWER 1" and "POWER 2" provide different power supply voltages and also a CAN bus for communication.

The connectors "DIV 1", "DIV 2", "DIV 3", and "EXTENSIONS I/O" deliver several digital input/output signals as well as pulse-width-modulated (PWM) outputs, CAN bus and the TTL levels.

For further details on these connections, please consult your local HAMILTON representative.

2.10 Power / Voltage

2.10.1 Basic ML STAR Line

Make sure that the instrument is connected to a 115 or 230 VAC (50 or 60 Hz) socket. The instrument automatically recognizes any voltage within that range, without user intervention.

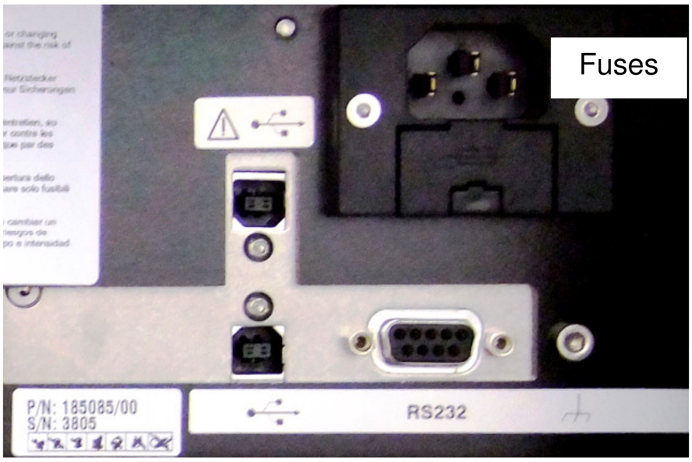
We recommend using an Uninterruptible Power Supply (UPS) for the instruments.

Ensure that the instrument is correctly grounded when connected to the power supply.

The main plug is on the left-hand side of the instrument at the front.

The fuses for the instrument are placed in the main power socket (see picture below).

Plug the main cables for the computer and the instrument into the same electrical outlet. Connect them only to a grounded outlet.



ATTENTION

Place the appropriate fuses (see chapter 7 "Technical Specifications") in the main power switch before switching on the instrument.

2.10.2 Needle Wash Station

The pump unit has its own power supply. The main plug is on the rear of the pump station (see picture below).

Ensure that the Needle Wash Station is correctly grounded when connected to the power supply. Connect the wash station only to a grounded outlet.

Ensure that the voltage selector at the pump station of the needle washer is correctly set before operating the needle wash station. The needle washer does not recognize the main voltage automatically.

The fuses for the instrument are located next to the main power switch (see picture below). The pump station has two fuses for the power supply which can be accessed by opening the cover above the main plug.

Voltage selector

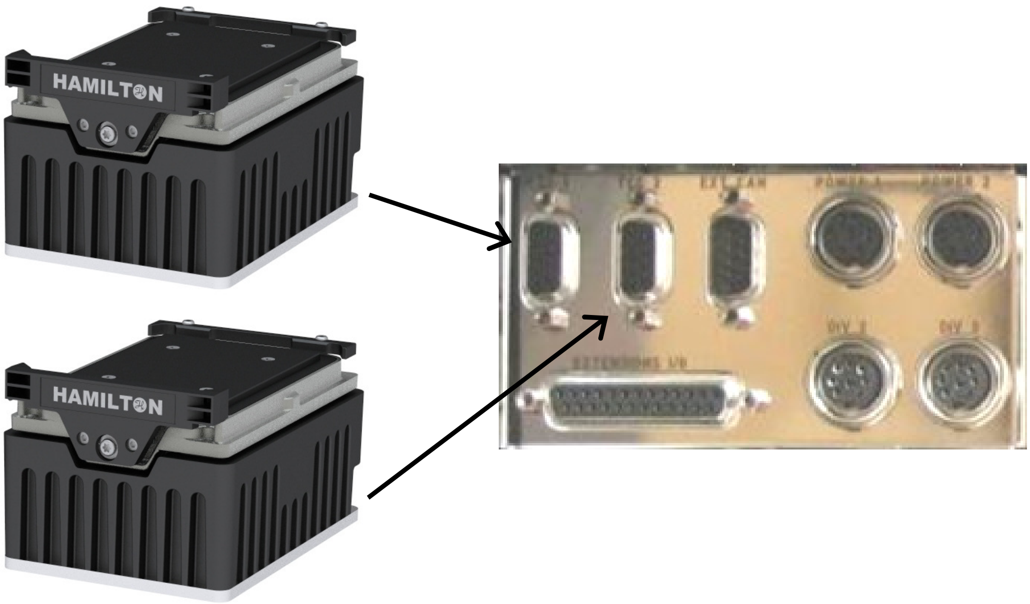
Fuses for power supply of wash station

The technical specifications regarding electrical power and fuses to be used for the wash station are listed in chapter 7.1.1 "CR Needle Wash Station".

2.10.3 Hamilton Heater Shaker

2.10.3.1 Up to 2 HHS

It is possible to integrate one or two HHS directly into a ML STAR instrument (via TCC connector, see picture below). The power and communication wire run inside the same cable. Both HHS are fully controlled by the ML STAR Software.



Two HHS connected to a ML STAR instrument

ATTENTION

Before connecting or disconnecting the HHS from or to a ML STAR instrument make sure that the ML STAR is switched off. *Only when the ML STAR is switched off it is safe to connect or disconnect the HHS. Not obeying this instruction may result in damage of the HHS.*

Do not touch the HHS by hand during run time and also not shortly after finishing a run, as it might be hot. Wait until the HHS has cooled down to room temperature.

NOTE

The HHS is intended to be used in combination with an ML STAR instrument only!

Do not use the HHS as a stand alone device.

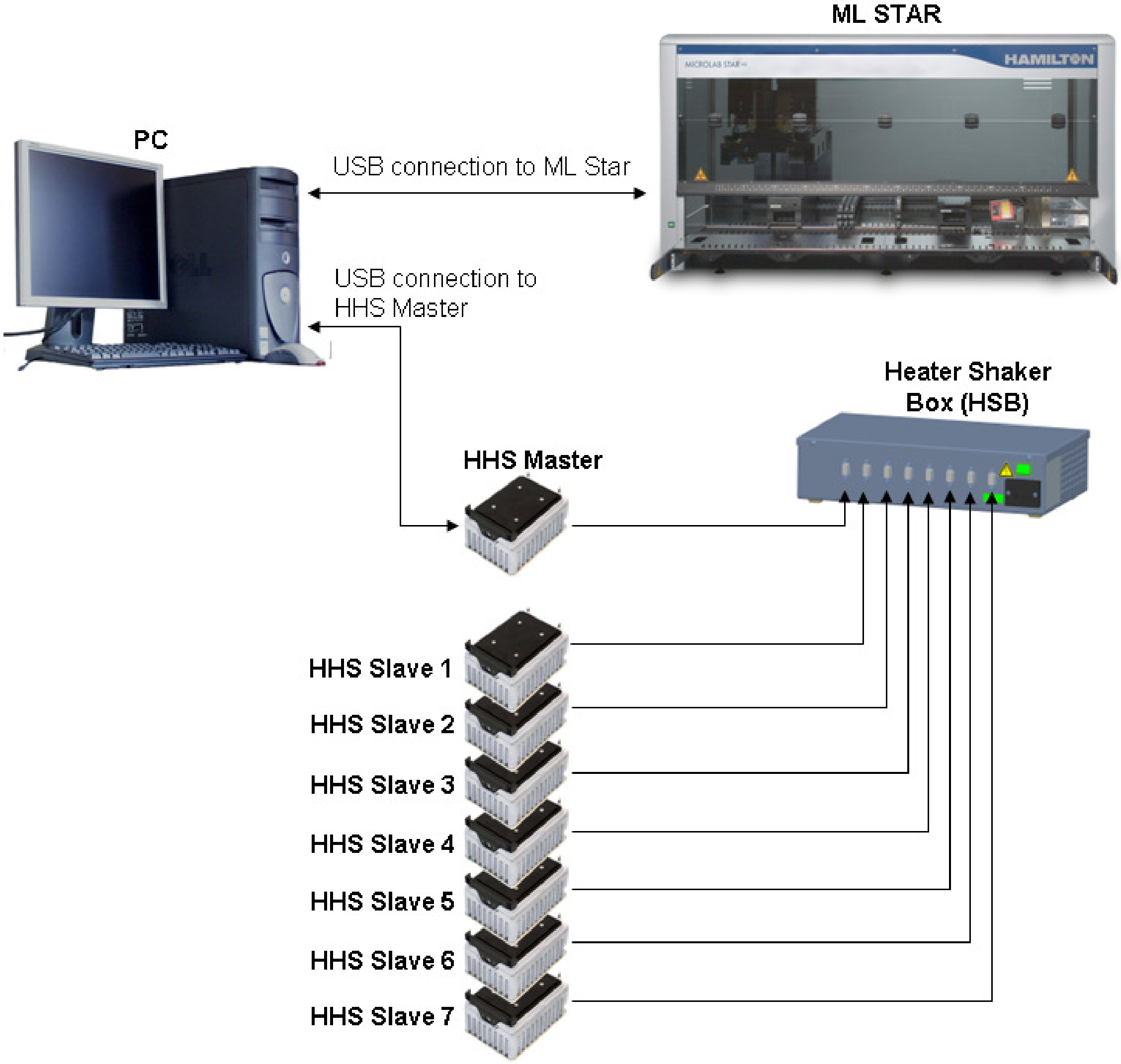
ATTENTION

When working with toxic chemicals or samples, which will be used in particularly sensitive tests, take into account the evaporation that may occur while the HHS is running.

Be aware that heating of liquids will affect the pipetting process, as well as shaking the labware can influence the precision of pipetting.

2.10.3.2 Up to 8 HHS

If more than two HHS are required, the Heater Shaker Box (HSB) is needed. One HHS (HHS master) is connected to the USB port of the PC and serves as master module. In addition to the master HHS, up to seven additionsl HHS (HHS slave 1 …. 7) can be connected to the HSB. The HSB serves a power supply and as signal distributor for all the HHS slaves.



ATTENTION

When working with samples, which will be used in particularly sensitive tests, take into account the evaporation that may occur while the HHS is running. Be aware that heating of liquids will affect the pipetting process, as well as shaking the labware can influence the precision of pipetting.

ATTENTION

When shaking labware, the velocity has to be adjusted to the type of labware and the volume within the labware to prevent cross-contamination due to spillages.

2.11 Disposal

After the life cycle of the instrument has reached its end, please contact your local Hamilton representative regarding disposing the instrument.

The European Community requires from manufacturers to organize the disposal and waste of electrical and electronic equipment (WEEE). For this reason, HAMILTON Bonaduz AG took part to organize the disposal of STAR Line products through a European disposal network. Please contact your local HAMILTON representative for further information about the recycling procedure.

Otherwise local disposal regulations are to be observed.

2.12 Training

Training in operation of the ML STAR instuments and general use of the VENUS Software will be provided by HAMILTON personnel upon initial set-up.

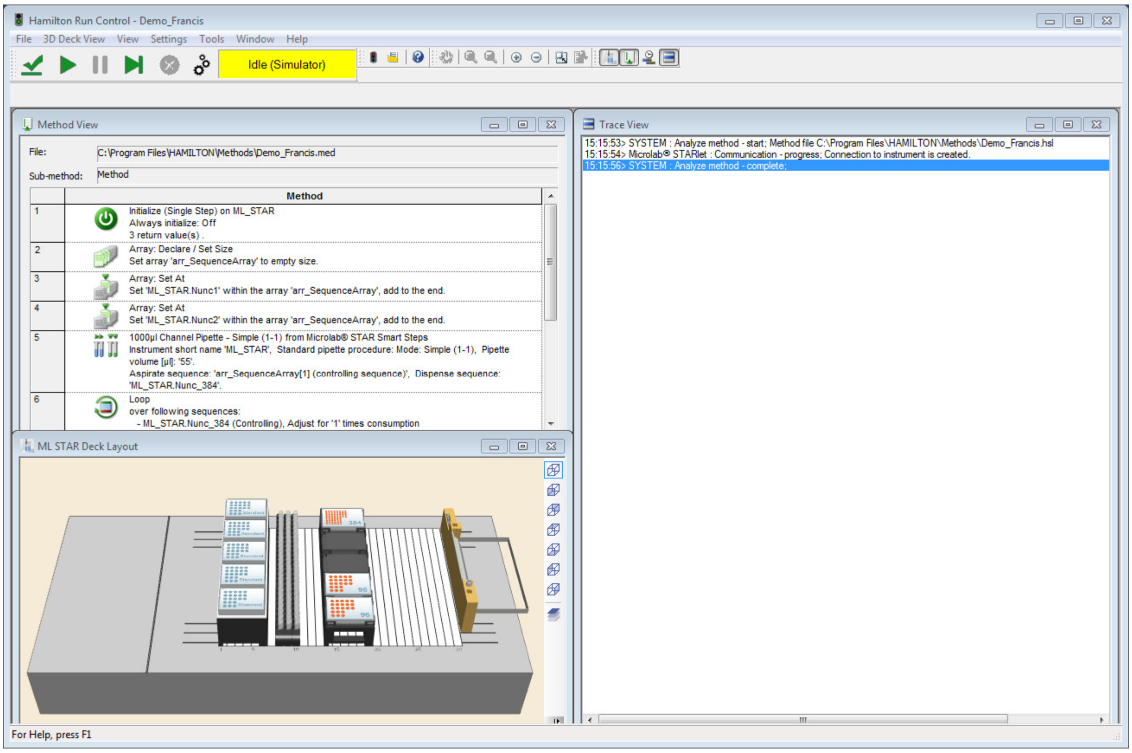
3 Routine Use

3.1 Loading the ML STAR instument

To run a method:

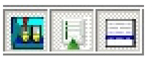
1. To access the Run Control, double-click the "Microlab STAR Run" shortcut on the desk top:

2. From the File menu of Run Control, select "Open", to open a method (*.med). Your method is now loaded:

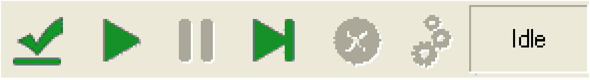


The free area you see after starting "Run Control" can display up to three windows:

- Deck View
- Method View
- Trace View

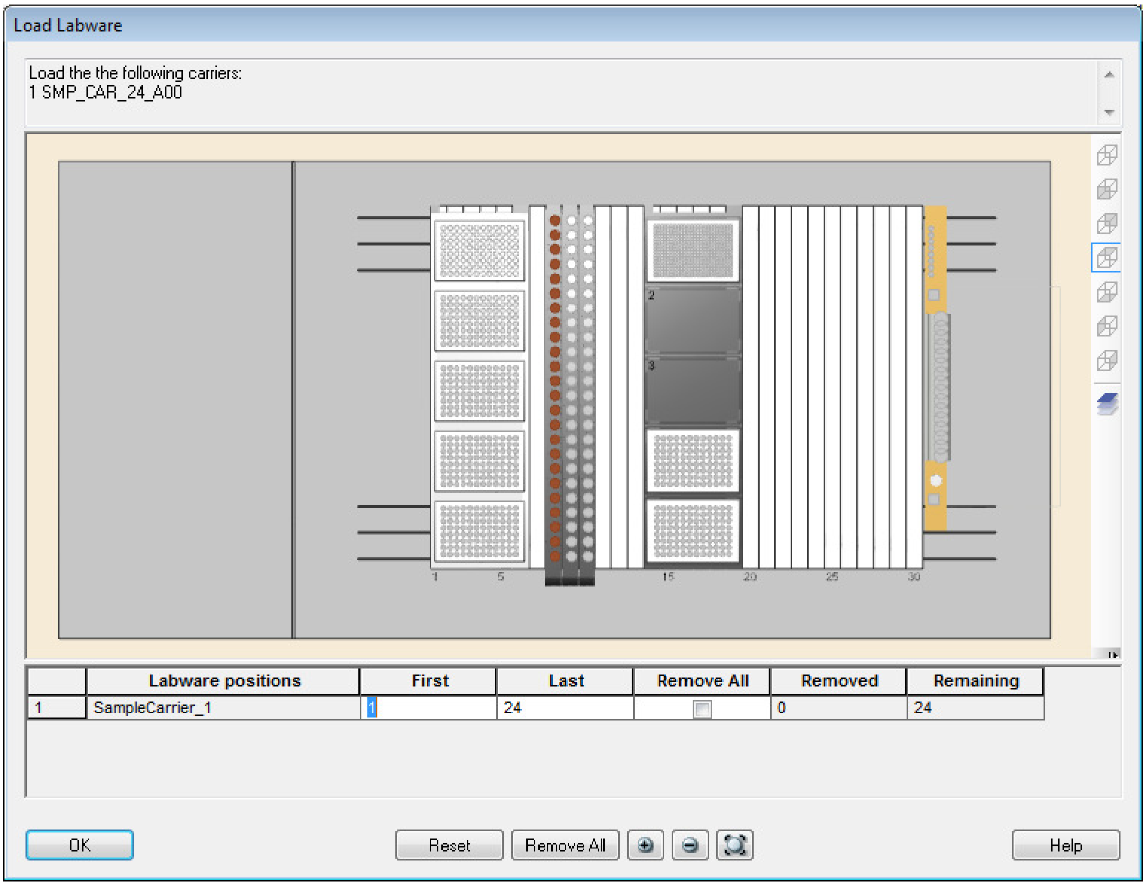
The windows may be enabled/disabled by means of the menu "Run Views" (click the desired menu entry) or by clicking on the appropriate icon in the "Run Views" Toolbar  (if enabled). Arrange the windows according to your preference using the menu "Window". In the window shown above, you can see the deck layout and the method log (trace) of a run as it is generated.

3. Press the [Start] button (the one which looks like a [Play] button)



to run the method. You will see that the steps in the method are traced to the log frame.

If a loading step is part of the method, a dialog appears that requests a reduction of the number of positions on e.g. the sample carrier, as well as a start position for the tips to be picked up. Both pieces of information are optional.



Loading Dialog

4. Enter the number of samples, tips or wells for this run, delete items graphically from the highlighted positions on deck (first select the appropriate item in the table), or accept the default.

   Generally, whenever the system finds a 'Load Carrier' command in the method, the operator is requested to place the carrier holding the appropriate labware on the Autoload tray. The correct position is highlighted by blinking LEDs.

ATTENTION

It is important to ensure that adequate amounts of the correct liquids are placed in the correct labeled containers.

When pouring reagents into the reagent containers, ensure that there is no foam on the surface of the liquid.

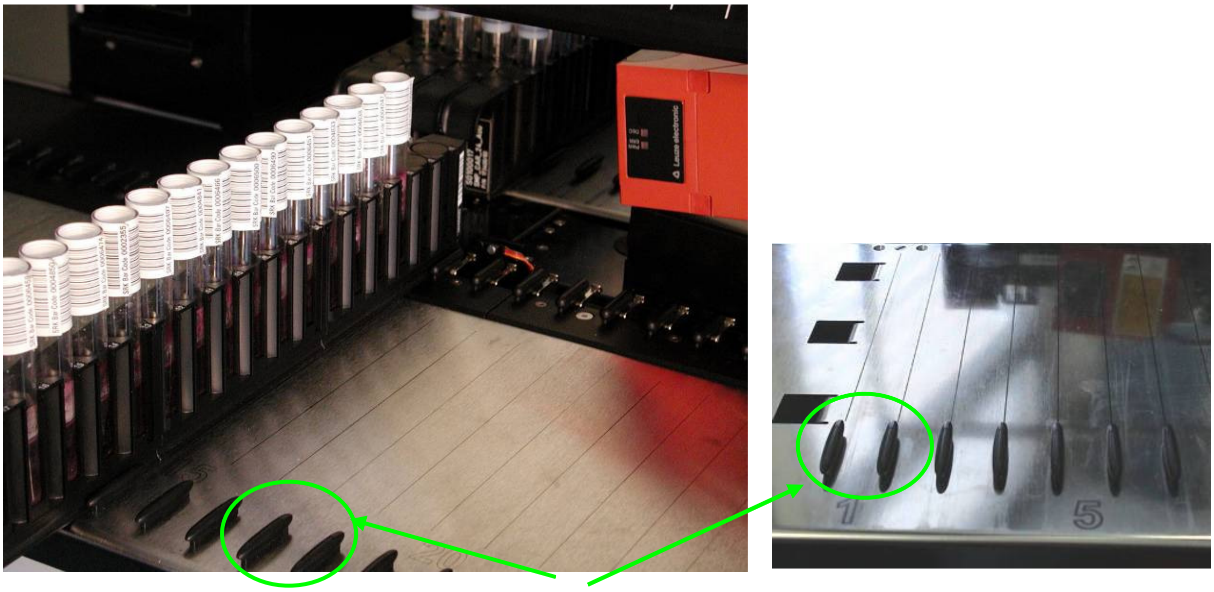
Do not overfill reagent containers: fill approx.10 mm below the top of the container.

Always use the proper labware (tips, microplates, tubes, etc.) corresponding to the definitions of the method's layout.

Position microplates correct such that well #A1 is placed according to the deck layout.

Handle any 1-track carrier (such as a sample carrier) with particular care, as this type of carrier can fall over and cause injury or contamination. Position it on the Autoload tray (see figure below), or place several carriers together to minimize this risk.

5. Insert the carriers into the tracks between the front and rear slide blocks of the Autoload tray until they touch the stop hooks on the far side of the tray.



Slide Blocks for Carriers

Stop Hooks for Carriers

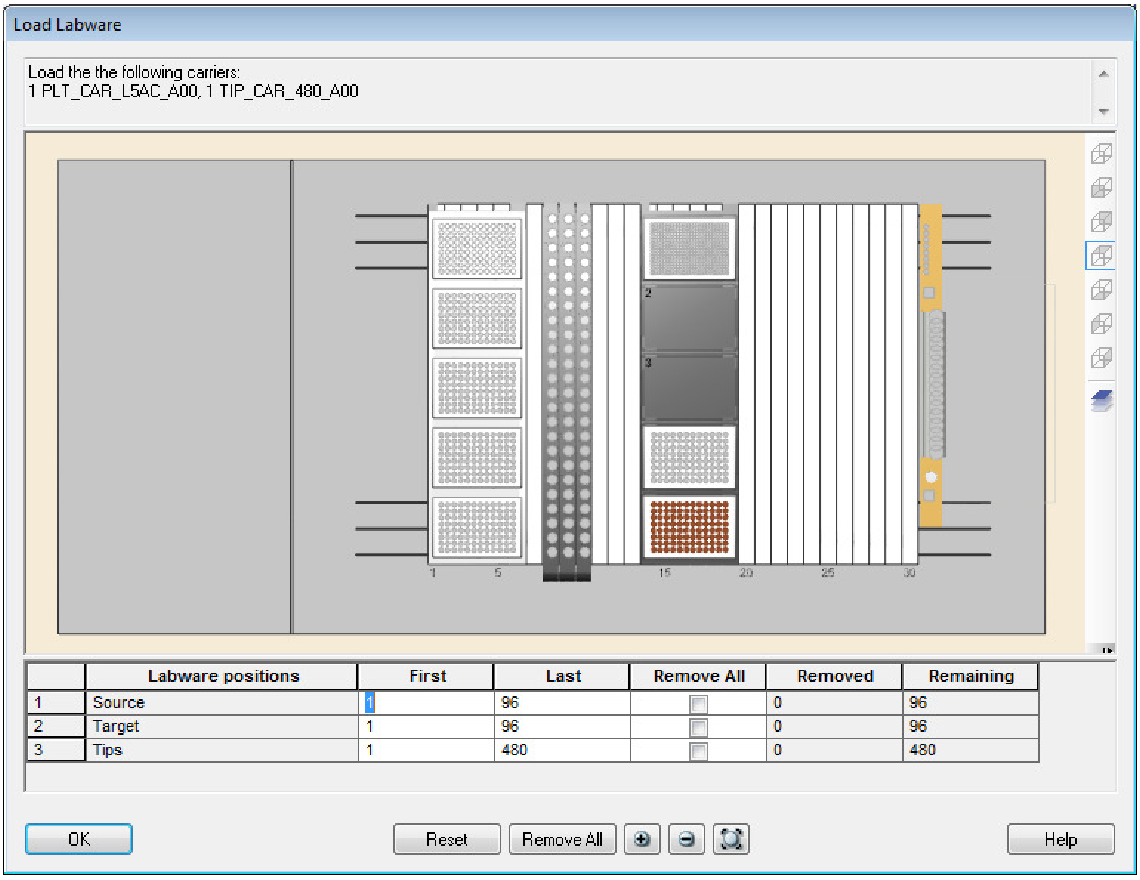
ATTENTION

Make sure the carriers are inserted completely as far as they will go on the loading tray.

6. Click **[OK]** in the dialog. The carriers are loaded onto the deck automatically by the 'Load Carrier' command in the method. During loading, the barcodes of carriers and labware are read and stored in a file.

   Or:

   Alternatively, load the carriers onto the defined positions of the Autoload tray before starting the method. Loading and barcode reading will then be performed without operator input. Still the dialog pops up:



7. You can enter the number of wells on e.g. the source plate for this run, delete wells graphically from the sequence or accept the default (copy the whole plate).

Manual Load Only:

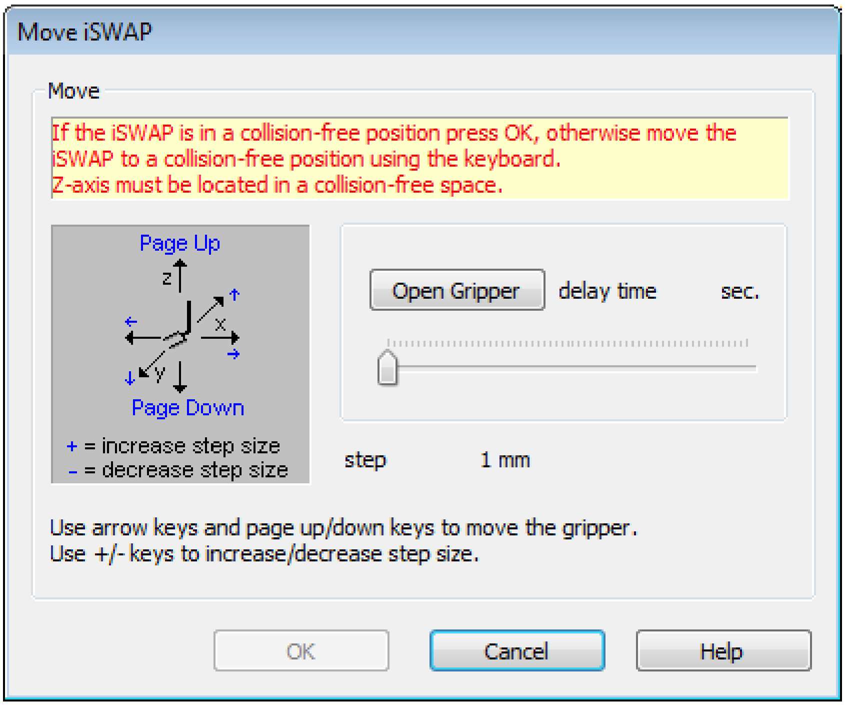
8. Load the deck with the carriers mentioned in the upper part of the dialog box (the two plate carriers and two tip carriers). Don't forget to place labware onto the correct positions.

Autoload Only:

9. Whenever the system finds a Load Carrier command in the method, the user is requested to feed the carrier holding the appropriate labware onto the autoload tray. The correct position is highlighted by LEDs on the instrument. Alternatively, all carriers can be placed directly in their correct positions on the autoload tray.
10. Click **[OK]** in the dialog box to start loading.

iSWAP only

11. Usually, at the first run of a day, the iSWAP needs to be initialized. There is also a possibility that there is still a plate in the gripper of the iSWAP. In case a plate is caught by the gripper, you can safely remove it.



Removing a gripped object:

Use the “+”/”-” keys to define the distance (e.g. 10mm) the gripper shall be opened to release the plate. Define a “delay time” moving the pointer of the slide bar. Calculate the time (e.g. 5 sec) you will require to walk from the PC to the instrument and place your hand underneath the plate. Open up the safety cover of the instrument and check if you can reach-out for the plate. If you cannot access the plate move the pipetting arm using the arrow keys and page up/down keys.

ATTENTION

Make sure the selected step size and moving direction will not crash with the iSWAP.

Always wear gloves for manual manipulations inside the instrument.

If there is enough time and space to hold the plate and the gripper parameters are set press the **[Open Gripper]** button.

Removing a gripped object:

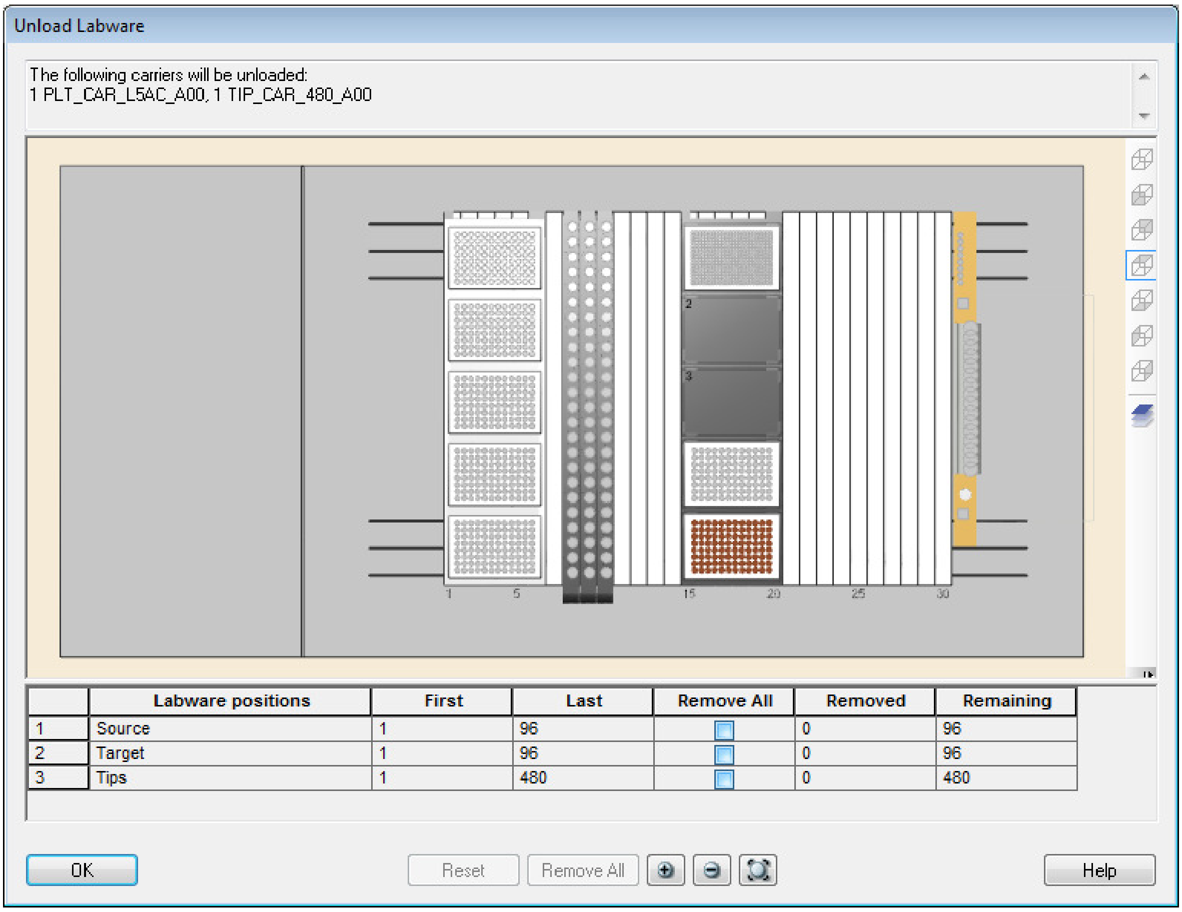
If the iSWAP cannot move up to the traveling height without a collision (e.g. shelving position where the iSWAP would collide to the shelf above it), it must be guided to a save area using the keyboard (arrow keys, page up/down keys, +/- keys).

ATTENTION

Make sure the selected step size and moving direction will not crash the iSWAP.

12. If the iSWAP is ready for initialisation press the **[OK]** button.

If an unload step is part of the method, you are requested to unload the carriers from the deck. The following dialog box opens:



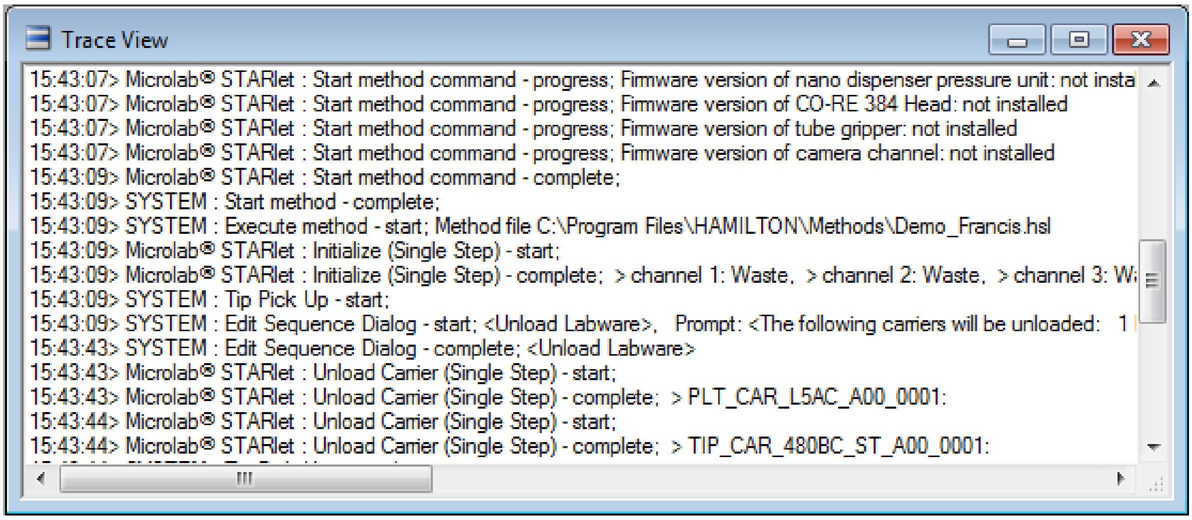
Manual Load Only:

13. Click **[OK]** in the dialog. Unload all the carriers manually. The unloading will be checked by the system.

Autoload Only:

14. Click **[OK]** in the dialog. The carriers are unloaded to the autoload tray.

The method is now finished. Method completion information is visible in the “Trace View” window.



Method Trace File

Every run creates a separate method trace file. The method trace file is stored under “{method namex32x}.trc” within the “...\LogFiles” directory. Each method trace contains the method name, and a unique number consisting of 32 hexadecimal digits, here represented by “x32x”, within the file name. The method traces are not overwritten or appended.

Communication Trace File

The communication trace file is created every day or when the instrument is switched on. A communication trace file is also generated under the “communication name” (Simulator, HxUSBComm) and current date, where the communication of each run of the method will be appended. If barcode information is generated, its data will be stored in a file labeled with the method’s name followed by **“*_Barcodes.txt”.**

NOTE

From time to time all unused method traces and com traces should be deleted from the hard disk.

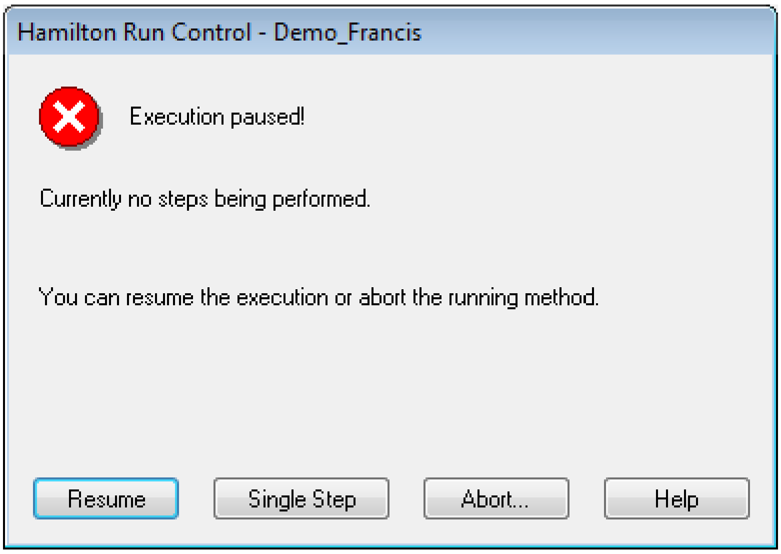
Single Step Mode

You can test your method using single steps. It is always possible to start a run in single step mode (like TipPickUp, Aspirate, Dispense, etc.)

1. Click the **[Single Step]** button, to start the method in single step mode.

   Or:

2. Pause a method at any time by clicking on the **[Pause]** button of Run Control. The step currently running will be terminated, and the following message box appears:



3. Click anywhere in the pause dialog to stop the beeping.

   During the pause, it is possible to open the front cover of the instrument.

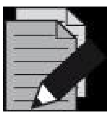
4. Before continuing the method, make sure the cover is closed again.
5. Resume the paused method by clicking **[Single Step]**,

   or

   resume method execution by clicking **[Continue]**, to run the method without further breaks

   or:

   abort method execution by clicking **[Abort...]**. You will be prompted to confirm the abort.



NOTE

An abort may cause the loss of data.
Aborted methods cannot be restarted again, unless explicitly programmed.

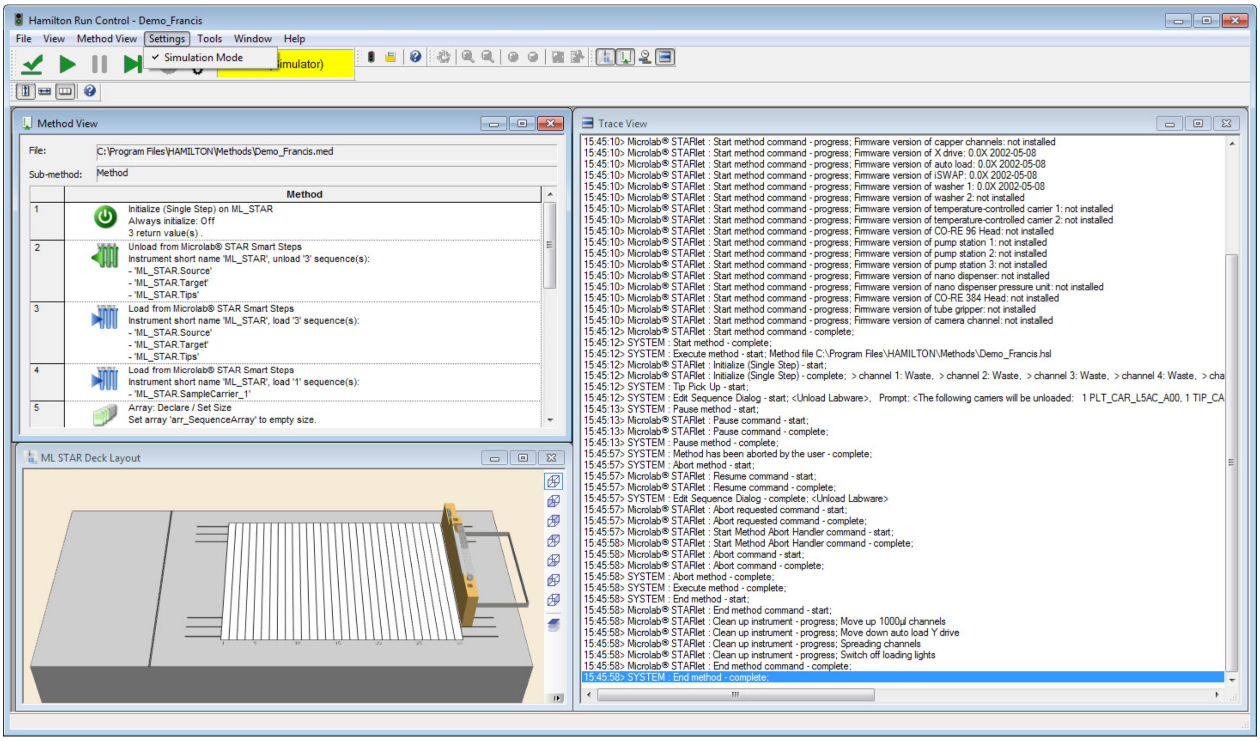
A fast abort can always be done by opening the front shield of the ML STAR instrument during a run.

*A method can be aborted even if **[Pause]** is active.*

*To prevent loss of data or bad pipetting result **[Pause]** will finish the proceeding step and stops then.*

3.2 Run Simulations

It is also possible to run a simulation instead of the actual run. It is recommended that you always simulate a newly created method first, before running it on the instrument. The run simulation is switched on in the menu "Settings":



Enable the simulation by clicking to **[Settings** / **Simulation Mode]**.

The run simulator checks the consistency of each (single) step of the system, together with minimum aspiration and dispense volume management.

The speed for simulations is adjustable. If the default speed (at maximum is best guess under real runs) is too fast the speed can be reduced by the simulator delay via the Configuration Editor. Refer to the Programmer's Manual.

3.3 Run-Time Error Handling

Before using the run time error handling, several types of problems causing errors have to be solved first. Among these are

- Syntax errors when programming in HSL (e.g. forgotten ";")
- Logical errors (e.g. "tip eject" before "pick-up", or "asp 20µl", "disp 100µl")
- Semantic errors (e.g. wrong pipetting pattern)
- Method/deck interaction errors (e.g. dispense 100µl into the first well of a 1536-well MTP)
- Liquid handling/application errors (e.g. droplets, foam, non-pipetted wells)
- User-related errors (e.g. wrong deck loading)

These problems *cannot* be handled by any run time error handling. They have to be solved in advance, while or after the programming. Refer to the Programmers Manual.

Problems that *can* be handled in run time are

- Not enough liquid
- Liquid level not found (if it occurs exceptionally)
- No tip picked up
- Clot detected
- Barcode unreadable (if it occurs exceptionally)
- Execution error (channel no. 1 has an error (e.g., not enough liquid), then channel nos. 2-8 have an execution error because they have been stopped before completion of the step)

NOTE

In principle, each channel may have one or more different types of errors at a time.

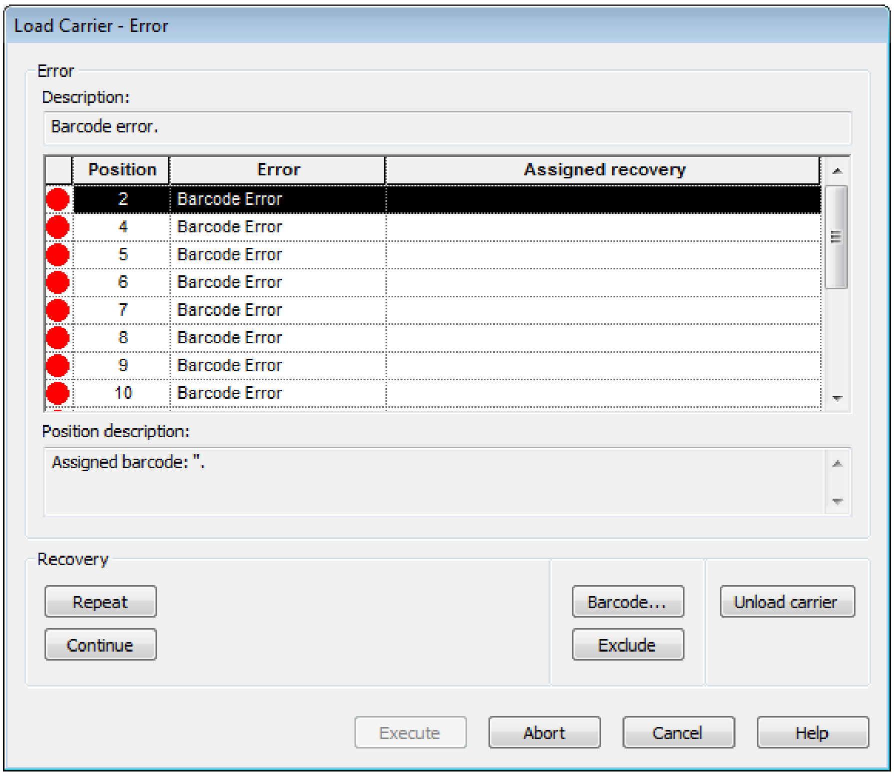
If all channels have the same error at the same time, a collective recovery can be made.

3.3.1 Examples

We now focus on some important examples. A detailed description is available in the Help-Menu. Click on **[Error Settings]** within the single step dialogs of the ML STAR Line-specific commands and then on **[Help]**.

Barcode Reading Error

In the case of an error, the process can be continued using the error handling procedure. If, for example, a barcode of a carrier cannot be read, a dialog window opens up:



The error recovery can be programmed to:

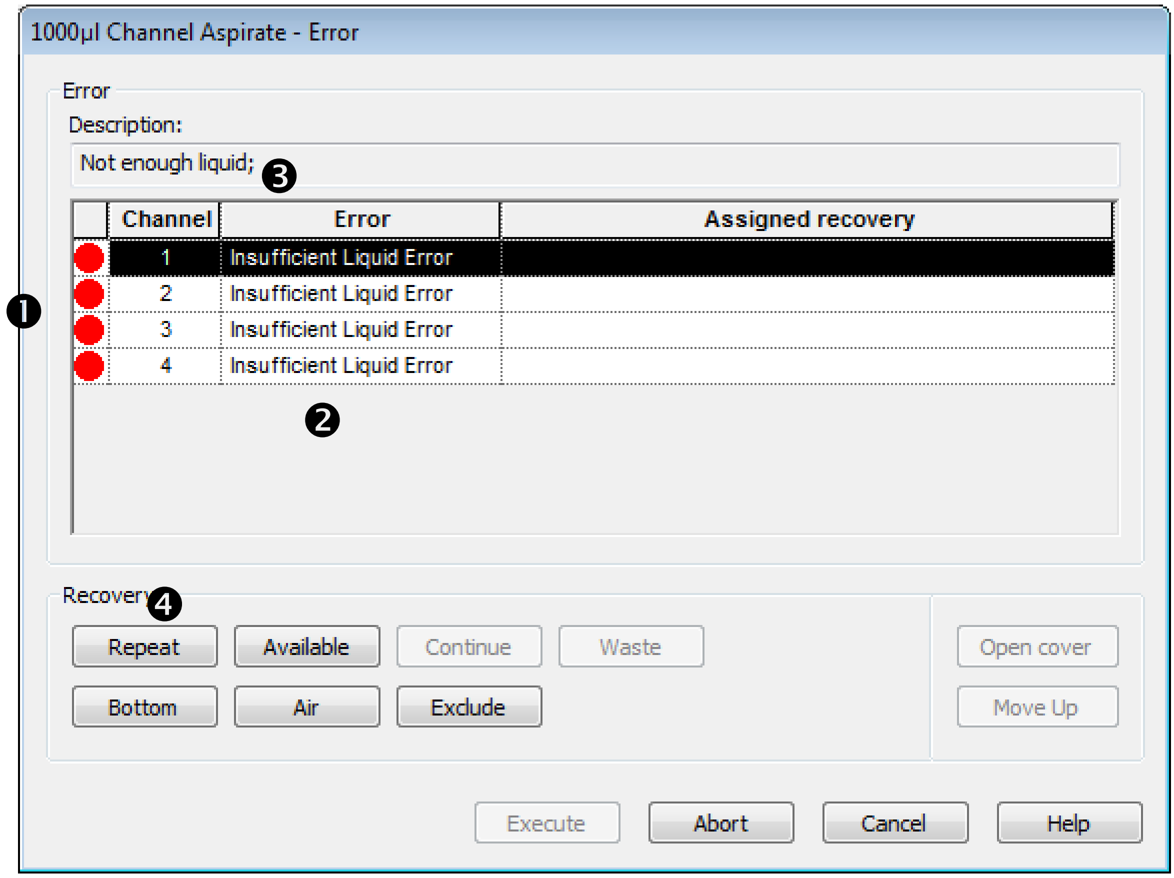
- **Continue** ignore the error message (here, failure of barcode reading)
  - **[Continue]** makes no sense in the case of a barcode reading error - at least a manual entry has to be made so that barcode data exists for further processing.
- **Repeat** try to read barcode once again.
  - Often a repetition of reading will solve the problem because the reading speed is reduced then. The selected action is displayed in the field **[Assigned recovery]**.
- **Barcode...** Enter barcode manually
  - Clicking **[Barcode]** opens an entry dialog box where a barcode can be entered (no entry is also allowed).

1. Assign a recovery option. The **[Execute]** button becomes active.
2. Click on **[Execute]**. The instrument proceeds with the selected recovery option.

Pipetting Error

If an error occurs with the pipetting channels, a dialog shows its error state and its recovery options for every single channel. (Different channels can have different errors.)

For example, in case of an LLD error such as no liquid in the container while aspirating, a window similar to the following pops up:



All pipetting channels which produced an error are listed with a red dot on the left ❶. A short error description ❷ is given. A detailed error description is shown at the top text field ❸ labeled "Description:" for the selected channel. There may be different errors which lead to the same short error description.

The **[Recovery]** ❹ frame buttons offer several possible actions to solve the problem:

- **[Repeat]**: executes the command which caused the error once again
- **[Available]**: aspirates the available volume from source and fills up the missing volume with air
- **[Continue]**: continues as if no error was recognized
- **[Waste]**: means that the erroneous tip is ejected to the waste and the channel is excluded.
- **[Bottom]**: activates the channel to move down to the bottom of the container, and the available volume is aspirated without LLD
- **[Air]**: means air is pipetted instead of liquid and the method will continue
- **[Exclude]**: allows you to disable any further action on the error-affected channel
- **[Move Up]**: this is not a real error recovery procedure, but useful e.g. to manually remove a clot. This causes the following actions (this action can be repeated):

- moves the barcode reader of the autoload (if present) to the very right
- moves the selected channel up by 10 mm

- **[Open cover / Close cover]**: this button does not start any error recovery actions either, but triggers the following actions:
  - **[Open Cover]**: enables you to open the front cover during error recovery
  - **[Close Cover]**: enables you to close the front cover before executing error recovery

You will need to assign an error recovery for every error. Selecting a channel followed by any possible recovery procedure assigns the selected recovery procedure to all error-affected channels with the same error.

If any recovery procedure is assigned to a channel (even one you do not wish to assign), the dot ➊ in the first column (see picture above) changes color to green.

Some recovery buttons are disabled, to prevent further faulty steps, e.g. an error-affected aspiration step cannot be "Continued", to prevent any later dispense with insufficient volume.

When the last faulty channel is assigned to a recovery procedure, the **[Execute]** button becomes active and the system can proceed.

In any case the method can be aborted without further recovery options.

3.4 Walk-Away (Predefined) Error Handling

The programmer can define a walk-away error handling which uses predefined default settings for different error situations. These settings can be customized for single steps and easy steps only. For SMART steps, the error recovery is fixed.

For every instrument-specific single step of your method, an individual error recovery can be defined. You can configure

- the appearance of the error recovery dialogs (which buttons are available)
- a timeout, after which the default recovery is carried out (the dialog automatically closes down)
- the default procedure (what should be executed if the timeout runs out)
- which error is flagged in the trace file.

For this purpose, every instrument-specific single step and easy step has an **[Error Settings]** button. To deal with the subject of error handling consult the Programmer's Manual.

4 Maintenance

Periodic maintenance routines need to be run in order to ensure safe and reliable operation of your ML STAR Instrument and the accessories.

Depending on how the VENUS Software is installed, these maintenance routines might be mandatory. If so defined – until they are completed, the user is reminded by a warning.

4.1 Maintenance Intervals

We recommend that you maintain your instrument at the following intervals:

- daily recommended before the instrument is shut-down.
- weekly recommended at the end of the week before the instrument is shut-down.
- six-monthly preventive service maintenance carried out by a service technician.

NOTE

If the operator decides not to run either daily or weekly maintenance before shut-down the instrument, these routines must be implemented at the next run start.

If any parts of the instrument, carriers or racks have become contaminated, the weekly maintenance procedure must be performed.

4.2 If Maintenance Fails

If an error is encountered during a maintenance procedure, try to rectify the problem and re-start the maintenance procedure. If you cannot rectify the error yourself, call your local service representative.

4.3 Preventative Maintenance

A brief preventive maintenance including volume verification should be carried out at regular intervals by a trained HAMILTON service engineer. A service agreement ensures regular maintenance and verification for a specified period of time. HAMILTON recommends that maintenance and verification take place twice a year.

4.4 Materials Required

- Disposable latex gloves
- Protective glasses
- Lab coat
- Paper towels
- Lint-free cloths or Q-tips
- Set of 8 Maintenance Needles - please refer to Appendix B, "Ordering Information"
- Ethanol (70%)
- De-ionized water
- MICROLAB™ Detergent & Disinfectant Kit (P/N 281242) (contains DECONEX 61 DR) - please refer to Appendix B, "Ordering Information" or Microcide SQ (only available in the USA).
- MICROLAB™ Disinfectant Spray Kit (P/N 281243) (contains DECONEX SOLARSEPT) - please refer to the Appendix B, "Ordering Information"
- MICROLAB™ Disinfectant STARTER Kit (P/N 281245) (contains DECONEX 61 DR and DECONEX SOLARSEPT) - please refer to the Appendix B, "Ordering Information"

ATTENTION

Use cleaning, disinfecting and decontaminating fluids in accordance with manufacturer's instructions. Do not use disinfecting materials which contain hypochlorite (Javel water, Chlorox) or bleaching fluids.

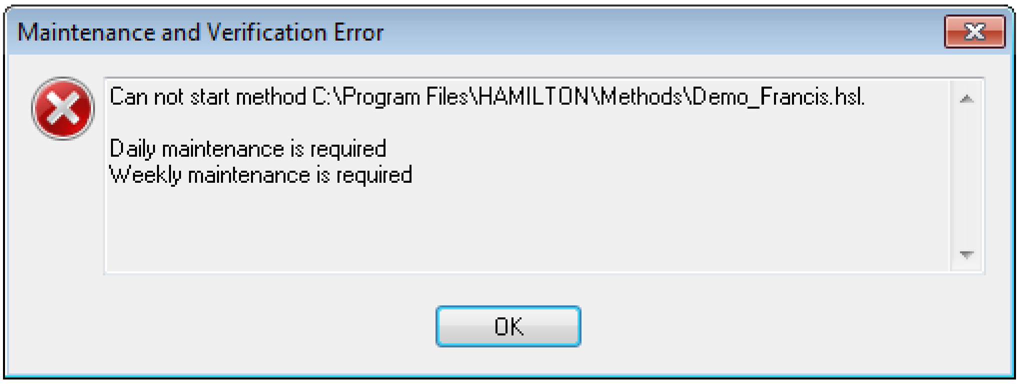
Prepare disinfectant fluids according to their labeling.

4.5 Instrument Maintenance Procedures

The operator will be guided by the VENUS Software through the regularly scheduled maintenance procedures.

Depending on the Maintenance configuration settings (*optional, warning* or *mandatory*) the instrument will display messages when starting up the instrument:

Displayed warning if maintenance needed, config. setting = warning



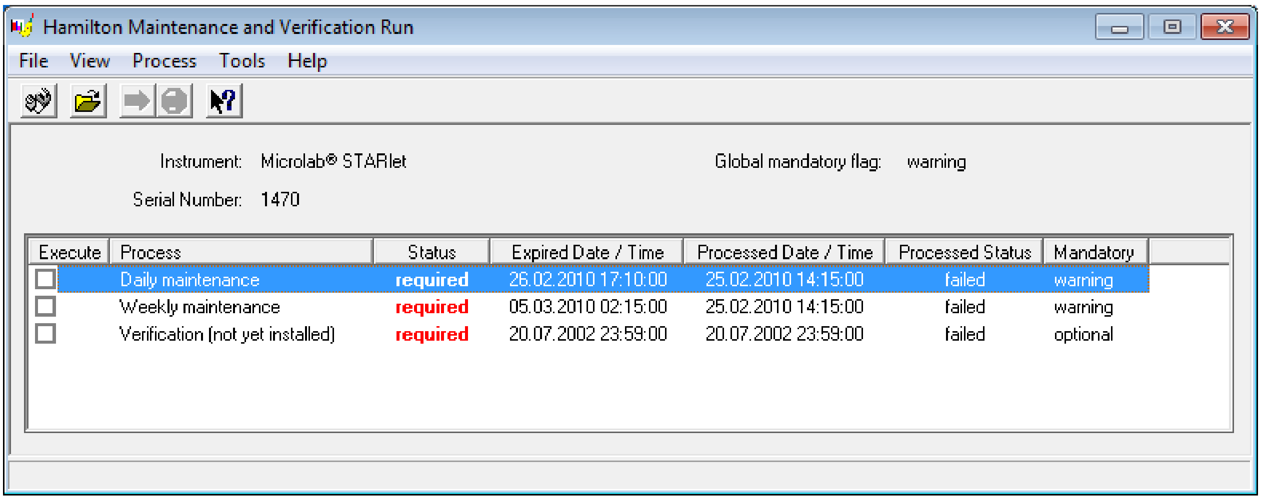
Displayed error if maintenance needed, config. setting = mandatory

To execute a maintenance procedure:

1. Double-click the following icon on the desktop:

In the main window "Maintenance and Verification Run", the process status information view lists all maintenance and verification processes for the connected/selected instrument.

Here the operator can start the necessary commands to perform all maintenance-related routines.



Main Window 'Maintenance and Verification Run'

2. Select the desired maintenance routine by clicking the specific check box and by pressing the **[Run Process]**  button. The VENUS Software will then issue on-screen instructions detailing all procedures required to perform the selected maintenance routine.

ATTENTION

Always wear disposable gloves during maintenance.

Do not clean the instrument in the vicinity of naked flames or devices which could create sparks. Do not use hot air blowers to dry the instrument. The liquids used for cleaning are flammable.

Maintenance work on iSWAP is not permitted, because adjustment may be lost. Clean iSWAP only where necessary after spillage and be careful of sharp edges on the gripper's pincers.

This manual provides indications regarding general disposal of waste. In addition, any regulations specific to the country of operation must be taken into account and observed.

Routine Completion

A maintenance routine is completed once the procedure has been fully implemented and the results are within the specifications.

Aborting Maintenance Procedures

Aborting a maintenance procedure will lead to a 'failed' status, and maintenance will need to be started again.

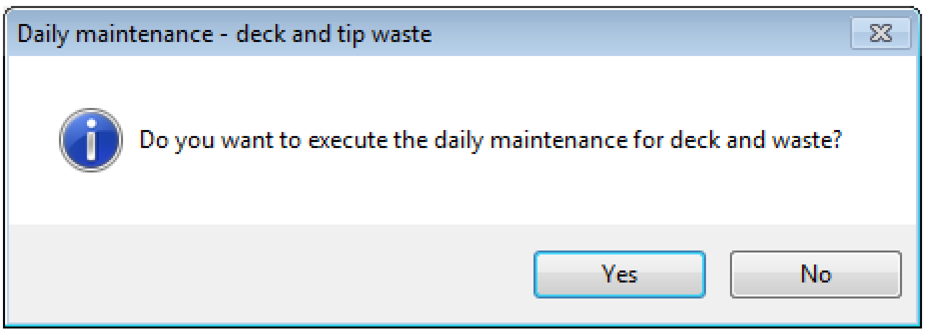
After aborting a maintenance procedure, switch off the instrument and switch it on again after 10 seconds.

4.5.1 Daily Maintenance

The following tasks belong to the daily maintenance:

- Checking if deck is clean
- Empty the tip waste
- Checking the tightness of the 5ml/1000µl pipetting channel
- Verifying the functioning of the cLLD (5ml/1000µl pipetting channel)
- Rinse procedure of the wash station(s)
- Checking if the CVS Vacuum system is working

After initialization of the instrument, the operator will be asked to execute the daily maintenance:



After clicking **[Yes]** the daily maintenance procedure will be started. Pressing **[No]** will abort the procedure.

The front cover (the hinged Plexiglas window that shields the instrument in front) can be opened for user intervention.

1. Once the maintenance procedure has been started the pipetting arm moves to the left side. The operator now has access to the deck to check if cleaning is needed or not:
    - If the deck is clean continue with the daily maintenance.
    - If the deck needs to be cleaned the daily maintenance can be interrupted. Instead of the daily maintenance carry out the weekly maintenance.
2. Continuing the daily maintenance procedure will lead the user to the next maintenance task. The tip waste needs to be emptied. Dispose of it with the rest of the laboratory's contaminated waste.

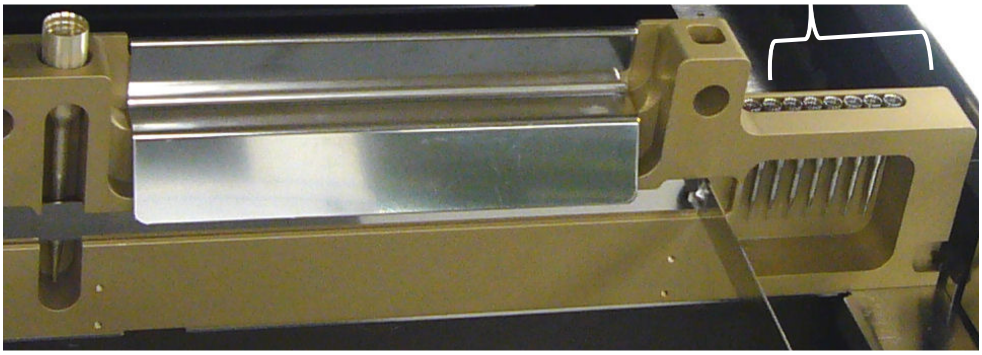
ATTENTION

The tip waste is always to be regarded as contaminated.

3. For the next steps the maintenance needles are required.

Maintenance needle for 5ml pipetting channels

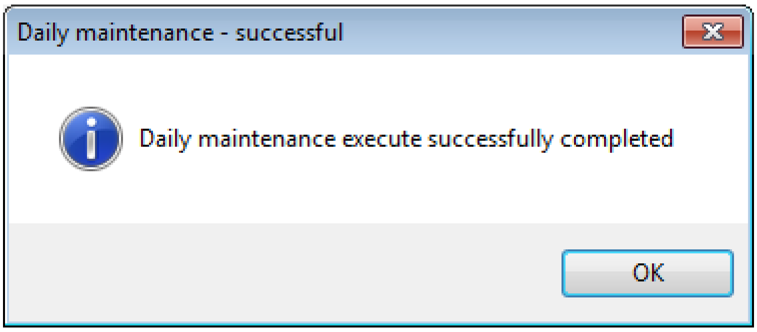
Maintenance needles for 1000µl pipetting channels



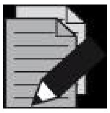
4. The procedure continues with the tightness check of the pipetting channels. The pipetting arm will travel to the right hand side to pick up the maintenance needles.

    Two checks are done with the pipetting channels, the over-pressure and the under-pressure check.
5. For the capacitive liquid level (cLLD) check the needles are picked-up again. One channel to the next is checked for the proper functioning of the cLLD.
6. If there is a CVS installed a rinse procedure is started. Refer to chapter 4.11 and the following "CVS Vacuum System.
7. If there is a needle wash station installed a rinse procedure is started. Refer to chapter 4.7 and the following "Needles".

8. The end of the daily maintenance is displayed:



The daily maintenance process status is saved on the instrument and a report file is created. Refer to chapter 4.5 "Printing a Report".



NOTE

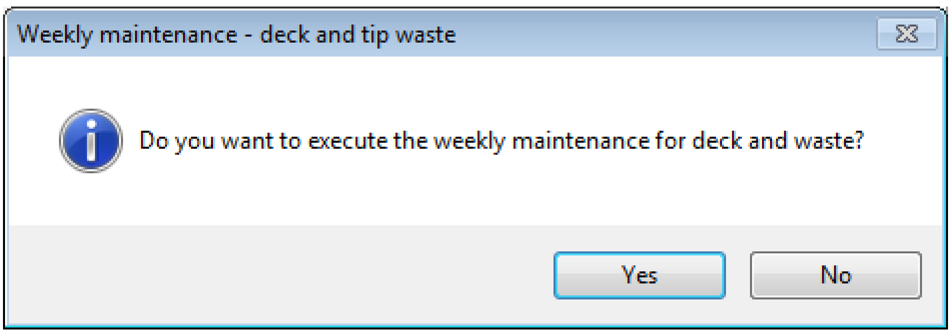
If any parts of the instrument, carriers or racks have become contaminated, the weekly maintenance procedure must be performed.

4.5.2 Weekly Maintenance

The following tasks are carried out by the weekly maintenance:

- Cleaning the deck and carriers
- Checking the condition of the carriers
- Empting and cleaning of the tip waste
- Checking the tightness of the 5ml/1000µl pipetting channel
- Verifying the functioning of the cLLD (5ml/1000µl pipetting channel)
- Maintenance procedure for the wash station(s)
- Cleaning of the pipetting head: stop disk, O-ring, tip eject sleeve
- Cleaning of the covers, autoload protecting ribbon
- Checking if the CVS Vacuum is working

After initialization of the instrument, the operator will be asked to execute the weekly maintenance:



When the instrument is initialized, the weekly maintenance program advises the user to unload the deck manually. If the Autoload option is activated, this step is carried out automatically.

1. Clean all carriers with Deconex Solarsept and leave them to dry. If they are heavily soiled, soak them afterwards in a solution of Deconex 61 DR (see the product data sheet for further information).

   Examine each carrier for scratches on the barcode and any signs of damage. If damage is apparent, replace with new carriers.

2. Continuing the weekly maintenance program will advice the autoload (if configured) to move to the right hand side of the instrument.

   Open the front cover and wipe the deck with a cloth saturated with Deconex Solarsept. The slide blocks in particular must be checked for cleanliness. Close the front cover.

   ATTENTION

   Do not spray directly at the Autoload unit or at electrical boards or connectors.

3. The next step of the maintenance procedure will advise the autoload (if configured) to move to the left hand side of the instrument. The tip waste needs to be emptied and cleaned. Dispose of it with the rest of the laboratory's contaminated waste.

   ATTENTION

   The tip waste, the tip eject plate, and the plastic bag are always to be regarded as contaminated.

   Remove the tip eject plate of the tip waste station and clean it: spray Deconex Solarsept directly onto the surface, and wipe. Remove the frame that holds the plastic bag in place, and discard the plastic in the laboratory's contaminated waste. Pull a new plastic bag over the frame and re-attach it. Put the clean tip eject plate back in place.

   ATTENTION

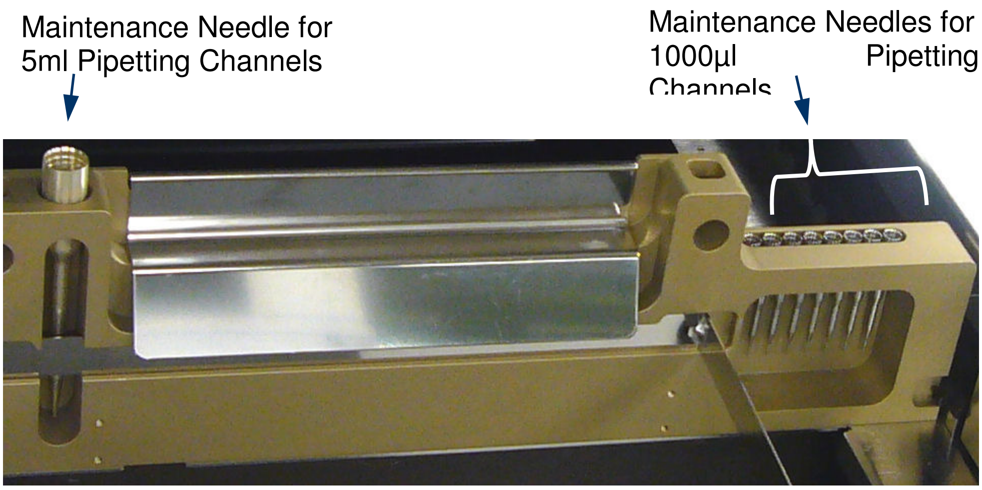
   Do not spray the maintenance needles.

4. If the Autoload option is configured: to prevent unreliable barcode reading, check the laser scanner window of the barcode reader and clean it with a lint-free cloth or Q-tips lightly soaked in Ethanol (70%).

   ATTENTION

   The laser scanner window must be completely dry and free from dust and fibers before the instrument can be reused.

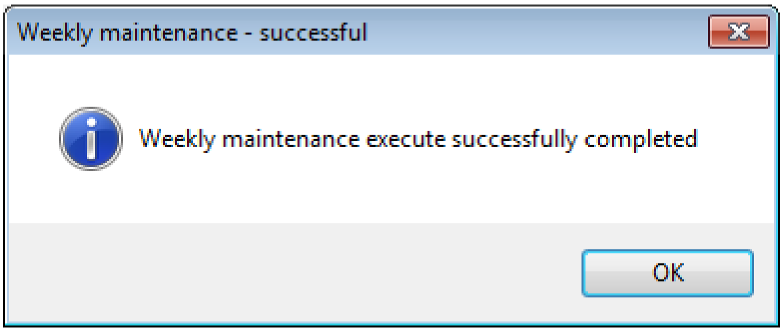
5. For the next steps the maintenance needles are required.



6. The procedure continues with the tightness check of the pipetting channels. The pipetting arm will travel to the right hand side to pick up the maintenance needles.

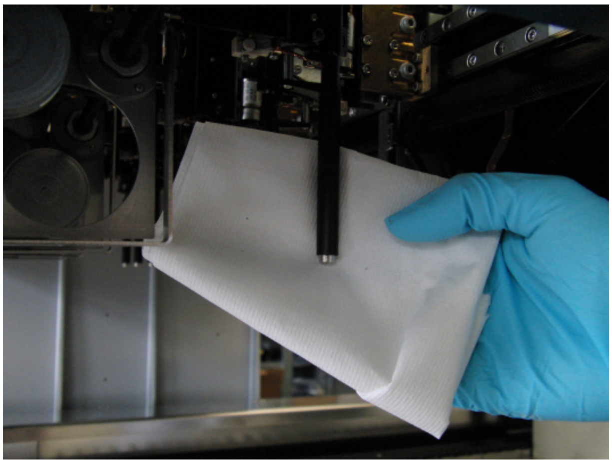
   Two checks are done with the pipetting channels, the over-pressure and the under-pressure check.

7. For the capacitive liquid level (cLLD) check the needles are picked-up again. One channel by the other is checked for the proper functioning of the cLLD.
8. If there is a CVS installed a rinse procedure is started. Refer to chapter 4.11 and the following.
9. If there is a needle wash station installed a weekly maintenance procedure is been started. Refer to chapter 4.7 "Needles".
10. The end of the weekly maintenance program is displayed:



   The weekly maintenance process status is saved on the instrument and a report file is created. Refer to chapter 4.6 "Printing a report".

11. Clean the tip eject sleeve (outer part of the pipetting channels) with a lint-free cloth soaked in DECONEX SOLARSEPT.

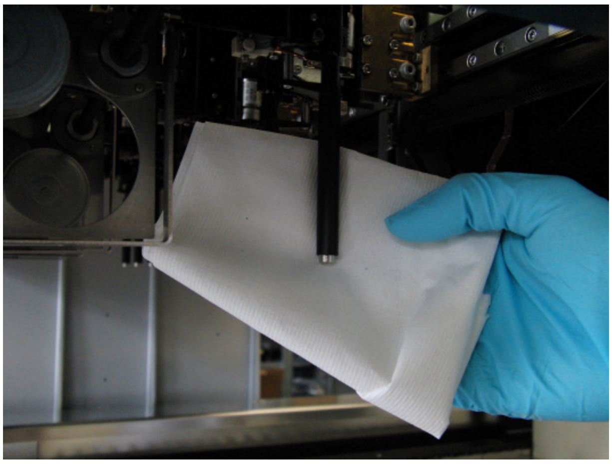


ATTENTION

Take care that no liquid gets inside the tip channel.

Whenever it is necessary to move Channels on the X-Arm, move them gently by pushing close to their Y-slide. Never force them as this may lead to damage. If possible, switch on the instrument as this will result in a smoother motion when Channels have to be moved on the X-Arm.

12. Clean the stop disk and the O-rings of the pipetting head (outer part of the pipetting channels) with a lint-free cloth soaked in DECONEX SOLARSEPT.



ATTENTION

Take care that no liquid gets inside the tip channel.

Whenever it is necessary to move Channels on the X-Arm, move them gently by pushing close to their Y-slide. Never force them as this may lead to damage. If possible, switch on the instrument as this will result in a smoother motion when Channels have to be moved on the X-Arm.

13. Spray the front and side cover with DECONEX SOLARSEPT and wipe dry.

14. Clean the autoload protecting ribbon with a cloth soaked in Deconex Solarsept, and wipe without exerting pressure.

ATTENTION

Do not spray directly at the Autoload unit or at electrical boards or connectors.

15. Clean the X-guide shaft behind the upper front cover with a dry cloth at least once a month.

NOTE

Carriers must be completely clean and dry before re-using.

4.6 Printing a Report

The maintenance process status can be printed. To print such a report:

From the "File" menu, select "Open Report". All maintenance and verification processes which are found in the default "Report Path" are listed.

- If necessary, change the report path using the browse button**[<...>]**.
- Select a report and press the **[Open]** button. The Report Viewer displays the selected report file.
- From the "File" menu, select **[Print]** to print the report file.

4.7 Needles

The recommended procedure for the maintenance of the needles - if the needles are badly soiled – is as follows:

Remove the needle(s) with the Needle Pickup tool (Needle Service Kit, see Appendix B "Ordering Information")

1. Clean the needles in an ultrasonic bath at 50°C containing Deconex 61 DR for 15 - 20 minutes
2. Rinse the needles with warm de-ionized water (50°C)
3. Put the needles back into the wash module.

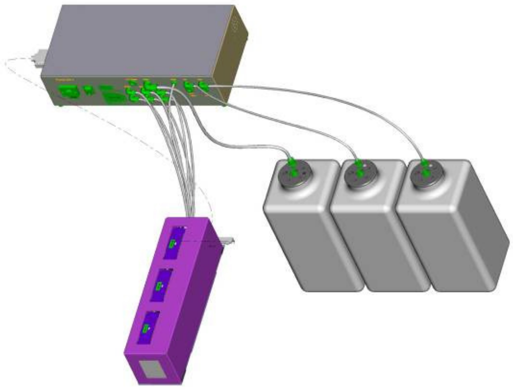
NOTE

HAMILTON recommends to replace steel needles every six month.

4.8 CR Needle Wash Station

HAMILTON recommends the following daily rinse routine and the weekly maintenance procedure to maintain the functionality of the Needle Wash Station. Excellent washing results can be achieved only with periodic maintenance.

If so configured in the VENUS Software, daily and weekly maintenance are required.

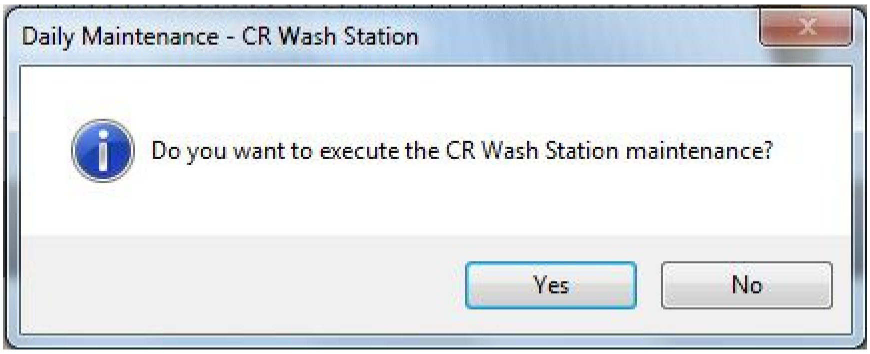


Before starting the procedure make sure the needles to be washed are placed in the Needle Wash Station.

4.8.1 Daily Maintenance

The daily maintenance procedure for the CR Needle Wash Station takes 15 minutes. The purpose of this procedure is to rinse the fluid path of the wash station. With this procedure, deposits inside the fluid path can be minimized. If the wash station is seriously soiled, the operator should start the weekly instead of the daily maintenance.

Empty any remaining liquid of the wash containers and the waste container. Partially fill the wash solution containers with de-ionized water. With the cap facing upwards, shake the containers in lengthwise direction for a few moments. Empty the containers.



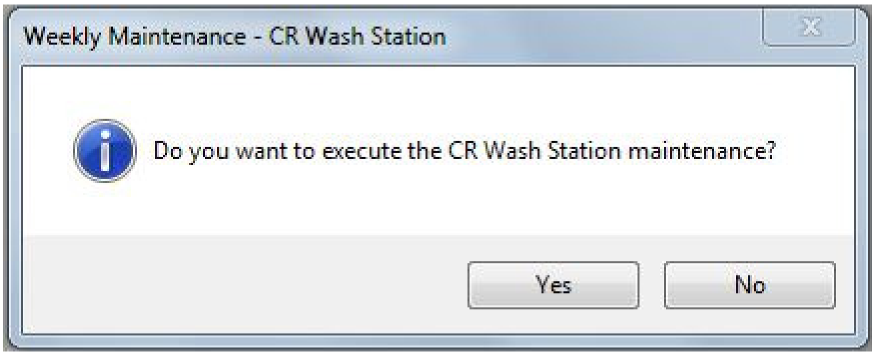
1. Once the maintenance procedure for the Needle Wash Station has been started, the software advises the operator to fill Wash Container 1 (red dot on the container lid) with 4 liters of de-ionized water and 40ml of Deconex 61 DR (p/n 281242).
2. 1x Wash step (Wash solution; Rinse time 60s, Soak time 1s, Flow rate 12ml/s, Draining time 12s).

3. The rinse procedure is interrupted by the maintenance program, to advise the operator to empty Wash Container 1 (red dot on the container lid) and to fill it up with 3 liters of de-ionized water.
4. 2x Rinse step (De-ionized water; Rinse time 60s, Soak time 1s, Flow rate 12ml/s, Draining time 12s).
5. At the end of the daily maintenance procedure, Wash Container 1 and the Waste Container need to be emptied.

4.8.2 Weekly Maintenance

The weekly maintenance procedure for the CR Needle Wash Station takes 40 minutes. The purpose of this procedure is to carry out periodical in-depth maintenance. The fluid path of the wash station and the needles are soaked in a special cleaning solution.

Empty any remaining liquid of the wash containers and the waste container. Partially fill the wash solution containers with de-ionized water. With the cap facing upwards, shake the containers in lengthwise direction for a few moments. Empty the containers.

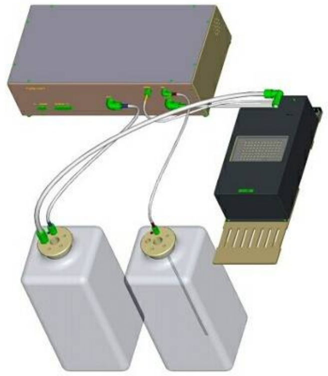


1. Once the maintenance procedure for the Needle Wash Station has been started, the software advises the operator to fill Wash Container 1 (red dot on the container lid) with 6 liters of de-ionized water and 60ml of Deconex 61 DR (p/n 281242).
2. 1x Wash step (Wash solution; Rinse time 60s, Soak time 300s, Flow rate 12ml/s, Draining time 12s).
3. 1x Wash step (Wash solution; Rinse time 60s, Soak time 1s, Flow rate 12ml/s, Draining time 12s).
4. The procedure is interrupted by the maintenance program to advise the operator to empty Wash Container 1 (red dot on the container lid) and to fill it up with 6 liters of de-ionized water. Partially fill the wash solution container with de-ionized water. With the cap facing upwards, shake the container in lengthwise direction for a few moments. Empty the container.
5. Fill up Wash Container 1 with 3 liters of de-ionized water.
6. 2x Rinse step (De-ionized water; Rinse time 60s, Soak time 1s, Flow rate 12ml/s, Draining time 12s).
7. At the end of the weekly maintenance procedure, Wash Container 1 and the Waste Container need to be emptied and let to dry.

4.9 Wash Station 96

HAMILTON recommends the following daily rinse routine and the weekly maintenance procedure to maintain the functionality of the Wash Station 96. Excellent washing results can be achieved only with periodic maintenance.

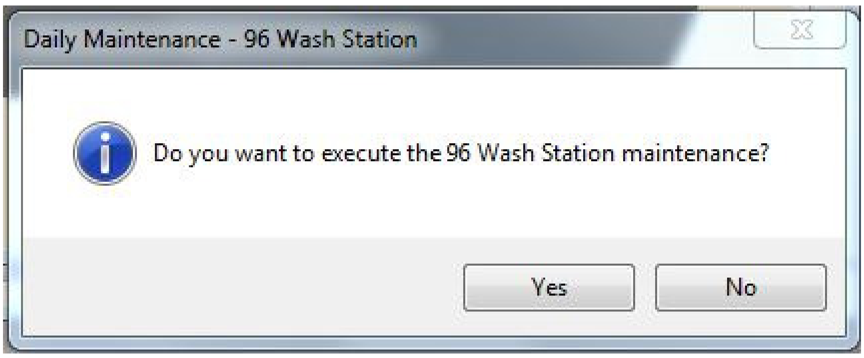
If so configured in the VENUS Software, daily and the weekly maintenance are required.



4.9.1 Daily Maintenance

The daily maintenance procedure for the Wash Station 96 takes 15 minutes. The purpose of this procedure is to rinse the fluid path of the wash station. With this procedure, deposits inside the fluid path can be minimized. If the wash station is seriously soiled, the operator should start the weekly instead of the daily maintenance.

Empty any remaining liquid of the wash container and the waste container. Partially fill the wash solution container with de-ionized water. With the cap facing upwards, shake the container in lengthwise direction for a few moments. Empty the container.



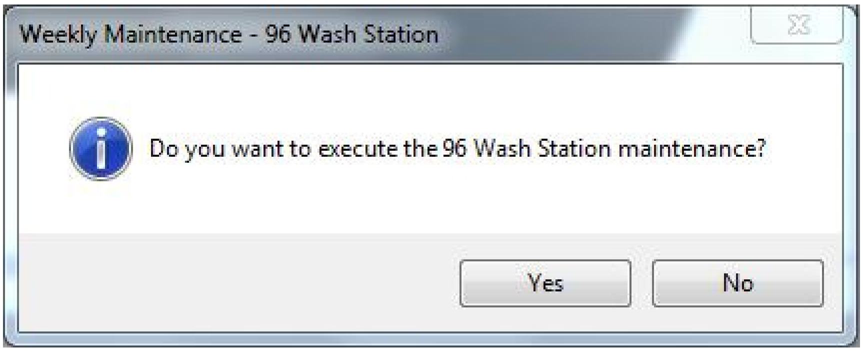
1. Once the maintenance procedure for the Wash Station 96 has been started, the software advises the operator to fill the Wash Container with 3 liters of de-ionized water and 30ml of Deconex 61 DR (p/n 281242).
2. Empty the waste container.
3. 1x Priming (Empty washer, refill after empty)
4. 1 x Rinse step (Empty washer, refill after empty,)
5. 1 x Empty washer

6. Empty the wash container. Partially fill the wash container with de-ionized water. With the cap facing upwards, shake the container in lengthwise direction for a few moments. Empty the wash containers. Fill the wash container with 3 liters of de-ionized water again.
7. 2 x Rinse step (Empty washer, refill after empty)
8. 1 x Empty washer
9. Empty the wash container and waste container
10. Empty any remaining wash solution. Partially fill the containers with de-ionized water. With the cap facing upwards, shake the containers in lengthwise direction for a few moments. Empty the containers and leave to dry.

4.9.2 Weekly Maintenance

The weekly maintenance procedure for the Wash Station 96 takes 20 minutes. To guarantee excellent wash efficiency, carry out the following steps at least once a week.

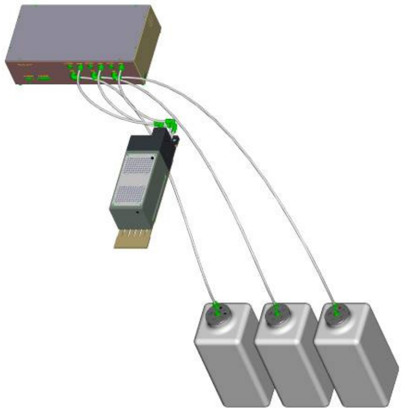
Empty any remaining liquid of the wash container and the waste container. Partially fill the wash solution container with de-ionized water. With the cap facing upwards, shake the container in lengthwise direction for a few moments. Empty the container.



1. Once the maintenance procedure for the Wash Station 96 has been started, the software advises the operator to fill the Wash Container with 3 liters of de-ionized water and 30ml of Deconex 61 DR (p/n 281242).
2. Empty the waste container.
3. 1x Priming (Empty washer, refill after empty)
4. 1 x Rinse step (Empty washer, refill after empty)
5. Soaking: Timer of 5 Minutes
6. 1 x Rinse step (Empty washer, refill after empty)
7. 1x Empty washer
8. Empty the wash container. Partially fill the wash container with de-ionized water. With the cap facing upwards, shake the container in lengthwise direction for a few moments. Empty the wash containers. Fill the wash container with 3 liters of de-ionized water again.
9. 2 x Rinse step (Empty washer, refill after empty)
10. 1 x Empty washer
11. Empty any remaining wash solution of the wash and the waste container. Partially fill the containers with de-ionized water. With the cap facing upwards, shake the containers in lengthwise direction for a few moments. Empty the containers and leave to dry.

4.10 Wash Station 96 or 384

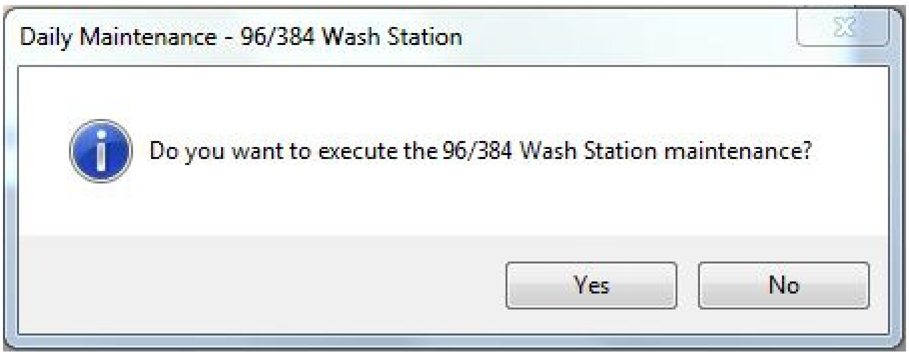
HAMILTON recommends the following daily rinse routine and the weekly maintenance procedure to maintain the functionality of the Wash Station 96/384. Excellent washing results can be achieved only with periodic maintenance.



4.10.1 Daily Maintenance

The daily maintenance procedure for the Wash Station 96/384 takes 10 minutes. The purpose of this procedure is to rinse the fluid path of the wash station. With this procedure, deposits inside the fluid path can be minimized. If the wash station is seriously soiled, the operator should carry out the weekly instead of the daily maintenance.

Empty any remaining liquid of the wash container and the waste container. Partially fill the wash solution containers with de-ionized water. With the cap facing upwards, shake the container in lengthwise direction for a few moments. Empty the containers.



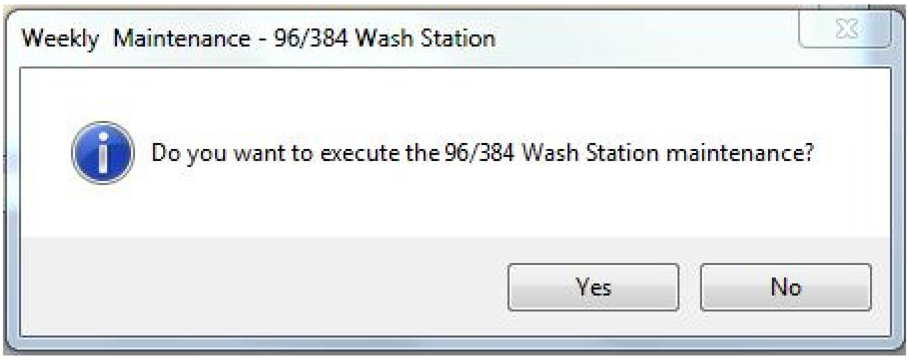
1. Fill the Wash Containers with 2 liters of de-ionized water and 20ml of Deconex 61 DR (p/n 281242).
2. Empty the waste container.
3. 1x Priming (Initialize washer, empty)
4. 2 x Rinse step for both wash chambers (Fill washer, empty)
5. Empty the wash containers. Partially fill the wash containers with de-ionized water. With the cap facing upwards, shake the containers in lengthwise direction for a few moments. Empty the wash containers. Fill the wash containers with 3 liters of de-ionized water again.

6. 2 x Rinse step for both wash chambers (Fill washer, empty)
7. Empty the wash containers and waste container
8. Empty any remaining wash solution. Partially fill the containers with de-ionized water. With the cap facing upwards, shake the containers in lengthwise direction for a few moments. Empty the containers and leave to dry.

4.10.2 Weekly Maintenance

The weekly maintenance procedure for the Wash Station 96/384 takes 20 minutes. To guarantee excellent wash efficiency, carry out the following steps at least once a week.

Empty any remaining liquid of the wash container and the waste container. Partially fill the wash solution containers with de-ionized water. With the cap facing upwards, shake the container in lengthwise direction for a few moments. Empty the containers.



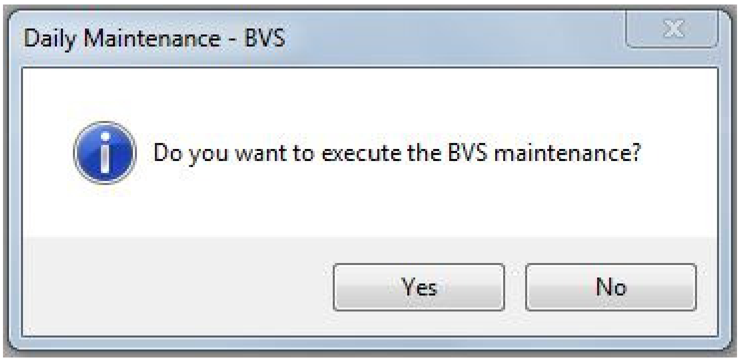
1. Fill the Wash Containers with 2 liters of de-ionized water and 20ml of Deconex 61 DR (p/n 281242).
2. Empty the waste container.
3. 1x Priming (Initialize washer, empty)
4. 1 x Rinse step for both wash chambers (Fill washer)
5. Soaking: Timer of 5 Minutes
6. 1 x Rinse step for both wash chambers (Empty washer, refill after empty)
7. 1x Empty washer
8. Empty the wash containers. Partially fill the wash containers with de-ionized water. With the cap facing upwards, shake the containers in lengthwise direction for a few moments. Empty the wash containers. Fill the wash containers with 3 liters of de-ionized water again.
9. 2 x Rinse step for both wash chambers (Fill washer, empty)
10. 1 x Empty washer
11. Empty any remaining wash solution of the wash and the waste containers. Partially fill the containers with de-ionized water. With the cap facing upwards, shake the containers in lengthwise direction for a few moments. Empty the containers and leave to dry.

4.11 CVS Vacuum System

Hamilton recommends maintaining the CVS Crystal Vacuum System following the daily and weekly maintenance procedures described below. Only with a periodical maintenance a long lifespan of the CVS Vacuum System can be achieved.

4.11.1 Daily Maintenance

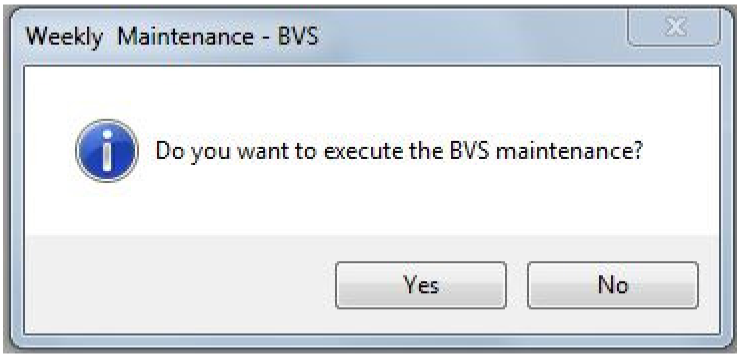
The daily maintenance procedure for the CVS Vacuum System takes 10 minutes. The purpose of this procedure is to rinse the tubing and the vacuum chamber. With this procedure, deposits inside the fluid path can be minimized. If the CVS is seriously soiled, the operator should start the weekly instead of the daily maintenance.



1. Fill 0.5 liters of de-ionized water with 5ml of Deconex 61 DR (p/n 281242) into the vacuum chamber of the CVS carrier. Use a soaking time of 1 Minute.
2. Evacuate the vacuum chamber
3. Fill 0.5 liters of de-ionized water into the vacuum chamber of the CVS carrier.
4. Evacuate the vacuum chamber
5. Empty the waste bottle. Partially fill the waste bottle with de-ionized water. With the cap facing upwards, shake the container in lengthwise direction for a few moments. Empty the waste bottle.

4.11.2 Weekly Maintenance

The weekly maintenance procedure for the CVS Vacuum System takes 15 minutes time. The purpose of this procedure is to rinse the tubing and the vacuum chamber. With this procedure, deposits inside the fluid path can be minimized.



1. Fill 0.5 liters of de-ionized water with 5ml of Deconex 61 DR (p/n 281242) into the vacuum chamber of the BVS carrier. Use a soaking time of 5 Minute.
2. Evacuate the vacuum chamber
3. Fill 0.5 liters of de-ionized water into the vacuum chamber of the CVS carrier.
4. Evacuate the vacuum chamber
5. Empty the waste bottle. Partially fill the waste bottle with de-ionized water. With the cap facing upwards, shake the container in lengthwise direction for a few moments. Empty the waste bottle.
6. Clean the Manifold Top with a lint-free cloth soaked in Deconex Solarsept. Rinse afterwards with de-ionized water and leave to dry before using the Manifold Top again.

ATTENTION

Do not soak the Manifold Top.

4.12 Temperature-Controlled Carrier (TCC)

Here is the recommended procedure for the daily/weekly maintenance of the TCC:

Clean all surfaces with a cloth soaked with DECONEX SOLARSEPT (see Appendix B "Ordering Information").

5 ML STAR instrument Decontamination

Here is the recommended procedure for decontaminating your ML STAR instrument:

Spray the front and side cover with DECONEX SOLARSEPT (see Appendix B "Ordering Information").

Open the front cover and spray the deck with DECONEX SOLARSEPT (see Appendix B „Ordering Information").

Remove the tip eject plate of the tip waste station and clean it.

Spray DECONEX SOLARSEPT directly onto the surface of the tip waste station.

Remove the frame that holds the plastic bag in place, and discard the plastic bag in the laboratory's contaminated waste. Put the tip eject plate back in place.

Clean the tip eject sleeve (outer part of the pipetting channels) with a lint-free cloth soaked in Deconex SOLARSEPT (see Appendix B "Ordering Information").

Clean all carriers with Deconex Solarsept (see Appendix B "Ordering Information ") and leave them to dry. If they are heavily soiled, soak them afterwards in a solution of DECONEX 61 DR (see the product data sheet for further information).

Perform needle maintenance (see section 4.7 "Needles").

Perform washer maintenance (see section 4 "Maintenance" and the following sections).

6 Verification

6.1 General Description

To perform the Verification of the ML STAR instrument "Field Verification 2" is used. The Field Verification 2 can only be performed by trained personnel.

For the 1000µl Pipetting Channels, the CO-RE 96-Probe Head and the CO-RE 384-Probe Head, a dye-pipetting procedure followed by gravimetric and photometric analysis is used to verify the Trueness and Precision. The 5ml Pipetting Channels are verified by a gravimetric approach.

Devices such as Heater Shaker, Cooler, Barcode Reader and Cover Savety can also be verified with the "Field Verification 2".

Note

This is an additional feature of the ML STAR Line, which is not included in the VENUS two Software.

Further information you will find in <u>Appendix B "Ordering Information" of this Manual or ask your local HAMILTON dealer.</u>

7 Technical Specifications

7.1 Basic ML STAR Line

Instrument Dimensions				
	STAR^{LET}	width: 1124 mm, height: 903 mm, depth: 795 mm (autoload: 1006 mm)		
	STAR	width: 1664 mm (1990 with 96/ 384-Probe Head), height: 903 mm, depth: 795 mm (autoload: 1006 mm)		
	STAR^{PLUS}	width: 2160 mm, height: 903 mm, depth: 795 mm (autoload: 1006 mm)		
Work Area Dimensions		Width (X):	Height (Z):	Depth (Y):
	STAR^{LET}	675 mm	195 mm	465 mm
	STAR	1215 mm	195 mm	465 mm
	STAR^{PLUS}	1705 mm	195 mm	465 mm
Weight		8 channels	96/ 384-Probe Head and 8 individual channels	
	STAR^{LET}	135 kg	150 kg	
	STAR	145 kg	160 kg	
	STAR^{PLUS}	205 kg	220 kg	
Deck Capacity				
	STAR^{LET}	30 tracks (T) allow combinations of: maximum of 30 tube carriers (1 T) holding 24 or 32 tubes per carrier maximum of 5 carriers (6 T) holding 5 tip racks or 5 plate positions per carrier		
	STAR	54 tracks (T) allow combinations of: maximum of 9 carriers (6 T) holding 5 tip racks or 5 plate positions per carrier		
	STAR^{PLUS}	82 tracks (T) allow combinations of: maximum of 11 carriers (6 T) holding 5 tip racks or 5 plate positions per carrier plus 16 T for the waste container and on-deck components. The 96-Probe Head or the 384-Probe Head can reach up to 9 carriers on the Deck and 65mm beyond the Deck (on the left side).		
Positional Accuracy		X-Y-Z positional accuracy of 0.1mm		
Tip Sizes		low volume: 10µl, intermediate volume 50µl, standard volume: 300µl, high volume: 1000µl		
Needle Sizes		low volume: 10µl, standard volume: 300µl, high volume: 1000µl, needles available only for 1000µl single channels		

Pipetting Specifications for Disposable Tips*	Disposable Tip Size	Volume	Trueness |R| (%)	Precision CV (%)
individual 1000µl pipetting channels	10µl	0.5µl	10.0%	6.0%
	10µl	1µl	5.0%	4.0%
	10µl	5µl	2.5%	1.5%
	10µl	10µl	1.5%	1.0%
	50µl	0.5µl	10.0%	6.0%
	50µl	1µl	5.0%	4.0%
	50µl	5µl	2.5%	1.5%
	50µl	50µl	2.0%	0.75%
	300µl	10µl	5.0%	2.0%
	300µl	50µl	2.0%	0.75%
	300µl	200µl	1.0%	0.75%
	1000µl	10µl	7.5%	3.5%
***Test criteria available upon request**	1000µl	100µl	2.0%	0.75%
	1000µl	1000µl	1.0%	0.75%
	For pipetting of less than 10µl HAMILTON recommends 10µl/ 50µl volume disposable tips to achieve highest pipetting precision.			
Pipetting Specifications for Needles*	**Needle Size**	**Volume**	**Trueness |R| (%)**	**Precision CV (%)**
individual 1000µl pipetting channels	10µl	1µl	5.0%	8.0%
	10µl	5µl	2.5%	2.0%
	10µl	10µl	1.5%	1.0%
	300µl	5µl	8.0%	8.0%
	300µl	50µl	2.0%	2.0%
	300µl	200µl	1.0%	1.0%
	1000µl	50µl	5.0%	3.0%
***Test criteria available upon request**	1000µl	100µl	3.0%	2.0%
	1000µl	1000µl	2.0%	1.0%
Pipetting Specifications for 5ml Disposable Tips*	**Disposable Tip Size**	**Volume**	**Trueness |R| (%)**	**Precision CV (%)**
individual 5ml pipetting channels	5ml	50µl	5.0%	2.5%
	5ml	500µl	2.0%	1.5%
***Test criteria available upon request**	5ml	1000µl	1.5%	1.0%
	5ml	5000µl	1.0%	0.5%

Pipetting Specifications for Disposable Tips*		Disposable Tip Size	Volume	Trueness |R| (%)	Precision CV (%)
1000µl CORE 96-Probe Head	Maximum pipetting volume: 1000µl	10µl	1µl	5.0%	5.0%
		10µl	5µl	2.5%	2.0%
		10µl	10µl	1.5%	1.5%
		50µl	1µl	5.0%	5.0%
		50µl	5µl	2.5%	2.0%
		50µl	50µl	1.5%	1.0%
		300µl	10µl	3.0%	2.0%
		300µl	50µl	1.5%	1.0%
		300µl	300µl	1.0%	1.0%
		1000µl	10µl	7.5%	3.5%
***Test criteria available upon request**		1000µl	100µl	2.0%	1.0%
		1000µl	1000µl	1.0%	1.0%
	For pipetting of less than 10µl HAMILTON recommends 10µl/ 50µl volume disposable tips to achieve highest pipetting precision.				

Pipetting Specifications for Disposable Tips*		Disposable Tip Size	Mode:	Volume	Precision CV (%)
50µl CORE 384- Probe Head	Maximum pipetting volume: 50µl	50µl	Surface:	0.1µl	8.0%
		50µl	Surface:	0.5µl	6.0%
		50µl	Surface:	1µl	3.5%
		50µl	Jet:	1µl	15.0%
		50µl	Surface:	5µl	3.0%
		50µl	Jet:	5µl	4.0%
		50µl	Surface:	10µl	2.0%
***Test criteria available upon request**		50µl	Jet:	10µl	3.0%
		50µl	Surface/Jet:	50µl	2.0%
Using the 50µl CORE 384-Probe Head as a 96-Probe Head	Maximum pipetting volume: 300µl	300µl Rocket		1µl	4.0%
		300µl Rocket		5µl	2.0%
		300µl Rocket		10µl	2.0%
		300µl Rocket		100µl	2.0%
		300µl Rocket		300µl	2.0%

Liquid Level Detection	Individual Channels:	Capacitive liquid level detection (cLLD) and pressure (pLLD) on aspiration, cLLD on dispense, minimum volume 10µl, depending on container type
	96-Probe Head:	Capacitive liquid level detection (cLLD)
	384-Probe Head:	Capacitive liquid level detection (cLLD)
Throughput	8 Channels:	To fill one 96-well microtiter plate with 100µl samples (new tip for each sample): 320s Aliquot reagent to a 96-well microtiter plate (<90µl per well): 60s
	96-Probe Head:	Replication of one 96-well plate, 100µl, with cLLD on aspiration: 35s (incl. new tips)
		Reformatting of four 96-well plates to one 384-well plate, 50µl, new tips, with cLLD on aspiration: 140s
	384-Probe Head:	Replication of one 384-well plate, 20µl, with cLLD on aspiration: 35s (incl. new tips)
		Reformatting of four 384-well plates to one 1536-well plate, 10µl, new tips, with cLLD on aspiration: 140s
Labware	all SBS standard plate types up to 1536 wells and most commercially available tube types	
Carriers	for all standard labware formats and according to customer requirements. The Nano-Option always comes with a Waste and Ultrasonic bath Carrier	

Operating Data	maximum power consumption	600 VA or 1000 VA (depending on configuration)
	voltage	115 VAC/ 230 VAC
	frequency	50 / 60 Hz ± 5%
	delayed action fuse 600 VA: 1000 VA:	 115V~: 6.3A (T6.3AL250) 230V~: 3.15A (T3.15AL250) 115V~: 10A (T10AL250) 230V~: 5A (T5AL250)
	Installation category	II
	Pollution Degree	2
	Temperature range	15°C – 35°C
	Relative humidity	30% – 85% (not condensating, indoors)
	Noise level	< 65 dBA (regarding EN27779)
	Altitude	max. 2000m above sea level
	Heat: the power consumed will be transferred to head	e.g. 600 or 1000 Watts of Heat = 600 or 1000 Joules/ Second
	Indoor use only	
Storage and Transportation	Temperature range Relative humidity	-25°C – +70°C 10% – 90% (not condensating, indoors)
Recommended PC	Pentium IV, ≥ 4 GB RAM, 250 GB Hard Drive, 16x DVD +/-RW , DirectX 250 MB graphic card, MICROSOFT Windows 7 Professional oder Windows XP Professional with Service Pack 3. (not included in shipment)	
Communication	USB, RS232	

7.1.1 CR Needle Wash Station

Specification	Needle capacity	24 per Wash Station
		8 per Chamber
	Wash time (typ.)	45 sec/cycle
	Liquid consumption (typ.)	100 ml/cycle
	Carry over (typ.)	1000µl Needle, 100µl: < $3*10^{-6}$
		300µl Needle, 100µl: < $2*10^{-6}$
		10µl Needle, 10µl: < $5*10^{-6}$
	Deck capacity	6 tracks
	Max. units	2 per Instrument
Operating Data	Maximum power consumption	140VA
	Nominal Voltage	100 VAC (-15%)
		240 VAC (+ 10%)
	Nominal Frequency	50 / 60Hz ± 5%
	Delayed action fuse	115V~: 3.15A (T3.15L250) 230V~: 1.6A (T1.6AL250)
	Installation category	II
	Pollution Degree	2
	Temperature range	15°C – 35°C
	Relative humidity	30% – 85% (not condensating, indoors)
	Altitude	max. 2000m above sea level
	Indoor use only	
Storage and Transportation	Temperature range	-25°C – +70°C
	Relative humidity	10% – 90% (not condensating, indoors)

7.1.2 96 Wash Station

Specification	Wash time (typ.)	94 sec/ 2 cycles
		164 sec/ 3 cycles
	Liquid consumption (typ.)	600 ml/ 2 cycles
		900 ml/ 3 cycles
	Carry over (typ.)	
	2 cycles	300µl Tips, 300µl: < $2.3*10^{-5}$
	3 cycles	300µl Tips, 300µl: < $2*10^{-6}$
	Deck capacity	8 tracks
	Max. units	3 per Instrument
Operating Data	Power consumption	41 V / 100VA (max.)
		supplied by the MICROLAB® STAR Line
	Temperature range	15°C – 35°C
	Relative humidity	30% – 85% (not condensating, indoors)
	Altitude	max. 2000m above sea level
	Indoor use only	
Storage and Transportation	Temperature range	-25°C – +70°C
	Relative humidity	10% – 90% (not condensating, indoors)

7.1.3 96 or 384 Wash Station Dual

Specification	Wash time (typ.)	67 sec/ 2 cycles (96x10 tip)
		71 sec/ 2 cycles (96x300 tip)
		73 sec/ 2 cycles (384x30 tip)
	Liquid consumption (typ.)	880 ml/ 2 cycles
	Carry over (typ.)	
	2 cycles	10µl Tips, 10µl: < $3*10^{-6}$
		300µl Tips, 300µl: < $5*10^{-6}$
		30 µl Tips: < $1.8*10^{-5}$
	3 cycles	30 µl Tips: < $1*10^{-6}$
	Deck capacity	6 tracks
	Max. units	3 per Instrument
Operating Data	Power consumption	41 V / 100VA (max.)
		supplied by the ML STAR Line
	Temperature range	15℃ – 35℃
	Relative humidity	30% – 85% (not condensating, indoors)
	Altitude	max. 2000m above sea level
	Indoor use only	
Storage and Transportation	Temperature range	-25℃ – +70℃
	Relative humidity	10% – 90% (not condensating, indoors)

7.1.4 iSWAP Specifications

Plate format	microtiter footprint plate height ≤ 43 mm		
Absolute positioning	accuracy	X,Y,Z=0.5mm	
	reproducibility	X,Y,Z=0.25	
Movement range (on a STAR 8/iSWAP instrument)	**Min. absolute Position**	**Max. absolute Position**	**remarks**
x	-206mm	+1578mm	at x_{min} 58mm space between plate and deck at x_{max} half plate on deck
y	-185mm	+605mm	
z	+100mm	+282mm	: iSWAP 182600 Rev. 00 – 02
	+0mm	+282mm	: iSWAP 182600 Rev. 03 and iSWAP Landscape 190220
Gripper opening	72mm	108mm	: iSWAP 182600 Rev. 00 – 03
	72mm	132mm	: iSWAP Landscape 190220
Gripping force		5 N – 16 N (default 9 N)	: iSWAP 182600 Rev. 00 – 03
		5 N – 16 N (default 9 N)	: iSWAP Landscape 190220
Transport mass		300g	filled deep-well plate
No restriction of random access range for 4-,8-,12-, and 16-channel ML STAR Line's.			
Operating Data	Temperature range		15℃ – 35℃
	Relative humidity		30% – 85% (not condensating, indoors)
	Altitude		2000m above sea level

7.1.5 Tube Gripper Specifications

<table>
<tr><td>Tube sizes</td><td></td><td colspan="3">Tube diameter from 8 mm to 20 mm
Tube height ≤ 120 mm</td></tr>
<tr><td>Positional Accuracy</td><td></td><td colspan="3">X-Y-Z positional accuracy of 0.1mm
(measured on tube gripper)</td></tr>
<tr><td rowspan="5">Movement range</td><td>x</td><td colspan="3">Reaches the same X-coordinate as the single Channels (e.g. on a STAR 8x 1000µl Pipetting Channel instrument)</td></tr>
<tr><td>y</td><td colspan="3">Reaches all tube positions of the sample carriers: SMP-CAR-32 and SMP-CAR-24 (on a ML STAR instrument with 6x 1000µl Pipetting Channel)</td></tr>
<tr><td rowspan="2">z</td><td>Min. absolute Position</td><td>Max. absolute Position</td><td>remarks</td></tr>
<tr><td>+54.2mm</td><td>+254.2mm</td><td>Measured from Deck to the gripping point on a tube</td></tr>
<tr><td></td><td colspan="3"></td></tr>
<tr><td rowspan="2">Gripper opening</td><td></td><td>Min. absolute Position</td><td>Max. absolute Position</td><td></td></tr>
<tr><td></td><td>5.5mm</td><td>22mm</td><td></td></tr>
<tr><td>Transport mass</td><td></td><td colspan="3">200g</td></tr>
<tr><td rowspan="3">Operating Data</td><td></td><td>Temperature range</td><td colspan="2">15°C – 35°C</td></tr>
<tr><td></td><td>Relative humidity</td><td colspan="2">30% – 85% (not condensating, indoors)</td></tr>
<tr><td></td><td>Altitude</td><td colspan="2">2000m above sea level</td></tr>
</table>

7.1.6 CO-RE Grip 1000µl Specifications

Labware format		microtiter footprint plate height ≤ 43 mm
Absolute positioning		accuracy X,Y,Z=0.5mm reproducibility X,Y,Z=0.25
Movement range		
	x	Track 1 – n (depending on instrument type)
	y	Depending on # of channels and used front channel
	z	Lowest position = 15mm over metal deck sheet
Gripping force		5 N – 16 N (default 9 N)
Transport mass		300g filled deep-well plate

7.1.7 CO-RE Grip 5ml Specifications

Labware format		microtiter footprint plate height ≤ 43 mm
Absolute positioning		accuracy X,Y,Z=0.5mm reproducibility X,Y,Z=0.25
Movement range		
	x	Track 1 – n (depending on instrument type)
	y	Depending on # of channels and used front channel
	z	Lowest position = 46mm over metal deck sheet
Gripping force		5 N – 16 N (default 9 N)
Transport mass		300g filled deep-well plate

7.1.8 Autoload Option: Barcodes and Reader Specifications

Carriers, containers, racks and tip racks can be identified by a barcode, which a reader, mounted on the Autoload slide, scans. The system must allow specification of ranges (barcode mask) for plausibility checking of barcode information.

7.1.8.1 Barcode Symbologies

The following barcode symbologies can be detected by the system:

- ISBT standard
- Code 128 (subset B and C)
- Code 39
- Codabar
- Code 2 of 5 Interleaved
- UPC A

For the highest reading safety HAMILTON recommends:

1. To use barcode type Code128 (subset B and C)
2. Disabling the unused barcode types in the configuration editor of the User Software (refer to the programmer's manual)
3. Defining a barcode mask via the labware editor of the User Software (refer to the programmer's manual)

7.1.8.2 Reading Accuracy

The rate of inaccurate readings of sample plates and container bar codes is less than 1 ppm.

The above-mentioned specification is valid under the following conditions:

- Bar code symbology module: ISBT standard
- Code density: 0.0065" (0.1651 mm)
- Print Quality see chapter 7.1.9.3 "Barcode Specifications"
- Recognized errors are defined as an accurate reading.

7.1.8.3 Barcode Specifications

Length of string	Maximum 20 characters excluding start, stop and check characters, depending on the code length (see label dimensions).	
Code Density, Tolerance	Minimum module width (X dimension) including a print tolerance: ≥ 0.0065" (0.1651 mm) Maximum module width (X dimension) including a print tolerance: ≤ 0.02" (0.508 mm) Best reading performance with X dimension ≥ 0.01" (0.254 mm)	
Check character	ISBT standard	One character
	Code 128	One character
	Code 39	None
	Codabar	None
	Code 2 of 5 Interleaved	None
	UPC A	One character
Quiet Zone	≥ 10 times the X dimension, but at least 3 mm.	
Print quality	The barcode print must be of a high quality. A printed barcode with an ANSI/ CEN/ ISO grade A or B is required. Offset, typographic, intaglio and flexographic printing are suitable. Mechanical dot matrix and thermo matrix printing are <u>not</u> suitable. The surface may be treated, sealed or plastic-coated.	

7.1.8.4 Sample Barcodes

Barcode specifications

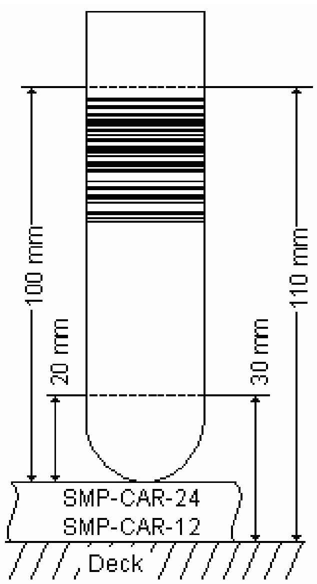
For general barcode specification see section 7.1.8.3

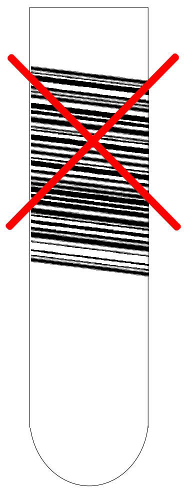
	Dimension	Min.	Max.
A	Label length	-	80 mm
B	Code length	-	74 mm
C	Quiet zone	3 mm	
D	Label width	12 mm	-
E	Code width	12 mm	-
F	Distance from code to label edge	-	1 mm

Positioning Barcode Labels

The label must be glued within a range of between 20mm to 100mm from the bottom of the tube.

The label must fit tightly at an angle of approximately 90° to the tube.





The label must fit tightly over its whole length.

7.1.8.5 Reagent Barcodes

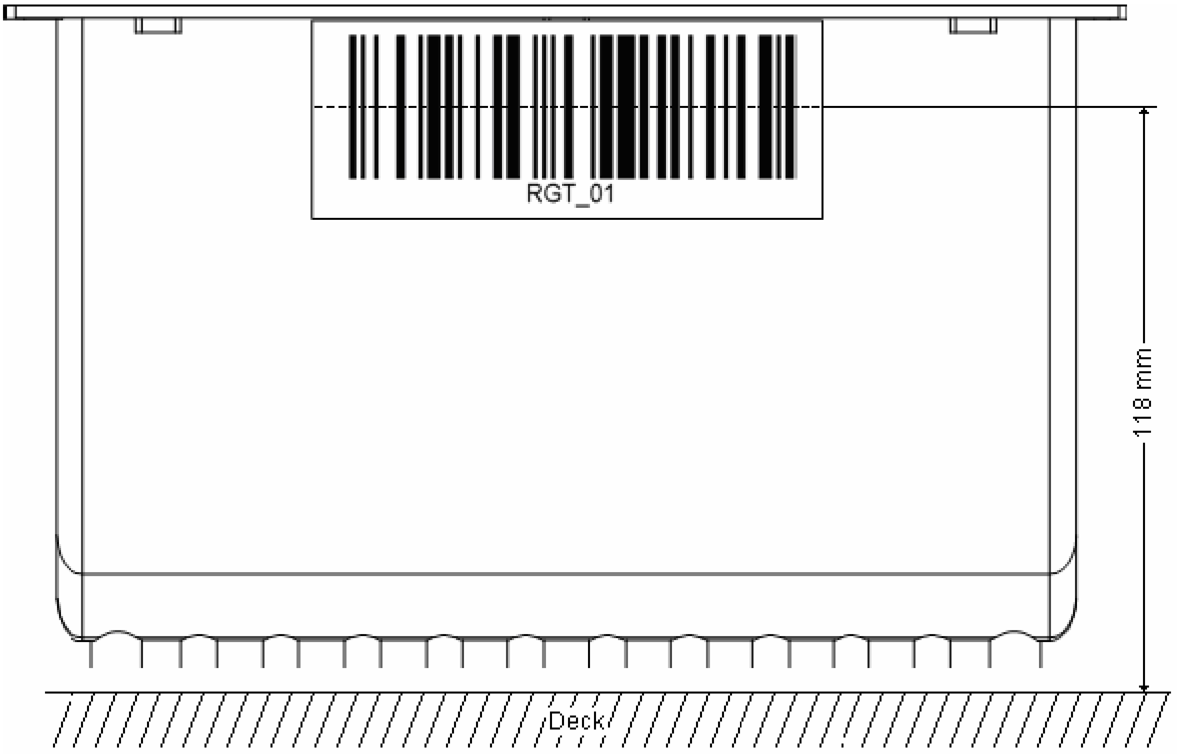
Proposed Barcode Mask:	
RGT_mm	**RGT**: Reagent
	_: Separator (underline)
	mm: Reagent number 1.....99

Label Specification

For general barcode specification see section 7.1.8.3

	Dimension	**Min**	**Max**
A	Label length	-	66 mm
B	Code length	-	60 mm
C	Quiet zone	3 mm	-
D	Label width	15 mm	-
E	Code width	12 mm	-
F	Distance from code to edge of label	-	1 mm

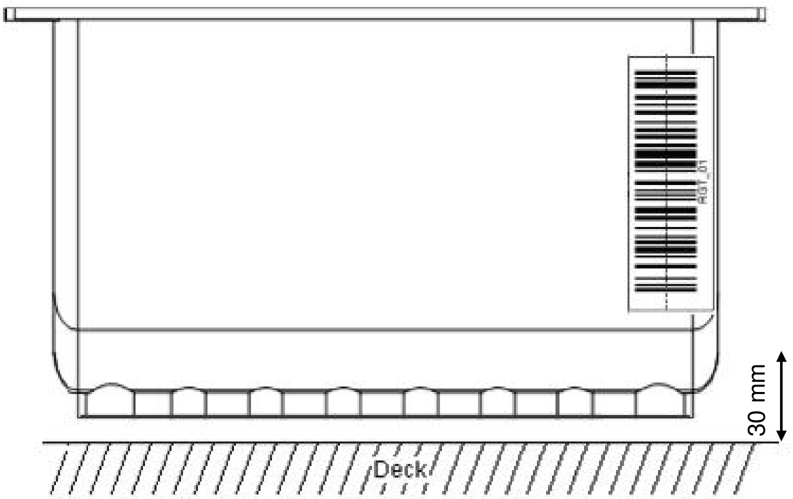
Positioning of Barcode Labels:



The label must be positioned on the upper edge, in the middle of the container.

The label must fit tightly over its whole length.

Positioning of Barcode Label on reagent trough 50ml:

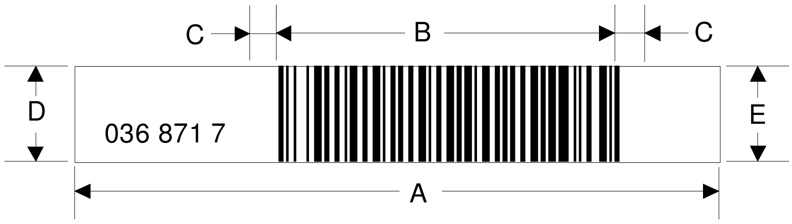


The label must be glued within a range of between 30mm to 110mm from the deck.

7.1.8.6 Plate Barcodes

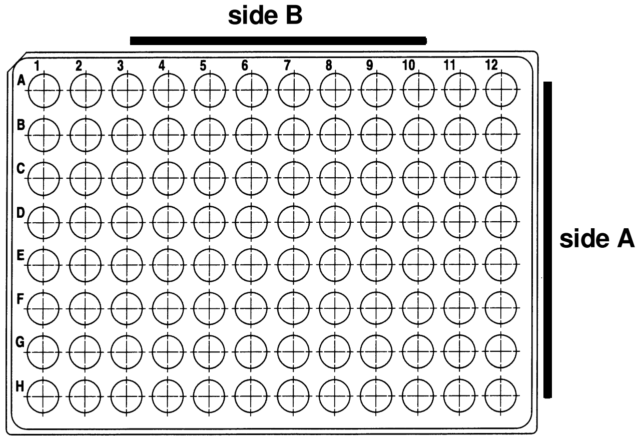
Label Specification

For general barcode specification see section 7.1.8.3



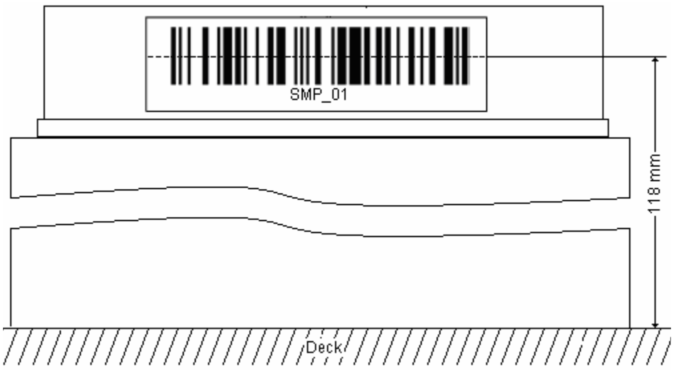
Dimension		Min	Max
A	Label length	-	66mm
B	Code length	-	30mm
C	Quiet zone	3mm	-
D	Label width	10mm	-
E	Code width	7mm	-
	Distance from code to label edge (if necessary)	-	1mm

Positioning Barcode Labels

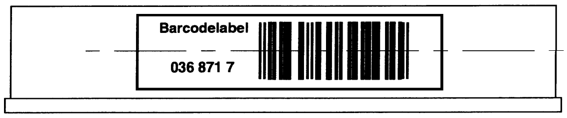


The plate barcode must fit on side A or side B of the plate.

The barcode must be positioned in the middle of the plate.



The barcode label must be centered and parallel to the edge of the plate.



The barcode label must not protrude above or below the edge of the plate.

Appendices

A Chemical compatibility

Chemical compatibility of polymers with respect to different materials

1.4310	Steel	PP	Polypropylene
EPDM	Ethylene-propylene-elastomer	PTFE	Polytetrafluorethylene
FPM	Fluoroelastomer	PVC	Polyvinylchloride
NBR	Acrylnitril-butadiene-rubber	PVDF	Polyvinylidenefluoride
PE	Polyethylene *)	FFPM	Per-Fluor-elastomer
PEEK	Polyetheretherketone	SI	Silikon
POM	Polyoxymethylene		

Effects (Key to codes in above table):

1 = no effect, little or no noticeable change

2 = slight corrosion or discoloration

3 = Moderate corrosion or other change in physical properties or dimensions; not recommended for continuous contact

4 = severe corrosion or physical change; prolonged contact not recommended

0 = No data

The table above for chemical compatibility is based on information from different manufacturers. The results refer to laboratory tests with raw materials. The outcomes with these materials are often liable to effects which cannot be observed under laboratory conditions (e.g. temperature, pressure, tension, chemical influences of substances, design features etc.). The results listed may be considered only as a guideline. In case of doubt we recommend significant tests. The chemical resistance is not sufficient for an evaluation of a particular material for a product. Particular regulations, e.g. explosion prevention in the case of flammable liquids, have to be taken into account.

Information resources:

- Otto Bürkle GmbH, http://www.buerkle.de/d2-1.htm
- Ahlborn-Kunststoffe, http://www.ahlborn-kunststoffe.at/gehr/chem_d.exe
- Prof. Dr. M. Häberlein, http://www.fbv.fh-frankfurt.de/mhwww/KAT/Bestaendigk/Bewert-Elast.htm

Chemical Resistance of CR Needle Wash Station

Chemical	1.4310	PE[1])	PP	PTFE	PEEK	FFPM	Over all resistance
Acetic acid, 20%	1	1	1	1	1	1	1
Acetic acid, glacial	1	1	1	1	1	1	1
Acetone	1	2	1	1	1	1	2
Acetonitrile	1	1	3	1	0	0	(3)
Ammonium hydroxide, 5%	1	1	1	1	1	1	1
Chloroform	1	3	3	1	1	1	3
Deionized water[2])	1	1	1	1	1	1	1
Dimethyl formamide	1	1	1	1	1	1	1
Dimethyl sulfoxide max. 30% (DMSO)	1	1	1	1	0	0	(1)
Ethyl acetate	1	2	1	1	1	1	2
Hexane	1	3	2	1	1	1	3
Hydrochloric acid, 20%	4	1	1	1	1	1	4
Isopropyl alcohol	1	1	1	1	1	1	1
Methanol	1	1	1	1	1	1	1
Methylene chloride	1	4	3	1	2	1	4
Nitric acid, 5-10%	1	1	1	1	1	1	1
Nitric acid, 70%	1	3	4	1	1	1	4
Phosphate buffer	1	1	1	1	0	1	(1)
Phosphoric acid, 85%	2	1	1	1	0	1	(2)
Potassium hydroxide conc.	1	1	1	1	1	1	1
Sodium acetate	1	1	1	1	0	1	(1)
Sodium borate	1	1	1	1	0	1	(1)
Sulfuric acid, 1-75%	2	1	1	1	2	1	2
Urine	1	1	1	1	1	1	1
Triethylamine	1	0	4	1	0	0	4
Toluene	1	3	3	1	1	1	3

[1]) *Storage Containers for Wash Liquids*

[2]) *the recommended electric conductivity should be < 1 µS/ cm*

Effects (Key to codes in above table):

1 = no effect, little or no noticeable change

2 = slight corrosion or discoloration

3 = Moderate corrosion or other change in physical properties or dimensions; not recommended for continuous contact

4 = severe corrosion or physical change; prolonged contact not recommended

0 = No data

Chemical Resistance of Wash Station 96/ 384

Chemical	1.4310	PE	PP	PTFE	PEEK	FFPM	EPT (EPDM)	Over all resistance
Acetic acid, 20%	1	1	1	1	1	1	1	1
Acetic acid, glacial	1	1	1	1	1	1	1	1
Acetone	1	2	1	1	1	1	1	2
Acetonitrile	1	1	3	1	0	0	3	(3)
Ammonium hydroxide, 5%	1	1	1	1	1	1	1	1
Chloroform	1	3	3	1	1	1	4	4
Deionized water[2])	1	1	1	1	1	1	1	1
Dimethyl formamide	1	1	1	1	1	1	1	1
Dimethyl sulfoxide max. 30% (DMSO)	1	1	1	1	0	0	1	(1)
Ethanol	1	1	1	1	1	1	1	1
Ethyl acetate	1	2	1	1	1	1	1	2
Formic acid 5%	1	1	1	1	1	1	1	1
Hexane	1	3	2	1	1	1	4	4
Hydrochloric acid, 20%	4	1	1	1	1	1	1	4
Isopropyl alcohol	1	1	1	1	1	1	1	1
Methanol	1	1	1	1	1	1	1	1
Methylene chloride	1	4	3	1	2	1	4	4
Nitric acid, 5-10%	1	1	1	1	1	1	3	3
Nitric acid, 70%	1	3	4	1	1	1	3-4	4
Phosphate buffer	1	1	1	1	0	1	1	(1)
Phosphoric acid, 85%	2	1	1	1	0	1	1	(2)
Potassium hydroxide conc.	1	1	1	1	1	1	1	1
Sodium acetate	1	1	1	1	0	1	1	(1)
Sodium borate	1	1	1	1	0	1	1	(1)
Sodium hydroxide 5%	1	1	1	1	1	1	1	1
Sodium hypochloride 10%	2L	1	1	1	0	1	1	2
Sulfuric acid, 1-75%	2	1	1	1	2	1	1-3	3
Toluene	1	3	3	1	1	1	4	4
Triethylamine	1	0	4	1	0	0	4	4
Urine	1	1	1	1	1	1	1	1

[2]) *the recommended electric conductivity should be < 1 µS/ cm*

1 = no effect, little or no noticeable change

2 = slight corrosion or discoloration

3 = Moderate corrosion or other change in physical properties or dimensions; not recommended for continuous contact

4 = severe corrosion or physical change; prolonged contact not recommended

0 = No data

B Ordering Information

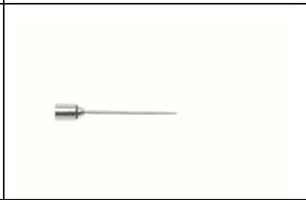
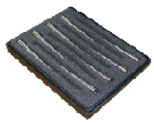
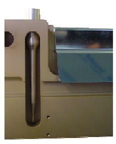
Part number	Description	
	4ml Tips	
184021	**4 ml CO-RE Tip** with Filter Conductive Case of 720 Tips (Blister Pack)	
184023	**4 ml CO-RE Tip** with Filter Conductive, sterilized Case of 720 Tips (Blister Pack)	
	5ml Tips	
184020	**5 ml CO-RE Tip** with Filter Conductive Case of 720 Tips (Blister Pack)	
184022	**5 ml CO-RE Tip** with Filter Conductive, sterilized, Case of 720 Tips (Blister Pack)	
	10µl Tips	
235900	**Low Vol. CO-RE Tips** without Filter Conductive Case of 5760 Tips (Blister Pack)	
235901	**Low Vol. CO-RE Tips** with Filter Conductive, Case of 5760 Tips (Blister Pack)	
235935	**Low Vol. CO-RE Tips** without Filter Conductive, sterilized Case of 5760 Tips (Blister Pack)	
235936	**Low Vol. CO-RE Tips** with Filter Conductive, sterilized Case of 5760 Tips (Blister Pack)	
235949	**Low Vol. CO-RE Tips** without Filter Conductive, in a Nested Tip Box Case of 11520 Tips (NTR)	
235983	**Low Vol. CO-RE Tips** without Filter Conductive, sterilized Case of 11520 Tips (NTR)	
235971	**Low Vol. CO-RE Tips** without Filter Conductive, sterilized, clear Tip Case of 11520 Tips (NTR)	

Part number	Description	
	50µl Tips	
235966	**50 µl Vol. CO-RE Tips** without Filter Conductive Case of 5760 Tips (Blister Pack)	
235948	**50 µl Vol. CO-RE Tips** with Filter Conductive Case of 5760 Tips (Blister Pack)	
235978	**50 µl Vol. CO-RE Tips** without Filter Conductive, sterilized Case of 5760 Tips (Blister Pack)	
235948	**50 µl Vol. CO-RE Tips** with Filter Conductive Case of 5760 Tips (Blister Pack)	
235947	**50 µl Vol. CO-RE Tips** without Filter Conductive, sterilized Case of 11520 Tips (NTR)	
235987	**50 µl Vol. CO-RE Tips** without Filter Conductive, sterilized Case of 11520 Tips (NTR)	
235964	**50 µl Vol. CO-RE Tips** without Filter Clear Tip Case of 11520 Tips (NTR)	
	300µl Tips	
235902	**Standard Vol. CO-RE Tips** without Filter Conductive Case of 5760 Tips (Blister Pack)	
235903	**Standard Vol. CO-RE Tips** with Filter Conductive Case of 5760 Tips (Blister Pack)	
235937	**Standard Vol. CO-RE Tips** without Filter Conductive, sterilized Case of 5760 Tips (Blister Pack)	
235938	**Standard Vol. CO-RE Tips** with Filter Conductive, sterilized Case of 5760 Tips (Blister Pack)	

300µl Tips		
235985	**Standard Vol. CO-RE Tips** without Filter Conductive, sterilized Case of 11520 Tips (NTR)	
235950	**Standard Vol. CO-RE Tips** without Filter Conductive, Case of 11520 Tips (NTR)	
235965	**Standard Vol. CO-RE Tips** without Filter Clear Tip Case of 11520 Tips (NTR)	
1 ml Tips		
235904	**High Vol. CO-RE Tips** without Filter Conductive Case of 3840 Tips (Blister Pack)	
235905	**High Vol. CO-RE Tips** with Filter Conductive Case of 3840 Tips (Blister Pack)	
235939	**High Vol. CO-RE Tips** without Filter Conductive, sterilized Case of 3840 Tips (Blister Pack)	
235940	**High Vol. CO-RE Tips** with Filter Conductive, sterilized Case of 3840 Tips (Blister Pack)	
384 / 50 µl Tips		
191102	**384-Head CO-RE Tips 50µl** without Filter Clear Tips Case of 3840 Tips (Blister Pack)	
191103	**384-Head CO-RE Tips 50µl** without Filter Clear Tips Case of 960 Tips (Blister Pack)	
235989	**384-Head CO-RE Tips 50µl** without Filter Conductive, sterile Case of 7680 Tips (NTR)	
235993	**384-Head CO-RE Tips 50µl** without Filter Conductive, sterile Case of 7680 Tips (NTR)	

Disposable Tips for CO-RE 384-Probe Head		
235974	**300µl Rocket Tips without filter** Conductive, Rack, containing 96 tips (4 channels to 1) Case of 4800 tips (Blister Pack)	

Disposable Waste Bags		
Part number	**Description**	
199201	**Plastic Chute "Biohazard" with Biohazard Labeling** (roll of 10 pcs), 700 x 600mm, PE This chute is used with Waste Container (P/N 281520)	
199202	**Waste Bag without Biohazard Labeling** (roll of 25 pcs), 460 x 500mm, PE Waste bags are used with Waste Hoop of ML STAR	
199203	**Waste Bags "Biohazard" with Biohazard Labeling** (roll of 25 pcs), 460 x 500mm, PE Waste bags with Biohazard Labeling	
185319	**Waste Chute without Biohazard Labeling** (roll of 10 pcs), 460 x 700mm, PE This chutes are used with Waste Container (P/N 281520)	
281520	**Waste Container "Biohazard"** (Pack of 10 boxes), bags are well to combustible This box is used with Waste Chute (P/N 199201 and P/N 185319)	

Steel Needles for 1000µl Single Channels		
Part number	**Description**	
235930	**1000UL NEEDLE SET CR** Set of 8x 1000 µl Needles, CR Wash Station	
235931	**300UL NEEDLE SET CR** Set of 8x 300 µl Needles, CR Wash Station	
235932	**10UL NEEDLE SET CR** Set of 8x 10 µl Needles, CR Wash Station	
182136	**SET OF 8 TEACHING NEEDLES** Used for the Maintenance to check the pressure tightness of the 1000 µl Pipetting Channels	
182176	**TEACHING NEEDLE** 1 Needle for 1000 µl Pipetting Channels, used for Labware teaching (Set of 8 -> 182136)	
184184	**TEACHING NEEDLE** 1 Needle for 5 ml Pipetting Channels, used for Labware teaching	

Reagent Containers		
Part number	**Description**	
137257	**REAGENT CONTAINER 100ml** Set of 20 Reagent Container 100ml for RGT_CAR_12R	
182703	**REAGENT CONTAINER 120ml** Set of 12 reagent troughs 120ml for carrier RGT_CAR_3R	
187297	**REAGENT CONTAINER 50ml** Set of 12 reagent troughs 50ml for carrier RGT_CAR_5R	

Liquids for maintenance		
Part number	**Description**	
289014	**HEAT EX. SOLUTION FOR TCC** Color-coded heat exchange liquid for TCC cooler	
281242	**Microlab™ Detergent & Disinfectant Kit** contents: 1 Beaker 100ml 3 Bottles DECONEX 61 DR	
281243	**Microlab™ Disinfectant Spray Kit** contents: 1 Spray Bottle 2 Bottles DECONEX SOLARSEPT	
281245	**Microlab™ Disinfectant Starter Kit** contents: 1 Beaker 100ml 1 Spray Bottle 1 Bottle DECONEX 61 DR 1 Bottle DECONEX SOLARSEPT	

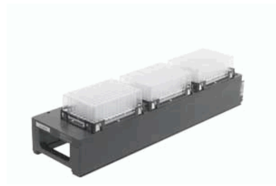
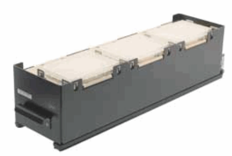
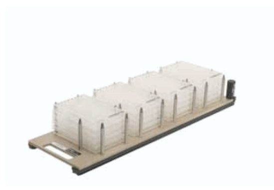
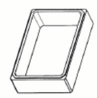
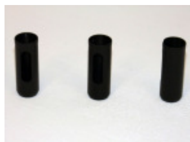
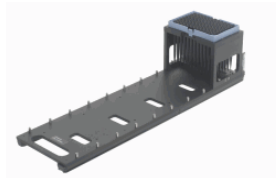
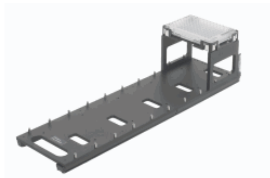
Plate Carrier		
Part number	**Description**	
182035	**PLT_CAR_L5PCR384** Carrier for 5x 384 PCR Plates	
182065	**PLT_CAR_P3AC** Carrier for 3x deep well plates, portrait orientation (6T)	
182070	**PLT_CAR_L5PCR** Carrier for 5x 96 well PCR plates (6T)	
182075	**PLT_CAR_P3MD** Carrier for 3x 96/384 well plates, portrait orientation (6T)	
182090	**PLT_CAR-L5AC** Carrier for 5 deep well plates or for 5 384 tip racks (e.g.384HEAD_384TIPS_30µl) (6T)	
191287	**PLT-CAR-L4HD** Carrier for 4x 1536 well plates (6T)	
182190	**PLT-CAR-P3HD** Carrier for 3 x 1536 well plates, portrait orientation (6T)	
182365	**PLT_CAR_L5MD** Carrier for 5x 96/384 well plates (6T)	

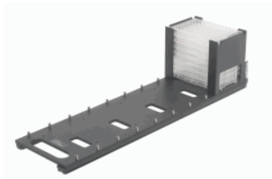
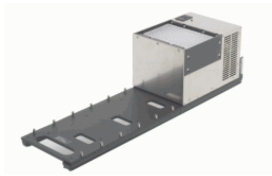
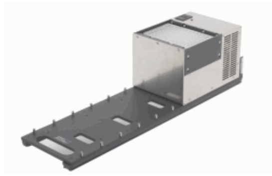
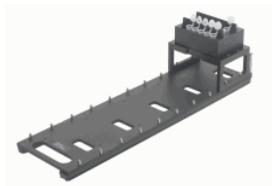
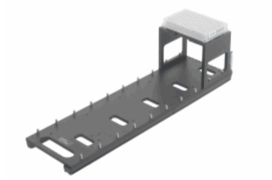
Plate Carrier		
Part number	**Description**	
185295	**PLT_CAR_L5AC PINNED** Carrier for 5x HAMILTON DWP (6T)	
185330	**PLT_CAR_L4ST (4x8 MTP)** Stacker carrier for 4 x 8 MTP (7T)	
185340	**PLT_CAR_L4ST (4x5 MTP)** Stacker carrier for 4 x 5 MTP (7T)	
182735	**PLT_CORE_COVER** Lid to cover MTP on standard carriers	
187223	**ANTI-EVAPORATIONS LID**	
182712	**FRAME FOR FILTER PLATE** To place filter plates on archive carriers	

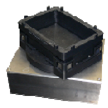
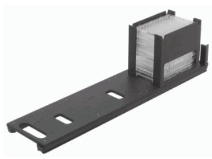
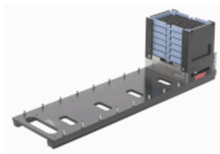
Reagent Carrier		
Part number	**Description**	
182080	**RGT_CAR_12R** Carrier for 12x reagent troughs 100ml (6T)	
185290	**RGT_CAR_3R** Carrier for 3x reagent troughs 130ml (1T)	
187239	**RGT_CAR_4R100** Carrier for 4x reagent troughs 100ml (1T)	
187299	**RGT_CAR_5R** Carrier for 5x reagent troughs 50ml (1T)	

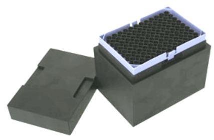
Tip Carrier		
Part number	**Description**	
182060	**TIP_CAR_288** Carrier for 3x 96 tip racks portrait for 12-channel (1000µl) MICROLAB® STAR Line or 3x 24 tip racks portrait for 6-channel (5ml) MICROLAB® STAR Line (4T)	
182085	**TIP_CAR_480** Carrier for 5x 96 tip (10µl, 50µl, 300µl, 1000µl) racks or 5x 24 tip (5ml) racks (6T)	
182390	**TIP_CAR_L384_A00** Carrier for 4x 96 tip racks for 16-Channel (1000µl) MICROLAB® STAR Line or 3x 24 tip racks portrait for 8-channel (5ml) MICROLAB® STAR Line (6T)	
182074	**TIP_CAR_NTR_A00** Carrier for Nestable Tip Racks (NTR) 6T; for 1 pack tray containing 1920x 10µl, 50µl, 300µl CO-RE Tips (5 stacks x 4 tip racks x 96 tips) or 7680x 50µl CO-RE Tips for 384-Probe Head (5 stacks x 4 tip racks x 384 tips)	
182040	**ADAPTER FOR TIP_CAR_480** Intermediate storage position for tips allows pick-up of a single tip/row/column using the CO-RE 96-Probe Head	
191055	**384 TIP SUPPORT** Intermediate storage position for tips allows pick-up of a single tip/row/column using the CO-RE 384-Probe Head The tip support fits into MultiFlex tip module (p/n 188160) or tip carrier (p/n 182085)	
191056	**ROCKET TIP SUPPORT** Intermediate storage position for tips allows pick-up of a single tip/row/column using the CO-RE 384-Probe Head The tip support fits into MultiFlex tip module (p/n 188160) or tip carrier (p/n 182085)	
182041	**DRIP PAN FOR TIP CARRIER**	

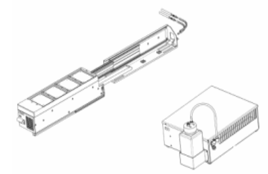
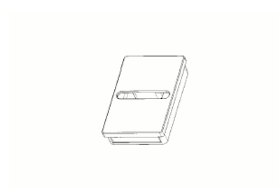
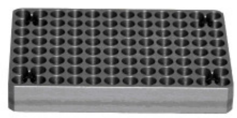
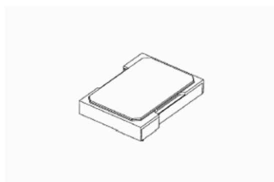
Sample Tube Carrier		
Part number	**Description**	
173400	**4 SMP-CAR-24** Set of 4x carrier for 24 tubes 14.5x60 – 18x120mm (1T)	
173410	**3 SMP-CAR-32** Set of 3x carrier for 32 tubes 11x60 – 14x120mm (1T)	
182045	**SMP-CAR-12** Carrier for 12 Falcon tubes 50ml (2T)	
182238	**EPI-INS-32L** Set of 32 inserts for 1.5ml Eppendorf cups in SMP-CAR-32	
182239	**EPI-INS-32S** Set of 32 inserts for 0.5ml Eppendorf cups in SMP-CAR-32	
187142	**24-FALCON-INS-15ML** Set of 24 inserts for 15ml Falcon tubes in SMP-CAR-24	
187350	**8 INSERTS FOR 1.5ML EPPENDORF CUPS IN SMP-CAR-32** Set of 8 inserts for 1.5ml Eppendorf cups in SMP-CAR-32	

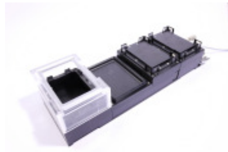
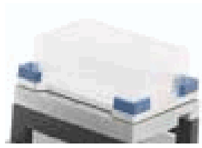
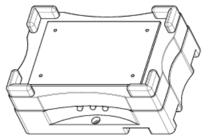
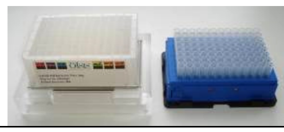
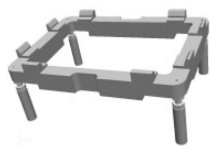
Multi-Flex Carrier		
Part number	**Description**	
188039	**MULTIFLEX CARRIER BASE (LANDSCAPE ORIENTATION)** Labware carrier base for up to 5 Multiflex modules	
188160	**MULTIFLEX TIP MODULE** Module to position a high-, standard-, low volume or 5ml tip rack (but not a 384 tip rack) The MultiFlex Carrier Base is not included.	
191420	**MULTIFLEX NTR 96 MODULE** Module to position a nestable tip rack (NTR) with standard (300µl), low volume (10µl) or 50µl tips The MultiFlex Carrier Base is not included.	
196371	**MULTIFLEX NTR 384 MODULE** Module to position a nestable tip rack (NTR) with 384 50µl tips The MultiFlex Carrier Base is not included.	
191425	**MULTIFLEX NTR4 MODULE** Module to position a stack of 4 nestable tip racks (NTR) with standard (300µl), low volume (10µl) or 50µl tips The MultiFlex Carrier Base is not included.	
188041	**MULTIFLEX MTP MODULE** Module to position 96-/384-well plates in SBS format / or flat reagent troughs	
188042	**MULTIFLEX DWP MODULE** Module to position a deep well plate / tube racks (MATRIX or MICRONICS) / NUNC reagent trough The MultiFlex Carrier Base is not included.	
188043	**CO-RE GRIPPER ON MULTIFLEX MODULE** Plate handling tool for plate transfer on the deck using two pipetting channels. Includes module with parking position for CO-RE gripper. The MultiFlex Carrier Base is not included.	

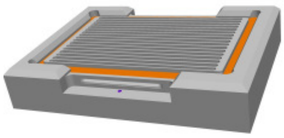
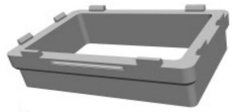
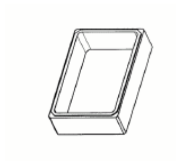
Multi-Flex Carrier		
Part number	**Description**	
188044	**MULTIFLEX STACKER MODULE (LANDSCAPE)** Module to use as passive plate hotel. Depending on plate height, up to 10 plates can be stacked on one position. The MultiFlex Carrier Base is not included.	
188045	**MULTIFLEX HEATING MODULE** Heating module (up to 60°C) including one adapter for one labware type. Temperature: Room temperature up to 60°C Temperature gradient: ± 1°C The MultiFlex Carrier Base is not included.	
188046	**MULTIFLEX COOLING MODULE** Cooling module (4 - 15°C) including one adapter for one labware type. Temperature: 4 – 15 °C. Temperature gradient: ± 1°C The MultiFlex Carrier Base is not included.	
188047	**MULTIFLEX REAGENT TROUGH MODULE** Module to hold six 50ml troughs The MultiFlex Carrier Base is not included.	
188048	**MULTIFLEX TUBE / CUP MODULE** Module to hold EPPENDORF, SARSTETT, NUNC tubes 0.5/ 1.5 / 2.0 ml with or without snap-lid in an passively cooled adapter The MultiFlex Carrier Base is not included.	
188049	**MULTIFLEX PCR PLATE MODULE 96** Module to position a 96-well PCR plate The MultiFlex Carrier Base is not included.	
188052	**MULTFLEX PCR PLATE MODULE 384** Module to position a 384-well PCR plate The MultiFlex Carrier Base is not included.	
188053	**MULTIFLEX CARRIER BASE (PORTRAIT ORIENTATION)** Labware carrier base for up to 3 Multiflex modules	

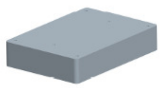
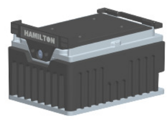
Multi-Flex Carrier		
Part number	**Description**	
188055APE	**PLATE TURNTABLE** Automated device to turn plates from landscape into portrait orientation or vice-versa The MultiFlex Carrier Base is not included.	
188057APE	**MULTIFLEX DOWNHOLDER MODULE** Module to hold down sealed plates The MultiFlex Carrier Base is not included.	
188058APE	**MULTIFLEX LID PARKING MODULE** Module to park the lid of the cooling or heating module The MultiFlex Carrier Base is not included.	
188059	**MULTIFLEX PLATE STACKER MODULE (PORTRAIT)** Module to use as passive plate hotel. Depending on plate height, up to 10 plates can be stacked on one position. The MultiFlex Carrier Base is not included.	
188114APE	**MULTIFLEX LIQUID DISPENSER TROUGH 8** Module to automatically refill a trough on the deck with fresh reagent. Compatible with 8-channel instruments. The MultiFlex Carrier Base is not included.	
188061APE	**MULTIFLEX TILT MODULE** Module to tilt plates on the y-axis (in landscape orientation) The MultiFlex Carrier Base is not included.	
188062APE	**MULTIFLEX TIP STACKER MODULE** Module to hold 4 standard volume or 6 low volume tip racks on one position The MultiFlex Carrier Base is not included.	
188063APE	**MULTIFLEX SEESAW MODULE** Module to shake bead-, cell-, or reagent solutions at a predefined speed in troughs or plates. Angle is adjustable up to 15°. 16-69 rpm adjustable in 4 steps. The MultiFlex Carrier Base is not included.	
188115APE	**MULTIFLEX LIQUID DISPENSER TROUGH 96** Module to automatically refill a trough on the deck with fresh reagent. Compatible with 96-channel instruments. The MultiFlex Carrier Base is not included.	

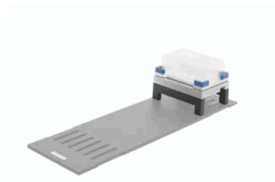
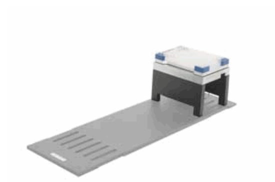
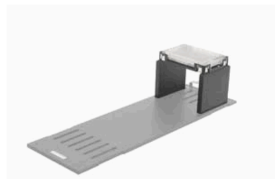
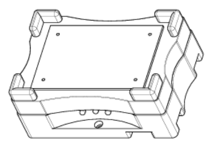
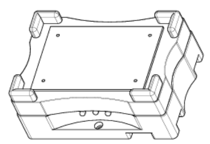
Multi-Flex Carrier		
Part number	**Description**	
188078APE	**MULTIFLEX STERILE TIP BOX MODULE** Module to store sterile tips on the deck.	
188082APE	**MULTIFLEX CORE-LID TOOL (CLT)** Suction cup that can picked up by a CORE- channel to thus move around lids	
182774	**STAR SHELF 4MTP/2AC PLATES** Shelving Unit for 4 MTP or 2 archive plates	

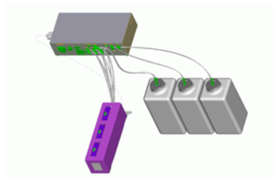
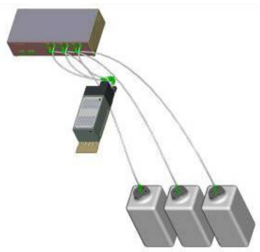
Temperature Controlled Carrier (TCC)		
Part number	**Description**	
182400	**TCC FOR ML STAR/PLATES** Temperature Controlled Carrier for MTP (no DWP)	
182436	**TCC-RGT-TUB** 100ml reagent tub with lid for TCC	
182532	**TCC-PCR-PLT-ADAPTERKIT 96** Set of 4 TCC-adapter for 96 well PCR plates	
182673	**TCC-PCR-PLT-ADAPTERKIT 384** Set of 4 TCC-adapter for 384 well PCR plates	
182734	**TCC-LID-ADAPTERKIT** Set of 4 lids to cover plates on the TCC	

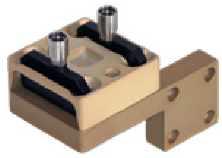
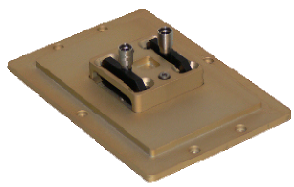
BVS / CVS Vacuum System		
Part number	**Description**	
199020 **190021**	**CVS VACUUM SYSTEM INCLUDING PUMP (230V)** **CVS VACUUM SYSTEM INCLUDING PUMP (115V)** Carrier with a vacuum box, one park position for manifold top and two plate positions including Vacuubrand Pump ME 4C VARIO with controller, pressure sensor and air-bleed valve	 
187143 **187149**	**TELESHAKER 220V FOR DWP** **TELESHAKER 115V FOR DWP** Maximal two shakers per CVS Carrier.	
187788	**SHAKER HEATER CAT SH10 FOR DWP (110/220V)** Shaker heater (Shaking: 200 - 1200 1/min, Orbital: 2mm, Temperature: from RT+5°C - 90°C) for standard 96 and deep well plates Maximal one shaker heater per BVS carrier.	
281539	**BVS 2-LITRE WASTE BOTTLE** 2 litres Waste Bottle for the BVS, including connectors	
281540	**BVS 4-LITRE WASTE BOTTLE** 4 litres Waste Bottle for the BVS, including connectors	
190034	**DWP KIT FOR BVS** Manifold Top for BVS and insert to adapt the BVS to the used collecting plate	 
186320	**BVS INSERT KIT** Adjustable insert to adapt the BVS to the used collecting plate	
190035	**BVS INSERT KIT FOR MTP** Insert to adapt the BVS to the used collecting plate	

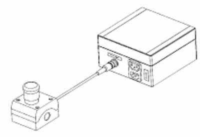
BVS / CVS Vacuum System		
Part number	**Description**	
190036	**BVS INSERT KIT FOR DWP** Insert to adapt the BVS to the used collecting plate	
190037	**BVS GRID I FOR MILLIPORE** Manifold Top for BVS incl. Grid I for Millipore Kits "Montage Plasmid Miniprep96 Kit" and "Montage PCR96 Cleanup Kit"	
186321	**BVS GRID II FOR MILLIPORE** Manifold Top for BVS incl. Grid II for Millipore Kits with 96/384 SBS plates with one filter	
186303	**BVS ADAPTER FOR MN DNA KIT** Insert to adapt the BVS to the used collecting plate	
182712	**FRAME FOR FILTER PLATE** To place filter plates on archive carriers	

Shaker Carrier		
Part number	**Description**	
187001	**PLT_CAR_L4_SHAKER** Template carrier with 4 positions for Hamilton Heater Shaker (optional: Shaker H+P, Shaker Heater CAT) and plate bases (7T)	
190755	**HEATER SHAKER BOX** Needed if more than two HAMILTON Heater Shakers are integrated (or the connectors “TCC1”, “TCC2” on the STAR are occupied by other devices). One heater shaker module is connected to a USB port of the PC and serves as master module for up to seven additional HHSs. The modules are connected to the external heater shaker box (HSB), which serves as power supply and as signal distributor.	
199013	**HAMILTON HEATER SHAKER SUPPORT BLOCK FOR MTP** Hamilton Heater Shaker support block for MTP. Used to raise the heater shaker if only MTPs are processed.	
199033	**HEATER SHAKER 3.0 MTP FLAT BOTTOM** Heater Shaker with 3.0mm shaking amplitude and flat bottom adapter (Shaking speed DWP/MTP: 100 - 2000/2500 rpm, Temperature control: RT+5°C - 105°C, max. loading: 300g)	
199034	**HEATER SHAKER 2.0MM FLAT BOTTOM** Heater Shaker with 2.0mm shaking amplitude and flat bottom adapter (Shaking speed DWP/MTP: 100 - 2000/2500 rpm, Temperature control: RT+5°C - 105°C, max. loading: 300g)	
199038	**HEATER SHAKER 2.0 NUNC DWP 96 2ML** Heater Shaker with 2.0mm shaking amplitude and flat bottom adapter (Shaking speed DWP/MTP: 100 - 1800/2400 rpm, Temperature control: RT+5°C - 105°C, max. loading: 300g)	
199039	**HEATER SHAKER 3.0 NUNC DWP 96 2ML** Heater Shaker with 3.0mm shaking amplitude and fitted adapter for plates (Shaking speed DWP/MTP: 100 - 2000/2500 rpm, Temperature control: RT+5°C - 105°C, max. loading: 300g)	
188318	**HEATER SHAKER 2.0MM MTP FLAT BOTTOM** Heater Shaker with 2.0mm shaking amplitude and fitted adapter for Abgene 96 deep well plates (Shaking speed DWP/MTP: 100 -2000/2500 rpm, Temperature control: RT+5°C - 105°C, max. loading: 300g)	

Shaker Carrier		
Part number	**Description**	
188319	**HEATER SHAKER 3.0MM MTP FLAT BOTTOM** Heater Shaker with 3.0mm shaking amplitude and fitted adapter for Abgene 96 deep well plates (Shaking speed DWP/MTP: 100 -1800/2400 rpm, Temperature control: RT+5℃ - 105℃, max. loading: 300g)	
187143 **187149**	**TELESHAKER 22OV FOR DWP** **TELESHAKER 115V FOR DWP** Shaker (Shaking: 100 - 2000 1/min, Orbital: 2mm) for standard 96 and deep well plates on the PLT_CAR_L4_shaker (7T)	
187295 **187296**	**TELESHAKER 220V FOR MTP** **TELESHAKER 115V FOR MTP** Shaker (Shaking: 100 - 2000 1/min, Orbital: 2mm) for standard 96 well plates (no DWP) on the PLT_CAR_L4_shaker (7T)	
187144	**DWP BASE FOR SHAKER CAR** Position for standard and deep well plates on the PLT_CAR_L4_shaker (7T)	
187292	**MTP BASE FOR SHAKER CAR** Positions for standard well plates (no DWP) on the PLT_CAR_L4_shaker (7T)	
187788	**SHAKER HEATER CAT SH10 FOR DWP (110/220V)** Shaker heater (Shaking: 200 - 1200 1/min, Orbital: 2mm, Temperature: from RT+5℃ - 90℃) for standard 96 and deep well plates on the PLT_CAR_L4_shaker (7T)	
187789	**SHAKER HEATER CAT SH10 FOR MTP (110/220V)** Shaker heater (Shaking: 200 - 1200 1/min, Orbital: 2mm, Temperature: from RT+5℃ - 90℃) for standard 96 well plates (no DWP) on the PLT_CAR_L4_shaker (7T)	

Wash Stations		
Part number	**Description**	
186360	**CR NEEDLE WASH STATION** Wash Station for all needle types, two wash liquids, Needle Wash in parallel to the pipetting	
190247	**96 WASH STATION DUAL** Wash Station for 96 disposable tips, CO-RE 96-Probe Head, 2 Wash Chambers	
190248	**384 WASH STATION DUAL** Wash Station for 96/384 disposable tips, CO-RE 384-Probe Head, 2 Wash Chambers	

CO-RE Gripper		
Part number	**Description**	
188066	**CO-RE GRIPPER WITH ATTACHMENT FOR WASTE BLOCK** Plate handling tool for plate transfer on the deck using two 1000µl pipetting channels. Includes parking position for attachment to waste block (waste block not included). Note: Order this gripper if your MICROLAB® STAR Line instrument is equipped with 1000µl pipetting channels only.	
184089	**CO-RE GRIPPER WITH ATTACHMENT FOR WASTE BLOCK (INSTRUMENTS WITH 1000UL AND 5ML CHANNELS)** Plate handling tool for plate transfer on the deck using two 1000µl pipetting channels. Includes parking position for attachment to waste block (waste block not included). Note: Order this gripper if your MICROLAB® STAR Line instrument is equipped with 1000µl and 5ml pipetting channels.	
184099	**CO-RE GRIPPER WITH ATTACHMENT FOR WASTE BLOCK (INSTRUMENTS WITH 5ML CHANNELS)** Plate handling tool for plate transfer on the deck using two 5ml pipetting channels. Includes parking position for attachment to waste block (waste block not included). Note: Order this gripper if your MICROLAB® STAR Line instrument is equipped with 5ml pipetting channels.	
186100	**CO-RE GRIPPER** Gripper tool for plate transport with pipetting channels, including parking position	

Emergency stop box		
Part number	**Description**	
186060	**EMERGENCY STOP BOX** Category 0 Emergency Stop Button	

Software	
Part number	**Description**
911004	**VENUS TWO BASE PACK 4.3.**
911095	**VENUS TWO DYNAMIC SCHEDULER 5.1**
911099	**VENUS TWO TADM FEATURE 5.1**
910111	**VENUS TWO BASE PACKAGE 4.3 FOR MICROLAB SWAP**
911122	**VENUS TWO DATA BASE PLUS 1.1**

C Regulatory Affairs

CE, CSA and UL conformity are maintained for MICROLAB® STAR Line. See the Declaration of Conformity for the instrument reproduced on the next page.

Radio Interference (USA and Canada)

This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to both Part 15 of the FCC Rules and the radio interference regulations of the Canadian Department of Communications. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses and can radiate radio frequency energy and, if not installed and used in accordance with the present user manual, may cause harmful interference to radio communications. Operation of this equipment in a residential area is likely to cause harmful interference, in which case the user will be required to correct the interference at his own expense.

Pursuant to the Canadian Radio Interference Regulations, ICES-001 Notice for Industrial, Scientific and Medical Radio Frequency Generators, this ISM apparatus meets all requirements of the Canadian Interference-Causing Equipment Regulations. Please note that this requirement is only for generators which operate at over 10,000 Hz.

In Vitro Diagnostics

The MICROLAB® STAR Line is *not* an In Vitro Diagnostic Device. The following text defines an "In Vitro Diagnostic Device" [from: Directive 98/79/EC of the European Parliament and of the Council of 1998-10-27 on in vitro diagnostic medical devices]:

'[...] *in vitro* diagnostic medical device' means any medical device which is a reagent, product, calibrator, control material, kit, instrument, apparatus, equipment, or system, whether used alone or in combination, intended by the manufacturer to be used *in vitro* for the examination of specimens, including blood and tissue donations, derived from the human body, solely or principally for the purpose of providing information:

- concerning a physiological or pathological state, or
- concerning a congenital abnormality, or
- to determine the safety and compatibility with potential recipients, or
- to monitor therapeutic measures.

Specimen receptacles are considered to be *in vitro* diagnostic medical devices. 'Specimen receptacles' are those devices, whether vacuum-type or not, specifically intended by their manufacturers for the primary containment and preservation of specimens derived from the human body for the purpose of *in vitro* diagnostic examination.

Products for general laboratory use are not *in vitro* diagnostic medical devices unless such products, in view of their characteristics, are specifically intended by their manufacturer to be used for in vitro diagnostic examinations; [...]'

D Declaration of Conformity

Each individual instrument includes a Declaration of Conformity.

E Glossary

ADC

Anti Droplet Control

Adjustment

Detailed positional setting for the hardware

Air displacement tip

Commercial pipetting tip

Aliquot

Aliquots are identical small volumes of liquid.

Aspirate

To draw up liquid into a pipetting device.

Autoload

Option and hardware assembly that enables automatic loading of the MICROAB® STAR Line. It consists of a loading head movable in Y direction, which draws the carriers into the MICROLAB® STAR Line and can read the barcodes on them.

AutoLoad tray

Hardware unit. The carriers can be placed on it and held outside the MICROLAB® STAR Line. The loading tray is attached to the MICROLAB® STAR Line, to support the automatic loading and unloading process.

Barcode Mask

The barcode mask defines the basic structure of a barcode. It is a pattern to which a barcode must conform. The assignment of a specific labware item can be done this way. The barcode mask can require a barcode to contain specific strings at fixed positions. It can contain wildcards, too.

Barcode Reader

Device for reading sample/plate Barcodes. Part of the Autoload Option.

Basic MICROLAB® STAR Line

Basic parts of the MICROLAB® STAR Line with pipetting arm and deck, to which the loading unit and the options can be added on.

Carrier

Unit for loading plates, tubes and tips on the MICROLAB® STAR Line deck. Can be used manually or, if possible, by the Autoload option.

Container

A container defines a tube, vessel or a single well of a plate.

Container identification

Barcode for the identification of a container. Serves for unambiguous identification of the vessel, e.g. a sample test tube.

Continuous loading

Refers to the loading of elements that can be manipulated onto the MICROLAB® STAR Line after processing has been started.

Deck

The work surface of the MICROLAB® STAR Line. It presents at the same time the greatest possible area, cf. *Work Area*. The placing of the carriers on it is defined by the tracks, as long as they are in the operating range of the pipetting area.

Deck Layout

A collection of Labware placed upon a Deck.

Dispense

To distribute quantities of liquid from a pipetting device.

Docking station

The long bar at the back of the MICROLAB® STAR Line Instrument for guiding the cables and tubing of accessories, as Wash Stations, TCC, etc.

Firmware

Lower Level program code that is carried out on the processors of the MICROLAB® STAR Line Instrument.

Front Cover

Protective covering for the MICROLAB® STAR Line Instrument, featuring a hinged front window made of transparent Plexiglas. With this option and assembly, the work surface of the MICROLAB® STAR Line is covered in such a way that it is shielded from user intervention and other outside influences (such as dust). At the same time, it protects the user from the movements of the MICROLAB® STAR Line.

Hardware error

Type of error that is caused by a technical problem of the hardware.

HSL

Hamilton Standard Language

Instrument

Hardware of the MICROLAB® STAR Line (mechanics, electronics, and firmware)

Instrument commands

The commands made available by the firmware for controlling the MICROLAB® STAR Line.

Labware

Refers to movable items to be placed on the MICROLAB® STAR Line deck, such as carriers, containers, or racks.

LIMS

Higher-level data processing system, generally known as Laboratory Information Management System, also LMS.

Liquid

Includes all kinds of liquids, among which are included reagents, controls, standards, wash fluids.

LLD (Liquid Level Detection)

Positive pipetting of liquid which may be achieved either by pressure, or capacitive signal detection and transfer.

Loading, unloading

The process by which plate, tube and tip carriers are brought on and off the MICROLAB® STAR Line deck. This can happen automatically by means of the Autoload Option, or manually.

MAD

Monitored Air Displacement: aspiration monitoring feature. During the aspiration process, the pressure within the pipetting channel is measured in real time.

Method

The method contains all instructions as to how the content of the source container is to be processed.

MICROLAB® STAR Line VENUS Software

Software to run the MICROLAB® STAR Line.

MTP (Microtiter plate)

In general we assume a plate with 96 Wells (8 x 12) 9 mm wide.

There are also plates with 384 Wells (16 x 24 / 4.5 mm), or others with a different size.

Pause

Interruption of processing. The current processing steps are completed.

Pipetting

Transfer of liquids, usually a defined volume, from one container to another.

Pipetting arm

Assembly equipped with the pipetting tool and/or plate handler, as well as the common X-drive.

Pipetting channel

Hardware including the function of picking up a tip, aspirating, dispensing, tip eject, liquid level detection and the Y/Z- movements.

Pipetting module

Firmware (-processor-program) which controls a pipetting channel, in which category are included the Y and Z pipetting movement, and the LLD.

Pooling

Pipetting of different liquids in one well:

1,2,3 .... to n, and n to 1,2,3,....

Processing step

Defines what must be carried out on the MICROLAB® STAR Line Instrument, as well as the location where it must be carried out and possible interaction with other system components or labware. The action is defined in accordance with the methods, the loading, and the tasks.

Pump station

Part of the needle/tip wash station. Its function is to pump wash liquid to and from the wash station.

Rack

Grouping of containers, as DWP, MTP, etc.

Rack identification

Barcode for Rack identification.

Random access

Means that every channel can access any place on the work area.

Run

Execution of the processing steps defined in the relevant method with the aim of processing one or more liquids and containers (e.g. MTP). The run is a series of timed commands, in order to carry out processing on the MICROLAB® STAR Line according to the processing plan. The run can be interrupted to load more elements. Then processing goes on according to a newly calculated processing plan, with the run being started again. Loading is not a part of the run.

Run abort

Cancelled run by the user or by the MICROLAB® STAR Line.

Run Visualization

Visualization of the current run, reporting the status of the MICROLAB® STAR Line.

Sample

Refers to a liquid in an unambiguously identified container which is to be processed.

Stacker

Storage unit for racks

TADM

Total Aspiration and Dispense Monitoring: The pressure inside each individual pipetting channel is monitored, during aspiration and dispensing.

Tip

Disposable tip for pipetting

Tip rack

Frame that hold the tips.

Tip waste

Container for ejecting tips.

Touch-off

Type of dispensing whereby the tip or needle approaches the bottom of the empty container so close as to allow the dispensed droplet to have simultaneous contact with the tip or needle and the container bottom.

Trace

Record of the status during processing.

Tube

A container for liquid, usually having a circular cross-section, and a cylindrical longitudinal section.

User

User of the software. Access rights for different types of users can be defined, such as operators, laboratory managers, etc.

Verification Kit

Balance, liquid, disposable tips to verify the function (volume check) of the MICROLAB® STAR Line pipetting heads.

Waste Container

A device on the MICROLAB® STAR Line deck to collect used disposable tips.

Well

The individual container of a MTP or DWP.

Well type

Geometrical shape of the well, such as U, V, or flat shape.

Wick side of container

Type of dispensing whereby the tip or the needle touches the side of a container and thus releases the droplet. This function is not available for MICROLAB® STAR Line instruments.

Work area

The area on the MICROLAB® STAR Line to which access is provided during processing. Elements to be pipetted or handled can be placed in this area.

Worklist

Information sent from outside the system, as to what method(s) is (are) to be executed on the MICROLAB® STAR Line, and with what liquid.